

The centre column of Rule 30: excluded periods, decimation of the Duhamel class, and the front of the diagonal defects

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Abstract

Let $c = (c_t)$ be the centre column of the elementary cellular automaton Rule 30 started from a single black cell. Wolfram's first Prize Problem asks whether c is eventually periodic; the second asks whether its density is $\frac{1}{2}$; the third asks whether computing c_t requires at least $O(t)$ computational effort. Wolfram names an algorithm below the direct $O(t^2)$ as a partial result; one running in $O(t^2/\log t)$ time is already in print, and it is not Problem 3 itself. We collect what this programme established about c , each statement labelled PROVED, EXACT (a finite computation with no fitting), MEASURED, KNOWN (in the literature) or OPEN. Part I proves that infinitely many period words are impossible as eventual periods of c , for every number-like configuration: every necklace all of whose cyclic zero-runs have length at least 19 (with 1-runs arbitrary; at least 17 for every necklace tested), every word with isolated ones and zero-runs of length at least 10, the words $0^m 11$ ($m = 6, m \geq 8$), $1^b 0^a$ ($b \geq 9$), $1^b v$ ($b \geq 29, |v| \leq 12$), and both constants; the method is a finite-state over-approximation of the right half whose reach is characterised exactly. At the boundary of that reach two facts are proved — a finite block that traps the interface, and a cap of six cells on straight influence into it — and what is left open there is whether the interface language over 01 contains an interface that is not eventually periodic: uncountability of that language would suffice, is not necessary, and no enumeration can reach it (positive entropy of that language is MEASURED on random right halves; its uncountability is OPEN). Part II writes c as a linear functional of the quadratic source of Rule 30 through the Rule 150 propagator, proves that this class of functionals is closed under the Cartier operators (so the 2-kernel of c is the image of an explicit orbit), gives the functional in closed form in the variable α with $y = \alpha/(1 + \alpha + \alpha^2)$, and shows that only the centre line and the chiral part of the source are visible to c ; certified computations give no Mahler relation of depth ≤ 12 and degree ≤ 8000 , and a 2-kernel census that is injective wherever it has been compared, giving $|\mathcal{K}(c)| \geq 138\,672$ pairwise distinct truncations from 2^{21} terms; controls show, however, that such a census measures counting room rather than aperiodicity, and the family it can certify from 2^ν terms is capped at $2^{\nu-4} - 1$ elements (about $n/16$), which closes the census route. Part III splits each left diagonal into a periodic background and a finitely supported defect, proves the reset formula, the exact degree formula, monotonicity of the defect front, an exact failure condition, and Theorem H*, which reduces the conjecture that the periodic texture never crosses the centre to two inequalities on the front, on cancellation depths and on stall runs, whose margins are measured and wide; it also re-derives, in the diagonal frame, the unboundedness of the diagonal periods; that result is KNOWN, being Gravner–Griffeath's Proposition 3.3 (a left-permutative edge cellular automaton has no periodic solution), stated there for Rule 30, and

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the argument given here is theirs in different coordinates. Part III also treats the existence of the front limit $\lim M_d/d$, which no theorem in the literature supplies for this edge, and reduces it twice: the near-diagonal inequality it needs follows if M stays within $O(N^\alpha)$ of its least concave majorant for any $\alpha < 1$ — strictly weaker than the constant we had been pursuing, since $\sum n^{\alpha-2}$ converges throughout that range — and, separately, M_N/N equals exactly $\sum_{k \geq 1} q_k(N)$ at every N , where $q_k(N)$ is the frequency of jumps of size at least k below N , so the existence of the limit reduces to the convergence of one increment’s distribution with a summable dominant. Part IV records the collapse pyramids over the column, with an exact identity for the traffic rules and a two-sided criterion, together with measurements showing every bounded-window predictor of c tested ($w \leq 22$) at chance, and a remark on why that statement must stay MEASURED. Part V asks what is provable about Problem 3. The window transformation monoid of Rule 30 contains A_5 and is therefore not solvable, placing the rule on the hard side of the Barrington–Thérien and Moore dichotomies. A rank lower bound on the prediction matrix does not reach the column. The rank is exactly 2 at every $n \leq 10$ for Rules 90, 150 and 105 and about 1.59^n for Rule 30 on that range (MEASURED), but Rules 22 and 54, whose centre columns are respectively eventually 0 and periodic with period 4, have rank over $\mathbb{F}_2$ of the same order as Rule 30’s or larger at every n computed (EXACT): the rank measures the prediction function over all inputs, not the single-seed column, and that route to Problem 3 is closed. A second control inside our own family — Rule 126, whose centre column agrees for every $t \leq 2^{14}$ with the Fredholm–Rueppel sequence, which has the perfect linear-complexity profile and whose t -th bit is computable in $O(\log t)$ — bounds what any complexity measurement can establish. No route to any of the three Prize Problems is claimed. Part VI states exactly where each route stops, and a separate section states why no measure-level argument can reach Problem 2.

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1 Conventions and labels

Rule 30 acts on configurations $x \in \{0, 1\}^{\mathbb{Z}}$ by

$$a(n+1, i) = a(n, i-1) \oplus (a(n, i) \vee a(n, i+1)), \quad a(0, \cdot) = x, \quad (1)$$

equivalently, over $\mathbb{F}_2$, $u_x^{t+1} = u_{x-1}^t + u_x^t + u_{x+1}^t + u_x^t u_{x+1}^t$. Part I uses the notation $a(n, i)$ of the exclusion paper; Parts II–IV use u_x^t , with $c_t = u_0^t$ the centre column and $u_x^0 = [x = 0]$ the single seed. A sequence is *eventually periodic* if it agrees with a periodic sequence from some index on.

Every statement carries one of five labels. **PROVED**: a written argument, machine-checked where finite, with the certificate identified. **EXACT**: an exact finite computation, reproducible from the script named. **MEASURED**: a statistic, a fit, or an out-of-sample test; never used as a hypothesis of a theorem. **KNOWN**: in the literature before this programme, cited. **OPEN**: not settled here. Nothing labelled **MEASURED** is used in a proof.

The three Prize Problems are: **P1**, c is not eventually periodic; **P2**, the density of ones in c tends to $\frac{1}{2}$; **P3**, computing c_t requires at least $O(t)$ computational effort (Wolfram’s wording, with t for his n ; Section 23). None is settled here. An algorithm below the direct $O(t^2)$ is the partial result Wolfram asks for, not P3 itself, and it is **KNOWN**: c_t is computable in $O(t^2/\log t)$ time [40].

2 Two tiers of equivalence, and which one is P1

Several distinct conditions on c are easy to conflate, and the routes in this paper divide according to which tier they aim at, so we set them out once. Write $F(x) = \sum_n c_n x^n \in \mathbb{F}_2[[x]]$.

Proposition 2.1 (**KNOWN**). *The following are equivalent, and each is the negation of P1:*

- (i) c is eventually periodic;
- (ii) F is rational over $\mathbb{F}_2(x)$;
- (iii) c satisfies a finite linear recurrence with constant coefficients;

- (iv) the subword complexity $p(n)$ is bounded;
- (v) the linear complexity of c is finite.

The following are equivalent to each other, and are strictly weaker:

- (a) c is 2-automatic;
- (b) F is algebraic over $\mathbb{F}_2(x)$;
- (c) the 2-kernel $\mathcal{K}(c)$ is finite.

The second tier implies $p(n) = O(n)$ and does not imply the first.

Proof. In the first tier, (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) is the standard correspondence between rational series and constant-coefficient recurrences, together with the observation that over a finite field a recurrence of order d has a finite state space and so forces a cycle; (i) $\Leftrightarrow$ (iv) is Morse–Hedlund; (v) restates (iii). In the second tier (a) $\Leftrightarrow$ (b) is Christol’s theorem [44] and (a) $\Leftrightarrow$ (c) is the Eilenberg characterisation [73]. The implication to $p(n) = O(n)$ is Cobham’s. Both non-implications are witnessed: Thue–Morse is 2-automatic and not eventually periodic, so the second tier does not imply the first; and the Fibonacci word has $p(n) = n + 1$, linear yet not automatic, so linear complexity does not imply the second tier either. $\square$

Two consequences worth keeping in view. First, the weakest hypothesis that settles P1 is that $p(n)$ be *unbounded* — superlinear growth is more than is needed, and neither Christol’s theorem nor any transcendence argument is required for that route, since Morse–Hedlund gives it directly. Second, and this is why the automaticity machinery of Part II aims at the second tier rather than the first: a proof that c is not 2-automatic gives P1 as a corollary, because the first tier implies the second, and the second tier is where the certified computations of Theorem 14.1 have purchase. What no computation can do is establish either tier from a prefix, by Remark 14.5: we compute $p(n) = 2^n$ exactly for $n \leq 16$ (EXACT) and that bounds no period beyond the range computed.

Part I

Excluded period words

*Source: rule30_excluded_words.tex (2026-09-09), reproduced with its labels prefixed **ex**:. The only input from the seed is Theorem 3.1; everything else holds for every number-like configuration.*

3 Setting and the two-column theorem

Rule 30 acts on configurations $x \in \{0, 1\}^{\mathbb{Z}}$ by

$$a(n+1, i) = a(n, i-1) \oplus (a(n, i) \vee a(n, i+1)), \quad a(0, \cdot) = x. \quad (2)$$

A configuration is *number-like* [65, Def. 2.4] if $x \neq 0^{\mathbb{Z}}$ and $x_i = 0$ for all $i < N$, some N . The single seed δ_0 is number-like.

For columns $i \leq j$ the *trace* is $\text{Tr}_{[i,j]}(x)(t) = (a(t, i), \dots, a(t, j))$. Write $c = \text{Tr}_0$ (the centre column), $u = \text{Tr}_1$, $\ell = \text{Tr}_{-1}$. A sequence s is *eventually periodic* if $s(t+p) = s(t)$ for all $t \geq T$, some $p \geq 1$, $T \geq 0$.

A word $w \in \{0,1\}^p$ is a *tail* of c if there is N with $c(t) = w[(t - N) \bmod p]$ for all $t \geq N$. If w is a tail so is every rotation of w ; statements below are about necklaces, and any rotation may be certified.

Theorem 3.1 (two-column theorem; KNOWN: Jen [12, 13]; Kopra [65, Thm. 3.5]). *Rule 30 is left permutive—(2) is a bijection in $a(n, i - 1)$ for fixed $a(n, i), a(n, i + 1)$ —and left spreading, since $001 \mapsto 1$. Consequently, for every number-like x and every i , the width-2 trace $\text{Tr}_{[i, i+1]}(x)$ is not eventually periodic.*

Proof (for completeness; the argument is Jen’s). Suppose $\text{Tr}_{[i, i+1]}$ is p -periodic for $t \geq T$. Solving (2) for $a(n, i - 1)$,

$$a(n, i - 1) = a(n + 1, i) \oplus (a(n, i) \vee a(n, i + 1)), \quad (3)$$

so Tr_{i-1} is p -periodic for $t \geq T$, and by induction every column to the left of i is p -periodic for $t \geq T$. Let $M \geq N$ be the position of the leftmost 1 of x . Since $x_j = 0$ for $j < M$ and $001 \mapsto 1$, $000 \mapsto 0$, the leftmost 1 moves one site left per step: the column at site $M - d$ is 0 for $t < d$ and 1 at $t = d$. For $d > \max(T + p, M - i - 1)$, so that the column at site $M - d$ is one of those shown to be p -periodic from T , this column takes the values 0 at time $d - p$ and 1 at time d , both $\geq T$, so it is not p -periodic there. $\square$

Only the zeros to the left of N are used, which is why the theorem holds for all number-like x and not just for finite-support ones. Kopra [65, closing remarks] observes that Rule 90 satisfies every hypothesis of Theorem 3.1 and has a periodic centre column from the seed; so no argument at the level of Theorem 3.1 alone can decide the centre column, and everything below uses the specific gate of Rule 30, the OR in (2). The eventually periodic structure of the *left* side of the seed’s evolution is described by Rowland [66] and Gravner–Griffeath [67]; nothing there constrains column 0.

4 Left reconstruction and the interface

Lemma 4.1 (PROVED). $\ell(n) = c(n + 1) \oplus (c(n) \vee u(n))$. *On a row with $c(n) = 1$, $\ell(n) = c(n + 1) \oplus 1$ does not depend on $u(n)$; on a row with $c(n) = 0$, $\ell(n) = c(n + 1) \oplus u(n)$.*

Proof. (3) at $i = 0$. $\square$

Fix a word w of length p with Z zero positions, and suppose c has tail w from row N . The *interface sequence* u^* is the sequence of values $u(n)$ over the rows $n \geq N$ with $c(n) = 0$, in order.

Corollary 4.2 (PROVED). *If x is number-like and c has tail w , then u^* is not eventually periodic.*

Proof. Suppose u^* is eventually periodic with period q (counted in zero-rows). The zero-rows occur at positions periodic in n with period p , Z per period, so u restricted to zero-rows is eventually periodic in n with period pq , and so is c . By Lemma 4.1, ℓ is eventually periodic with period pq . Then $\text{Tr}_{[-1, 0]} = (\ell, c)$ is eventually periodic, contradicting Theorem 3.1 at $i = -1$. $\square$

So to exclude w it suffices to show that every configuration with $c = w^\infty$ (from some row) has an eventually periodic interface. This is a statement about the fibre of the trace map over w^∞ and makes no reference to the seed: the seed enters only through Theorem 3.1.

5 The K -window automaton

Fix $K \geq 1$. The *window* at row n is $W(n) = (a(n, 1), \dots, a(n, K)) \in \{0, 1\}^K$, so $u(n) = W(n)_1$. By (2),

$$W(n+1) = \Phi(c(n), W(n), f(n)), \quad f(n) = a(n, K+1),$$

where $\Phi(c, w, f)_i = w_{i-1} \oplus (w_i \vee w_{i+1})$ with $w_0 = c$ and $w_{K+1} = f$. For $S \subseteq \{0, 1\}^K$ define

$$\begin{aligned} \text{step}(S, c) &= \{\Phi(c, w, f) : w \in S, f \in \{0, 1\}\} \quad (\text{the cell } K+1 \text{ is unknown and left free}); \\ Z_u(S) &= \text{step}(\{w \in S : w_1 = u\}, 0) \quad (\text{a zero-row on which the interface bit is } u); \\ O(S) &= \text{step}(S, 1) \quad (\text{a one-row}). \end{aligned}$$

Lemma 5.1 (soundness; PROVED). *For any configuration x and any row N , put $S(N) = \{0, 1\}^K$ and $S(n+1) = Z_{u(n)}(S(n))$ if $c(n) = 0$, $S(n+1) = O(S(n))$ if $c(n) = 1$. Then $W(n) \in S(n)$ for all $n \geq N$; in particular $S(n) \neq \emptyset$.*

Proof. Induction on n . If $W(n) \in S(n)$ then $W(n+1) = \Phi(c(n), W(n), f(n))$; on a zero-row $W(n)_1 = u(n)$, so $W(n)$ lies in the selected subset; in either case $W(n+1) \in S(n+1)$. $\square$

The automaton is an over-approximation: it forgets everything right of column K and remembers only a set of possible windows. What it never does is discard a window that the true configuration occupies. That the method is an over-approximation rather than a decision procedure is not a weakness of this construction but a necessity: deciding whether a given temporal sequence of states can be generated in $|s|$ steps — which is exactly the question Part I asks of a candidate period word — is NP-complete, for an automaton Green constructs for the purpose [33]. So no complete and efficient test for this question can exist in general, and a sound incomplete criterion with a characterised reach is the right kind of object to look for.

6 The period graph and the exclusion criterion

For a word w define the finite directed labelled multigraph G_w :

- vertices: the subsets $S \subseteq \{0, 1\}^K$ reachable from $\text{ALL} = \{0, 1\}^K$ by the edges below;
- edges: from S , for each u -word $(u_1, \dots, u_Z) \in \{0, 1\}^Z$, apply the letters of w in order, using Z_{u_j} at the j -th zero position and O at each one position; if the result T is nonempty put an edge $S \rightarrow T$ labelled $(u_1, \dots, u_Z)$. Distinct labels are distinct edges even when their targets coincide.

Lemma 6.1 (PROVED). *If x is number-like and c has tail w from row N (N chosen so that row N is the start of w), then u^* , cut into blocks of Z , is the label sequence of an infinite path in G_w starting at ALL .*

Proof. Lemma 5.1 with the period structure of w ; every state along the path is nonempty, so every edge exists. $\square$

Lemma 6.2 (single-cycle criterion; PROVED). *Suppose every strongly connected component C of G_w that contains at least one edge has exactly as many edges (with multiplicity, both ends in C) as vertices. Then the label sequence of every infinite path in G_w is eventually periodic.*

Proof. G_w is finite, so an infinite path eventually stays inside one strongly connected component C , which then contains an edge. Strong connectivity gives every vertex of C at least one outgoing edge inside C (a one-vertex component with an edge has a self-loop); with $\# \text{edges} = \# \text{vertices}$ each vertex has exactly one. So C is a single directed cycle with one label per step, and once the path is in C its labels are periodic with period $|C|$ blocks. $\square$

Theorem 6.3 (exclusion criterion; PROVED). *If for some K the graph G_w satisfies the hypothesis of Lemma 6.2, then w is not a tail of the centre column of any number-like configuration, in particular not of the single seed.*

Proof. Lemmas 6.1 and 6.2 make u^* eventually periodic; Corollary 4.2 forbids it. $\square$

Remark 6.4. The criterion is one-sided. If G_w fails the hypothesis at some K , nothing follows: the over-approximation may branch where the true fibre does not. Only an exact computation on the fibre (Section 9) decides that a word is genuinely beyond the method. The hypothesis is equivalent to every recurrent component having Perron root exactly 1, but the proofs rest on the combinatorial count, which is exact.

Lemma 6.5 (transfer in the length of a run of ones; PROVED). *Let $w_b = v 1^b$. Suppose that in G_{w_B} , for every vertex S and every admissible label, the state T reached after v and B one-rows satisfies $O(T) = T$. Then $G_{w_b} = G_{w_B}$ (same vertices, same labelled edges) for every $b \geq B$.*

Proof. Induction on $b \geq B$. From any vertex and any label, the state after v and b one-rows is a T with $O(T) = T$, so after $b + 1$ one-rows it is $O(T) = T$: same target, same label. Starting from ALL the two graphs are generated identically, and the O -fixed property persists. $\square$

Hence one certificate for w_B , together with the O -fixed check, certifies all w_b with $b \geq B$.

Remark 6.6 (what kind of object the exclusion criterion is). The graph G_w is the accessible subset automaton of a *partial nondeterministic* transition system, and the criterion applied to it is the Perron-root condition of Remark 6.4. It is not synchronisation, for three independent reasons: nothing collapses to a single state; the empty set is unreachable, since $S \neq \emptyset$ forces some $Z_u(S) \neq \emptyset$; and the interface bits are tests on the state rather than free input letters, so although mortal words exist in the free monoid none is realisable, because $u(n) = 1$ with $c(n) = 0$ forces $u(n + 1) = 1$. The quantifier differs too: a Černý-type bound asserts that *some* short word synchronises, while the exclusion needs a conclusion for *every* word. Where the comparison is legitimate — the deterministic window map does synchronise — the bound is vacuous, the exact reset thresholds being 3, 4, 9, 11, 14, 17 for $K = 2, \dots, 7$ against the quadratic $(2^K - 1)^2$, too weak by a factor of 949 at $K = 7$.

7 Words with a long run of zeros

The exclusions in this section were obtained by a direct argument on the fibre, recorded because it exhibits the mechanism. An *impulse* is a row with $c = 1$ preceded and followed by rows with $c = 0$; *zero drive* is a run of rows with $c = 0$. Write v, y, z for columns 2, 3, 4.

Lemma 7.1 (quench; PROVED). *Let x be any configuration. Suppose c has ≥ 10 consecutive zero-rows, then an impulse at row t_0 , then zero drive. Then, for as long as the zero drive continues,*

(i) $u(t) = 1$ for all $t \geq t_0 + 7$;

- (ii) $(v(t), y(t)) \neq (0, 0)$ for all $t \geq t_0 + 11$;
- (iii) $(u, v, y)(t) = (1, 0, 1)$ for all $t \geq t_0 + 12$.

The state at a row depends only on c at earlier rows, so the conclusions hold at the row of the next impulse as well.

Proof. Nesting. A row under zero drive for $a \geq 10$ steps is a row under zero drive for exactly 10 steps started from its own predecessor at step $a - 10$; so any fact holding for every configuration with pre-run exactly 10 holds for pre-run ≥ 10 .

Three finite facts. With $c = 0^{10}10^{20}$ and every other cell of row 0 free, the following are unsatisfiable (certificates `drive2.py`, `freeze.py`; CaDiCaL 1.5.3 and Glucose 4 agree): $u = 0$ at $t_0 + 7$; $(v, y) = (0, 0)$ at $t_0 + 11$; $(v, y) = (0, 0)$ at $t_0 + 12$. Every cell in columns ≤ 3 at rows $\leq T$ depends only on row-0 cells at columns $\leq T + 3$, so the instance width is sufficient.

Monotonicity. Under zero drive $u(t+1) = u(t) \vee v(t)$, so $u = 1$ persists; with the first fact, (i).

Chain. Suppose $u(t-1) = u(t-2) = 1$ and $(v, y)(t) = (0, 0)$, with $t-2 \geq t_0 + 7$. From $v(t) = u(t-1) \oplus (v \vee y)(t-1)$ we get $(v \vee y)(t-1) = 1$. From $y(t) = v(t-1) \oplus (y \vee z)(t-1) = 0$: if $v(t-1) = 0$ then $y(t-1) = 1$ and $y(t) = 1$, contradiction; so $v(t-1) = 1$, whence $(v \vee y)(t-2) = 0$, i.e. $(v, y)(t-2) = (0, 0)$. Descending by 2 from any $t \geq t_0 + 13$ lands on $t_0 + 11$ or $t_0 + 12$, both excluded. This is (ii).

Freeze. For $t \geq t_0 + 12$, $v(t) = 1$ would need $(v \vee y)(t-1) = 0$ with $t-1 \geq t_0 + 11$, excluded. So $v(t) = 0$, and $(v \vee y)(t) = 1$ gives $y(t) = 1$. With (i), (iii). $\square$

Lemma 7.2 (block step; PROVED). *An impulse landing on $(u, v, y) = (1, 0, 1)$, followed by zero drive, gives $u = 0, 0, 1$ at the next three rows.*

Proof. From (2): $u(+1) = 1 \oplus (1 \vee 0) = 0$, $v(+1) = 1 \oplus (0 \vee 1) = 0$, $y(+1) = 0 \oplus (1 \vee z) = 1$; $u(+2) = u(+1) \vee v(+1) = 0$, $v(+2) = u(+1) \oplus (v \vee y)(+1) = 1$; $u(+3) = u(+2) \vee v(+2) = 1$. $\square$

Theorem 7.3 (isolated ones, zero-runs ≥ 10 ; PROVED from EXACT inputs). *Let w be a necklace in which every 1 is isolated and every cyclic zero-run has length ≥ 11 . Then w is not a tail of the centre column of any number-like configuration. The same holds with zero-runs ≥ 10 (one further finite fact, `run10.py`), and for the word 0000000001 (§7.1).*

Proof. Take any configuration with $c = w^\infty$ from some row. The first impulse is preceded by ≥ 10 zero-rows, so Lemma 7.1 freezes (u, v, y) at $(1, 0, 1)$ from 12 rows after it; the next impulse is ≥ 12 rows later and lands on $(1, 0, 1)$. By Lemma 7.2 the interface on the following block reads $0, 0, 1, 1, \dots$ ($u = 1$ is then preserved by monotonicity until the next impulse), and the next impulse again finds $(1, 0, 1)$ by Lemma 7.1 applied afresh. By induction on blocks u^* is eventually periodic with the period of w ; Corollary 4.2 concludes. $\square$

7.1 The word 0000000001

Zero-run 9 is one short of Lemma 7.1. Here the fibre was enumerated exactly: writing τ_j for the number of leading zeros of u on the j -th block of zero-rows, every admissible τ -sequence is, after at most three symbols, an alternating 1212... spine that either continues forever or is eventually absorbed into 2222..., and this is proved by five bounded-window facts (11, 2221, 21221 never occur; 1221 only at positions ≤ 1 ; symbols outside $\{1, 2\}$ only at positions ≤ 2) which reduce, by the observation that the row at time p is itself a phase-0 configuration of the fibre, to an enumeration at depth 6. Every admissible interface is therefore eventually periodic and Corollary 4.2 applies. Details in [80, §2].

Theorem 7.4 ($K = 14$ family certificate, EXACT; see Section 10). *Every necklace all of whose cyclic zero-runs have length ≥ 19 , with 1-runs of arbitrary length, is excluded for every number-like configuration (certificate `cover14_19_48`, Section 10). At $K = 14$ every necklace tested with all cyclic zero-runs ≥ 17 is excluded as well, and at $K = 16$ so is every necklace tested whose cyclic zero-runs all equal some $Z \geq 15$, of the shapes $0^Z 1^a$ with $a \leq 5$ and $0^Z 1^a 0^Z 1^b$ with $a, b \leq 3$ (EXACT for each such word; Remark 9.15); the family-level statement at 17 rests on a product construction we have not rebuilt (see the sweep below) and is OPEN as a family statement. At zero-runs 16 the necklaces with 1-run 3, and with the 1-run pair $(3, 3)$, are not certified at $K = 14$ ($0^{16} 1^3$ is certified at $K = 16$).*

Two routes bear on this statement and they stop in different places, which is worth separating. By forcing — exact SAT on the fibre — the interface bit fails to be forced at blocks 4 and 5 already at zero-runs 18 with the 1-run pair $(3, 3)$, so that route halts at 19. By the single-cycle criterion of Theorem 6.3, applied word by word at $K = 14$, every necklace tested is excluded down to 17 (the family statement at 17 being OPEN, as above), and the obstruction at 16 is the 1-run shape 1^3 rather than the zero-run length.

The two theorems say: a long run of zeros in the period word quenches the interface. The threshold 10 (for isolated 1s) comes from the finite facts of Lemma 7.1; below a run of 5 the state $(v, y) = (0, 0)$ can be sustained indefinitely by the right context and no quench occurs.

8 Words with a long run of ones

The same mechanism operates from the other side: a long run of ones also quenches the interface. These results are certificates of Theorem 6.3 at $K = 10$, produced by two independent implementations of Sections 5–6 (`kauto3.py`, `s3_fam1b0a.py`), which agree on every word both cover ($1^b 0$ for $b = 9, \dots, 12$; $0^m 11$ for $m = 8, \dots, 11$), and spot-checked by a third (`kwin.py`, Appendix) on $1^9 0$, $1^{12} 0$, all rotations of $1^8 0$ and of $0^m 11$ for $m = 6, 7$, and $0^m 11$ for $m = 3, 4, 5, 8, 9$, with identical verdicts on every word both cover. “Certified” means the graph G_w was built and every recurrent component has $\#edges = \#vertices$; “transfer” means the O -fixed check of Lemma 6.5 at $B = 29$.

Theorem 8.1 (EXACT certificate of Theorem 6.3). $1^b 0$ is excluded for every $b \geq 9$.

Certificate. $b = 9, \dots, 34$ individually (26 graphs; for $b = 9$ the graph has 3 states and 3 edges); transfer at $B = 29$ gives all $b \geq 29$. $\square$

Theorem 8.2 (EXACT certificate of Theorem 6.3). $1^b 0^a$ is excluded for every $b \geq 9$ and every $a \geq 1$.

Certificate. For each $a = 1, \dots, 19$: $b = 9, \dots, 34$ individually and transfer at $B = 29$ (494 graphs, 19 transfer checks). For $a \geq 19$, Theorem 7.4. $\square$

Theorem 8.3 (EXACT certificate of Theorem 6.3). *For every word v with $|v| \leq 12$ that begins and ends with 0, the word $1^b v$ is excluded for every $b \geq 29$.*

Certificate. $b = 29$ with transfer, for all 2048 such v . $\square$

Theorem 8.4 (EXACT certificate of Theorem 6.3). $0^m 11$ is excluded for $m = 6$ and for every $m \geq 8$.

Certificate. $m = 6$ (all eight rotations at $K = 10$, e.g. 63 states, 89 edges for 00000011; also at $K = 12$), $m = 8$ (39 states, 59 edges), $m = 9, \dots, 20$ individually; for $m \geq 19$, Theorem 7.4. The word $0^7 11$ has a branching recurrent component in every rotation at $K = 10$ (6 vertices, 10 edges), $K = 12$ and $K = 14$ (14–16 vertices, 22–26 edges), and $m \leq 5$ branch at $K = 10$; these are not decided by the method at these K . $\square$

Remark 8.5 (sharpness). $1^8 0$ has a branching recurrent component at $K = 10$ (8 vertices, 15 edges) and at $K = 12$ (21 vertices, 39 edges), and it is uncertified at every K up to 16. By Remark 6.4 this proves nothing about the word, and $b = 9$ is not a threshold: $1^7 0$ is certified at $K = 10$ (Remark 9.15), so $1^8 0$ is a hole rather than the edge of the family, and the lower bound in Theorems 8.1–8.2 is only where the whole family is certified. The non-monotonicity in Theorem 8.4 ($m = 6$ certified, $m = 7$ not, at every window tried) is a hole of the same kind. It does not mirror the forcing of the interface after mixed zero-runs $0^{m_1} 1 0^{m_2} 1$, which is monotone in each run length separately (MEASURED: SAT, exact per cell, $6 \leq m_1, m_2 \leq 15$); an earlier reading of that forcing as a non-monotone landscape compared cells differing in both run lengths and is WITHDRAWN. Both holes, $1^8 0$ and $0^7 11$, branch in every rotation and the branching grows with K ; whether they are excluded by some other route is open.

Theorem 8.6 (the constants; PROVED). *The centre column of a number-like configuration is neither eventually 0 nor eventually 1.*

Proof. If $c(t) = 0$ for $t \geq T$ then, by (2) at $i = 1$, $u(t+1) = u(t) \vee v(t)$ for $t \geq T$: u is monotone, hence eventually constant, so (c, u) is eventually periodic, contradicting Theorem 3.1. If $c(t) = 1$ for $t \geq T$ then by Lemma 4.1 $\ell(t) = c(t+1) \oplus 1 = 0$ for $t \geq T$, so (ℓ, c) is eventually periodic, contradicting Theorem 3.1. $\square$

9 The boundary of the method

Theorem 6.3 uses the seed only through Theorem 3.1. Its reach is therefore exactly the set of words w for which *every* configuration with centre column w^∞ has an eventually periodic interface. We now show this set excludes the words that matter.

Proposition 9.1 (the fibre over a centre column; PROVED). *For every sequence $c \in \{0, 1\}^\mathbb{N}$ and every right context $r = (r_1, r_2, \dots) \in \{0, 1\}^\mathbb{N}$ there is exactly one configuration x with $\text{Tr}_0(x) = c$ and $a(0, i) = r_i$ for all $i \geq 1$. In particular every sequence is the centre column of some configuration, and the interface u^* is a function of (c, r) .*

Proof. Put $a(n, 0) = c(n)$ and $a(0, i) = r_i$. The cells $a(n, i)$ with $i \geq 1$ are determined forward by (2) from row $n - 1$ at columns ≥ 0 , all of which are already defined; the cells with $i \leq -1$ are determined by (3) from the two columns to their right. The rule (2) holds at every column $i \geq 1$ by construction and at every column $i \leq 0$ because (3) is (2) solved for its left input. Uniqueness: every cell has been determined. $\square$

Note that not every pair (c, u) is realised: (2) at column 1 forces $u(n+1) = \neg c(n)$ whenever $u(n) = 1$. The interface is an output of the fibre, not a free input.

Let $L(w) \subseteq \{0, 1\}^\mathbb{N}$ be the set of interface sequences u^* over all configurations with $c = w^\infty$ from row 0, and $L_k(w)$ its set of prefixes of length k (zero-rows). By compactness of $\{0, 1\}^\mathbb{Z}$ and continuity of the trace, $L_k(w)$ coincides with the set of length- k interface words realisable at finite depth, which is what the enumerations compute.

Proposition 9.2 (boundary; PROVED). *If $L(w)$ is uncountable—in particular if $|L_k(w)|$ grows exponentially in k —then there is a configuration with $c = w^\infty$ whose interface is not eventually periodic, and Theorem 6.3 cannot exclude w at any K .*

Proof. Eventually periodic sequences are countable, so an uncountable $L(w)$ contains one that is not; it is realised by a configuration by definition of $L(w)$. Lemma 6.1 uses only Lemma 5.1, which holds for every configuration, so this configuration’s path in G_w has a non-eventually-periodic label sequence, and the hypothesis of Lemma 6.2 fails at every K . If $|L_k(w)| \geq \lambda^k$ for some $\lambda > 1$ then $L(w)$, being closed, has positive topological entropy and is uncountable. $\square$

Proposition 9.3 (the entropy of the fibre is a limit, not a fit; PROVED). $|L_{m+n}(w)| \leq |L_m(w)| |L_n(w)|$ for all $m, n \geq 1$. Consequently $h(w) := \lim_k \frac{1}{k} \log |L_k(w)|$ exists and equals $\inf_k \frac{1}{k} \log |L_k(w)|$.

Proof. The language is factorial, so if $u \in L_{m+n}$ then its length- m prefix lies in L_m and its length- n suffix in L_n ; the map $u \mapsto (\text{prefix}, \text{suffix})$ is injective because u is the concatenation of the two. Hence $|L_{m+n}| \leq |L_m| |L_n|$, i.e. $k \mapsto \log |L_k|$ is subadditive, and Fekete’s lemma [97] gives the limit and its identification with the infimum. $\square$

Corollary 9.4 (the exponential clause is not computationally dischargeable; PROVED). *For no word w can a finite enumeration establish the hypothesis $|L_k(w)| \geq \lambda^k$ of Proposition 9.2. By Proposition 9.3 the entropy is the infimum of $k^{-1} \log |L_k(w)|$, so every computed value is an upper bound on $h(w)$ and none is a lower bound: the infimum over all k may be zero whatever the first thousand terms show.*

Remark 9.5 (what this costs, and where it points). The corollary is worth stating because the exponential clause reads like the practical branch of Proposition 9.2 and is in fact the unreachable one. It applies uniformly, not just to the hard case: the comparison necklaces 0000011111 (nine exact terms, ratio $\rightarrow 2.0$) and 0000000101 (ratio ≈ 1.55) look exponential and are MEASURED so, and by the corollary they cannot be proved so by extending those enumerations — the natural next move, and a wasted one. For 01 the situation is worse in a second and independent way, the measured ratio being 1.094 at $k = 60$ and falling, so there the counts do not even point where the hypothesis needs them to; the stream measurement of Remark 9.6 does, but at depths ($k \approx 700$) no enumeration reaches. Consequently uncountability of $L(w)$, for 01 or for any other background, has to come from a *construction* rather than from counting: an explicit infinite family, or a self-embedding that generates one. That is a narrow target but a definite one, and it is the only route to uncountability that Proposition 9.2 leaves open. Whether $L(01)$ is uncountable remains OPEN. Uncountability is sufficient for the conclusion of Proposition 9.2, one interface that is not eventually periodic, but not necessary: the orbit closure of $z = 1010^210^31\cdots$ consists of the shifts of z , the points 0^a10^∞ and 0^∞ , so it is a countable closed shift-invariant set, and z is not eventually periodic. Nor does superpolynomial growth force uncountability: the sequences with finitely many 1s in which, if there are m of them, consecutive 1s are at distance at least m form a countable closed shift-invariant set in which every point is eventually periodic, with $\sum_m \binom{k-(m-1)^2}{m}$ words of length k , a superpolynomial count that is 3334829 at $k = 60$ against $|L_{60}(01)| = 103220$ (PROVED; folklore-type examples).

Remark 9.6 (random right halves: the interface of 01 has positive entropy as measured; MEASURED, EXACT and PROVED parts as marked). The counts to $k = 60$ are dominated by words anchored at row 0. Twenty random right halves driven by $(01)^\infty$ for up to 400000 rows settle, after a transient of a few hundred letters, either into one of the periodic tails already known to

be realisable ($((14)^\infty$ and 1^∞ in run-length coding; 2 of 20) or into a stationary stream of gap letters 2 and 4 only, the 2s isolated and at rate 0.155–0.172 in every seed and every time window, with no coarsening over 200 000 letters; the picture is a boundary layer about four cells wide next to the column with ordinary Rule 30 chaos behind it (MEASURED; ledger 10.238d D). Every one of 342 804 sampled windows of length 36 of these streams lies in the exact $L_{36}(01)$, rebuilt independently by SAT-driven search with the counts of Section 22 letter for letter (EXACT; 10.238d A, D). The conditional block entropies of the late stream fall slowly, 0.384 to 0.366 bits per gap letter at block lengths 10 to 20, and extrapolate to about 0.35 bits per gap letter, 0.07 per letter of the interface, a growth rate near 1.05 (MEASURED; 10.238d E); with $U = 24^3$ and $V = 24^5$, the two commonest late blocks, every one of the 32 sequences of five blocks is realisable after a free head of 20 letters and again after 60, and 116 of the 128 sequences of seven blocks finished within the solver’s time, all realisable and none failed (EXACT; 10.238d F). So $L(01)$ has positive entropy by every measurement here, the growth fits above describe the row-0 transient ($2^{0.07k}$ overtakes $\exp(5.91k^{0.28})$ only near $k \approx 700$), and the zero-entropy expectation of the earlier text is withdrawn; uncountability remains OPEN, since what a proof needs is the free choice at every depth. Two facts say why that resists proof (ledger 10.239d). The rightmost cell at which two right-half evolutions with the same boundary differ moves right by exactly one cell per row, forever (PROVED: the cell to its right reads the difference through its XOR input while its OR inputs agree), so Rule 30 has no localised particle on any background and a free choice cannot be engineered as a glider in a periodic right half; and right halves with random cells on columns $1..N$ only ($N = 10$ and 100) give the same stream in 15 of 16 runs, so restricting to number-like right halves changes nothing as far as computed (MEASURED). Finally, every minimal forbidden space-time block of Rule 30 has width at most 2, except the twelve blocks of width 3 and height 2 that are the rule table itself: a block is globally admissible iff the rule holds at every interior cell and its rightmost two columns form an admissible two-column trace, by left-permutivity on the left and because the right neighbour is read only where the cell is 0 (PROVED; ledger 10.236d B), so no wider excluded block adds a constraint beyond the two-column trace; the Perron roots of the two-column Rauzy graphs descend 3.000, 2.618, 2.500, 2.521, 2.480, $\dots$, 2.342 at heights 2–12 without a plateau (EXACT; 10.236d C).

Proposition 9.7 (a branching criterion for uncountability; PROVED). *Call $u \in L(w)$ a branch word if $u0$ and $u1$ both lie in $L(w)$. If there is a legal u such that every legal word extending u has a branch word among its extensions, then $L(w)$ is uncountable, and hence by Proposition 9.2 some configuration with $c = w^\infty$ has an interface that is not eventually periodic.*

Proof. Recursively: u has a branch word u_0 among its extensions, and u_00, u_01 are both legal; each of those again has a branch word among its extensions. The resulting binary tree of branch words has $2^{\aleph_0}$ infinite paths, every prefix of every path is legal, and distinct paths give distinct sequences, so $|L(w)| \geq 2^{\aleph_0}$. Note the hypothesis already forces every legal extension of u to extend further, so no separate appeal to the absence of dead ends is needed. $\square$

Remark 9.8 (why the criterion cannot be made uniform). The converse fails — isolated points may accumulate on the perfect kernel, so an uncountable $L(w)$ need not contain a word all of whose extensions branch; the exact characterisation is the usual one, that the tree contains a perfect subtree, whose nodes need not be all the extensions of any word. What is worth recording is that the criterion cannot hold *uniformly* here. If every legal word had a branch word within B further symbols then $|L_{k+B}| \geq 2|L_k|$, so $|L_{mB}| \geq 2^m$ and $h(w) \geq (\log 2)/B > 0$. For 01 the measured ratio is 1.094 at $k = 60$ and falling, which read as $h = 0$ would mean that no uniform

B exists and the delay before branching is unbounded; the stream measurement of Remark 9.6 reverses that reading — the late stream has positive measured entropy and branches within a bounded number of blocks, and it is the row-0 transient that delays. The criterion may still be of the non-uniform shape that resists proof, since the transient must be crossed first, which is consistent with the problem’s difficulty rather than a defect of the criterion. It also sharpens what to measure: not whether branching occurs, but how the worst-case delay before it grows with depth. MEASURED, to depth 130 on a random sample: a branch that is forced for the 32 levels from 28 to the end of the enumeration resumes branching at 66 and then branches at every period-5 boundary to 121, so apparent forcing inside a finite window is a horizon artefact and not isolation.

Proposition 9.9 (a finite block traps the interface; PROVED). *In the right half-plane with column 0 pinned to $(01)^\infty$, every configuration whose row 0 begins with*

$$B = 1001100010010100\,0101$$

in columns $1, \dots, 20$ has interface identically $(10)^\infty$, whatever the cells beyond column 20 are.

Proof. A state is the block of columns $1, \dots, 20$ together with the row parity; its successor depends only on the state and the value of column 21, since column 0 is the parity. From $(B, 0)$, the set R_t of blocks reachable at row t over *all* input sequences becomes periodic after 21 rows with period 8; the 188 states on the cycle form a set closed under both inputs, and at every even row column 1 takes a single value across R_t . Two independent implementations agree on every number (`s3_trap_repro.py`; `agent_reach/proof/p3_wall_trap.py`), and 1 000 random tails evolved for 4 000 rows all give $(10)^\infty$. $\square$

Remark 9.10 (what the trap does and does not settle). Three consequences. (i) The push-forward of any product measure on row 0 to the interface has an atom: a random row 0 enters this trap with frequency about 0.034 (MEASURED, 3 000 trials, always within 100 rows). So no argument of the form “no finite cylinder determines the interface” is available, and uncountability of $L(01)$, if it holds, must be established despite the atoms. (ii) The trapped word is nevertheless *not* an isolated point of $L(01)$: $(01)^k 00$ and $(01)^k 01$ are both legal for $k \leq 60$ (EXACT), so $(01)^k 0$ is a branch word in the sense of Proposition 9.7, whose hypothesis the trap leaves untouched. (iii) The contrast with Gravner and Griffeath’s Proposition 3.3 [67] — a left-permutive edge automaton has no periodic robust solution against a *zero* edge, Rule 30 included — is that the pinned column here contains 1s, and it is the 1s that build the wall: with a free or all-zero column 0 no such block exists (EXACT, straight zero diagonals of length 80 are then realisable). What is not claimed: that every configuration whose interface no modification of far cells can change is a trap of this kind (MEASURED only, on every instance found; a configuration shielded only against single far flips need not be one, the 1^∞ crystal of Remark 9.12(i) being an example), or anything about uncountability itself, which remains OPEN.

Theorem 9.11 (the diagonal cap; PROVED). *In the right half-plane with column 0 pinned to $(01)^\infty$, $a(t, 0) = t \bmod 2$, let $g(N)$ be the largest l for which some configuration has $a(N - k, k) = 0$ for $k = 1, \dots, l$: the longest run of zeros ending at column 1 on the anti-diagonal $t + i = N$, which is also the longest straight chain of influence into the interface cell $a(N, 1)$, since $\partial(l \oplus (c \vee r))/\partial r = 1 \oplus c$. Then*

(i) $g(N) = 6$ for every even $N \geq 58$;

- (ii) $g(N) \leq 15$ for every even $N \geq 30$; with the exact values $g(N) = N$ for even $N \leq 28$ and $g(N) = 15$ for even $30 \leq N \leq 56$ (EXACT, each even $N \leq 150$ separately, `s3_runcap.py`), g on the even diagonals is the step function $N, 15, 6$;
- (iii) $g(N) = 2$ for every odd $N \geq 5$, with $g(1) = 1$ and $g(3) = 3$.

Proof. Three finite facts and a shift. (a) *The triangle.* The cells $a(N-l, l), \dots, a(N-1, 1)$ are computed from row $N-l$, columns $0, \dots, l$, alone, and column 0 there is the parity of $N-l$; so which patterns of columns $1, \dots, l$ produce the run is a brute force over 2^l cases that depends on N only through its parity (`s3_cap_verify.py`). For even N : the run of 7 occurs iff row $N-7$ carries 1111110 in columns $1, \dots, 7$; the run of 6 iff row $N-6$ carries 000000 in columns $1, \dots, 6$; the run of 16 iff row $N-16$ carries 0100100100100110 or 1001001001001000 in columns $1, \dots, 16$; in each case these are all the patterns. For odd N : the run of 2 occurs iff the odd row $N-2$ carries 10 in columns $1, 2$, the run of 3 iff the even row $N-3$ carries 110 in columns $1, \dots, 3$, and no pattern of row $N-4$ produces a run of 4. (b) *The shift.* The right half-plane is an autonomous one-sided automaton driven by the pinned column, which has period 2; so row 2 of any configuration is a legal row 0, and a pattern reachable at row $r+2$ is reachable at row r . (c) *The certificate.* On the 200-row light-cone instance, 1111110 in columns $1, \dots, 7$ is reachable at row 49 and unreachable at row 51, by Cadical153 and by Glucose4 (rows 53, 61, 99 unreachable as controls; `s3_cap_upper.py`). By (b) it is unreachable at every odd row ≥ 51 , so by (a) no even diagonal $N \geq 58$ carries a run of 7, hence none longer: $g(N) \leq 6$. Reachability at row 49 gives the run of 7 at $N = 56$, where the exact table has $g = 15$. In the same way both 16-patterns are unreachable at row 14, and one of them is reachable at row 12 as $g(28) = 28$ requires (`s3_cap_plateau.py`); so no even diagonal $N \geq 30$ carries a run of 16, which with the exact values is (ii). (d) *The construction.* On the ring of 728 cells Rule 30 has a cycle of minimal period 14 whose cell 0 equals the phase parity and whose cells $1, \dots, 6$ read 000000 at phase 12 (`agent_reach/cap/ring_cycle.txt`, re-verified cell by cell). Unrolled to the half-line, column i being cell $i \bmod 728$, and started at the even phase $(18-N) \bmod 14$, it evolves as the ring does, its column 0 agreeing with the pinned column, and it is at phase 12 on row $N-6$; by (a) diagonal N then carries a run of 6. Direct simulation finds a run of exactly 6 for every even $N \in [8, 3000]$ and for $N = 5\,000, 10\,000, 20\,002, 30\,000$. With (c) this is (i). (e) *Odd diagonals.* The pattern 110 in columns $1, \dots, 3$ of an even row has no predecessor: column 0 is 1 on the row above, so $a(\cdot, 1) = 1$ forces that row's columns $1, 2$ to be 0, then $a(\cdot, 2) = 1$ forces its column 3 to be 1, and then $a(\cdot, 3) = 1$, not 0. Hence no odd diagonal $N \geq 5$ carries a run of 3. The crystal of (d) carries 10 in columns $1, 2$ at phase 5; started at the even phase $(7-N) \bmod 14$ it is at phase 5 on row $N-2$, so every odd diagonal $N \geq 5$ carries a run of 2 (exactly 2 by simulation for every odd $N \in [5, 3001]$ and for $N = 10\,001, 30\,001$; `s3_cap_odd.py`). With $g(1) = 1$ and $g(3) = 3$ from the free row 0, this is (iii). $\square$

Remark 9.12 (what the cap says). (i) The witness in (d) is a temporally periodic half-plane of row period 14 and spatial cycle 728 whose interface is the gap word $(14)^\infty$, the crystal certified in Section 22, and it carries the longest straight chain of influence that remains available once $N \geq 58$. The opposite extreme, a block through which no influence passes, is the trap of Proposition 9.9, and it is not one of the crystals of Section 22: it is a finite block of 20 columns whose reachable set cycles with row period 8 whatever lies beyond column 20. The 1^∞ crystal (row period 4, spatial cycle 7) carries the trap's interface, $(10)^\infty$ up to phase, but is not a wall: single flips at columns 21–90 never reach its interface within 4000 rows, while random modifications of columns 21–120 (each cell flipped with probability $\frac{1}{2}$) reach it in 54 of 300 trials,

against 0 of 300 for the trap (MEASURED, `agent_reach/proof/p13_cycle7_shield.py`). (ii) A far-right perturbation that reaches the interface at a row $N \geq 58$ therefore cannot arrive along a straight chain; it must pause (a vertical step needs the right-neighbour gate to be 0, as $\partial(l \oplus (c \vee r))/\partial c = 1 \oplus r$), so its front moves left at speed strictly below 1. MEASURED on extracted single-flip fronts: about 0.3, with pauses and retreats, the bursts into column 1 never exceeding the cap (ledger 10.220–10.222). (iii) The cap is a property of the 1s in the pinned column, not of Rule 30’s diagonals: with column 0 free or all zero, straight zero diagonals of length 80 exist (EXACT), and with column 0 pinned to 1^∞ or $(01)^\infty$ they do not. (iv) We found no statement of this kind in print (ledger 10.224, every source opened): the Lyapunov and damage-spreading literature for Rule 30 — Wolfram’s measured leftward speed [88, p. 949], the defect shapes of Baetens and Gravner [26], whose Open Problem 9 asks for a damage-spreading theory for periodic states — concerns free configurations, and Gravner and Griffeath’s edge automaton [67] a zero edge. The cap is claimed as new, in the narrow form stated.

Remark 9.13. EXACT: for $w = 01$ the inequality was checked on all 1770 ordered pairs with $m + n \leq 60$ with no violation, which also fixes the indexing of the count list ($|L_1| = 2$, $|L_{44}| = 21\,389$, $|L_{60}| = 103\,220$; earlier working notes labelled the same list from $k = 2$, which is off by one and is corrected here). The sequence $\frac{1}{k} \log_2 |L_k|$ is monotone decreasing from $k = 5$ on, through 0.519 at $k = 16$, 0.327 at $k = 44$ and 0.278 at $k = 60$, so $h(01) \leq 0.278$ bits. The open question is therefore not whether the entropy exists but whether this monotone sequence tends to 0. Every growth law that fits the computed range predicts that it does, and that conclusion is the one part of the fitting that is robust: for $k^{1.9} \exp(0.87\sqrt{k})$, $\frac{1}{k} \log |L_k| \sim 1.9 \frac{\ln k}{k} + \frac{0.87}{\sqrt{k}} \rightarrow 0$, and the same holds for $\exp(c \ln^2 k)$ and for $k^a \exp(ck^\gamma)$ at every $\gamma < 1$. Which of them it is is undetermined (Section 22), and no finite depth decides the limit in any case. The submultiplicative bound itself is Meier and Staffelbach’s observation for the analogous count in the finite-ring setting [82, eq. (10)], where they use it only to conclude that their attack cost grows at most exponentially.

Measured. $|L_k(01)| = 2, 3, 5, 8, 12, 17, 25, 36$ for $k = 1, \dots, 8$; $|L_k(0000000101)|$ grows by a factor 1.55 per period; $|L_k(0000011111)|$ by a factor tending to 2 (MEASURED, nine exact terms). For 01, in run-length coding the alphabet is $\{1, 2, 3, 4\}$, the pairs $(1, 2), (2, 1), (2, 3)$ are forbidden in the recurrent part, and the minimal forbidden words are unbounded [80, §11], so the language is not of finite type. It has at least 3 591 distinct follower sets (EXACT, by complete enumeration), so any sofic presentation would need more than 3 591 states; that it is not sofic at all follows from the growth dichotomy for regular languages — polynomial or exponential, never intermediate — and is therefore conditional on the measured intermediate growth, exactly as stated in the list of what is not proved; the stream measurement of Remark 9.6 points instead to exponential growth beyond the row-0 transient, under which that inference lapses and only the follower-set bound stands. An earlier draft wrote “its follower sets are pairwise distinct”, which asserts an infinite statement that no enumeration reaches; that wording is WITHDRAWN in favour of the bound. Exact counts for 01 reach depth 60 ($|L_{44}| = 21,389$, $|L_{45}| = 23,809$, $|L_{46}| = 26,482$, $|L_{60}| = 103,220$, ratio $|L_k|/|L_{k-1}|$ equal to 1.112 at $k = 46$ and 1.094 at $k = 60$, and falling), which is consistent with zero entropy for the words anchored at row 0 that dominate these counts (Remark 9.6); the only bound proved from these counts is $h(01) \leq 0.278$ bits. Its growth is superpolynomial on that range, and *which* superpolynomial function it is remains OPEN: no fixed degree fits, since the local exponent $d \log |L_k|/d \log k$ rises from 3.13 at $k = 12$ to 5.22 at $k = 56$ without levelling, and the phrase “degree 5–6” used in earlier working notes reports that exponent *on the window* rather than naming a family; it is withdrawn. Section 22 gives the out-of-sample table and says exactly which families the data does and does not separate;

two gap laws stated in earlier working notes (444 never; the $2^a 42^b$ rule) were found false on re-enumeration and are withdrawn. Uncountability of $L(01)$ is not proved; positive entropy of $L(01)$ is MEASURED on the stream of Remark 9.6, not proved. Subject to that, Proposition 9.2 places 01, 001, 0011 and every word without a run of ≥ 9 ones or ≥ 10 zeros outside the reach of any argument that uses the seed only through Theorem 3.1.

Remark 9.14 (relation to the cryptanalysis of Rule 30). Rule 30 was proposed as a keystream generator, and the attack literature studies exactly the objects of this section under other names; we record the correspondence because several of our facts are theirs. Meier and Staffelbach [82] write the left reconstruction as their equation (4), $s_{i-1}(t) = s_i(t+1) \oplus (s_i(t) \vee s_{i+1}(t))$, and observe that “given the values of the cells in two adjacent columns, this allows the values of all cells in a triangle to the left of the temporal sequence to be reconstructed”, which they call *completion backwards* (the same equation and, nearly verbatim, the same sentence are already in Wolfram’s CRYPTO ’85 abstract [87], which credits the approach to C. and R. Feynman); their conclusion that “knowledge of the seed is equivalent to knowledge of one of its adjacent sequences” is the reduction used throughout this section, and their *right adjacent sequences* are our interface words, so $L_k(w)$ is their count. Three further items match. Their *completion forwards* statement, that any choice of the sites to the right has a completion consistent with the given temporal sequence, is the absence of dead ends in $L(w)$. Their proof that for an all-zero temporal sequence the right adjacent sequence “can only consist of k digits 0 followed by $n - k + 1$ digits 1”, so that the count grows only linearly, is the right-hand companion of the constant-trace fibre classified from the left by Condrey [91], and predates it by thirty-five years. And their Table 1 is not merely comparable to our enumeration, it is the same computation: their “completion forwards” is a half-line Rule 30 driven on the left by a prescribed boundary column, which is the object we enumerate, and all sixteen of their published means were reproduced exactly here (a first attempt with the wrong boundary convention returned 2.7 where they report 41.86, which is how the correspondence was pinned). Their averages are taken over random boundary columns, whereas $L_k(w)$ fixes the column, so the two are the ensemble and the single-word versions of one quantity; the local exponent of their means rises linearly in $\ln N$ with no bend, the same signature our own counts show, so the right-admissible count is not polynomial in either setting. The same literature also has the leftward spreading speed: Meier and Staffelbach note, after Wolfram, that the effect of a bit change propagates to the left with speed roughly $\frac{1}{4}$, and to the right with speed 1 by the partial linearity of the rule [82], and Wolfram’s CRYPTO ’85 extended abstract [87] gives these as Lyapunov exponents, 0.25 on the left and 1 on the right; so the leftward speed whose values in [88] are discussed in Remark 17.24 was published well before [88]. Finally, the $\frac{3}{4}$ that recurs in our measurements — the rate at which the interface bit is the complement of the previous centre bit — is the agreement probability of the best affine approximation to the Rule 30 gate, $\alpha = \frac{3}{4}$ in Koç and Apohan [83], and the reason their attacks work is that the gate lacks first-order correlation immunity and has low nonlinearity [89]. None of that literature addresses eventual periodicity: its complexity statements are experimental, with no closed-form exponent.

The reach of Theorem 6.3 is thus, in principle, the set of words whose interface language contains only eventually periodic sequences (Section 9); zero entropy is necessary for that but not sufficient (Remark 9.8), and Theorems 7.3–8.6 show that this set contains the families listed there, each of which has a sufficiently long run of one symbol. The words with no long run—those whose interface can be driven freely by the right context—are the ones that carry the difficulty of the Prize Problem, and for them the seed must enter through something other than

Theorem 3.1. The Rauzy graphs of the exact interface languages of ten short residual words (01, 0011, 001, 011, 0001, 0111, 00011, 00111, 01011, 00101), aligned to the period, branch at every depth computed, 24–26 rows, with Perron ceilings above 1 throughout and, for all but 00011, descending without settling, so none is excluded by the cycle criterion at that depth and each shows the fingerprint of 01 (EXACT; ledger 10.236d E).

Remark 9.15 (the criterion is an ω -regular emptiness test, and what that buys). The exclusion criterion has a standard logical form, and naming it converts a word-at-a-time computation into a family-at-a-time one.

Read the K -window subset automaton as a Büchi automaton over the interface alphabet: the criterion of Lemma 6.2 asks whether some infinite path has a label sequence that is not eventually periodic, which is an ω -regular emptiness question. So a *regular family* of candidate period words can be handled in one finite computation, by taking the product of the subset automaton with a deterministic safety automaton recognising the family and applying the same $\#edges = \#vertices$ test to the product — though not literally: whenever the family automaton must allow a choice of letter, as it must for the zero-run family, the product also branches on that choice, and no component is then a single cycle (see below). That is the route by which a family threshold of 17 for Theorem 7.4 was first proposed. On every necklace tested the lowering from 19 to 17 holds word by word (EXACT), but the family-level product certificate at 17 has not been reproduced, as recorded below, so the family statement at 17 is OPEN. Lemma 6.5 is absorbed into it as a one-state family automaton.

A sweep of the other families, and one methodological fact that governs all of them, follows below. The construction was validated three ways before being relied on: the window map was unit-tested against direct Rule 30 evolution; the published state and edge counts of Sections 5–6 were reproduced, including $|F| = 62$ at step 13; and the family automaton was compared against the word-by-word graph G_w on all 801 necklaces of length at most 12 at $K = 8$ and $K = 10$, with no disagreement. An independent implementation of the criterion, working at the letter level rather than one edge per choice of all interface bits in a period, confirms the boundary directly: at $K = 14$ every necklace we tested with all zero-runs at least 17 is excluded, while at zero-runs 16 the words with 1-run 3 and with the pair (3, 3) are not.

Two things came out of re-running the criterion across the other families of Part I, and the first is methodological. *The criterion is not monotone in K .* A larger window is a finer over-approximation, so states that were identified become distinguished, and a component that was a single cycle can acquire branching; the word $0^{16}1^3$ is not certified at $K = 12$ or $K = 14$ and is certified at $K = 16$, while $0^{15}1^3$ is certified at $K = 14$ but not at $K = 12$. Since a certificate at any single K is a complete proof — the soundness lemma holds for every K — the right procedure is to take the smallest certifying K and never to read a failure at one K as final. Every threshold below is therefore an upper bound on the true one.

The sweep itself. For the zero-run family, individual necklaces with all cyclic zero-runs equal to Z and 1-runs up to length 5, together with the pairs (a, b) for $a, b \leq 3$, are all certified at $K = 16$ down to $Z = 15$; at $Z = 14$ the shape 1^3 and the pair (3, 3) resist, and they still resist at $K = 18$. What is independently reproduced and what is not should be separated here. The per-word results above are ours and were obtained from a second implementation: at $K = 16$ every necklace we tested with all zero-runs at least 15 is certified, and the obstruction at 14 survives $K = 18$, so it appears genuine rather than a limit of computation. The *family*-level statement — one certificate covering the family with 1-runs unbounded, which is what would lower the certified threshold from 19 to 17 — rests on a product construction we have not been

able to rebuild. Two reconstructions of it fail: the single-cycle test cannot apply to the product, because the choice of letter branches and so no component is a single cycle; and the forcing test on the product's recurrent part leaves a residue that does not vanish, though it is strikingly small and degrades smoothly, one unforced state at $Z = 18, 19, 20$, then 7, 45 and 3206 at $Z = 17, 16, 15$ with $K = 14$. So our reconstruction differs from the reported one in a way we have not identified; until the product certificate is rebuilt, the family statement at 17 is OPEN and the certified family threshold is the 19 of `cover14.py`. The two recorded holes do not move: $0^7 11$ and $1^8 0$ are uncertified at every K up to 16. Two smaller facts worth recording because they show the families are not intervals: $1^7 0$ is certified at $K = 10$ although the published family begins at $b \geq 9$, and $0^7 1$ is certified at $K = 12$ while $0^8 1$ is not, so the isolated-ones family is not monotone in the zero-run length either.

The ceiling of the method is worth stating in the same language, because it explains why the family route cannot be pushed further. The criterion is expressible as a sentence of monadic second-order logic over one successor, but its *conclusion* — ultimate periodicity of the interface sequence — is not ω -regular. So the method can quantify over a regular family of candidate periods, and can never quantify over all periods at once. The gap between the two is exactly the gap between Part I's theorems and P1.

10 Reproduction

All certificates are computations with a stated input and a yes/no criterion.

- Lemma 7.1, finite facts: `drive2.py`, `freeze.py`, `cross.py` (Glucose 4 cross-check); requires `python-sat`.
- §7.1: `probe.py` with `satL.py` (prints $|L| = 10, 21, 31, 38, 47, 54$ at depths 1–6, and checks all five bounded-window facts on every admissible τ -sequence at each depth: 11, 2221 and 21221 never occur, 1221 only at positions ≤ 1 , symbols outside $\{1, 2\}$ only at positions ≤ 2 — all five hold at every depth to 6, and the 38 depth-4 sequences are printed in full).
- Theorems 8.1–8.4: `python s3_fam1b0a.py 10 1 19 7 34`; `python s3_thm9.py 10 29 1 12`; `python kauto3.py <word> 10 <m>`; independent check `python kwin.py <word> ...` (Appendix), which prints the state and edge counts and the failing components if any.
- Theorem 7.4: family-level certificate `cover14_19_48` ($K = 14$, fixed-point coverage over 1-runs ≤ 48), reproduced in full on 2026-09-17 by `cover14.py 14 19 48` (log `cover14_19_48.out`, 53 minutes): the zero-run map from the full window set reaches its fixed point at step 28 with $|F| = 556$, the 1-run map at step 47, and for every 1-run length $a = 1, \dots, 48$ the next zero-run is forced for all $m_1, m_2 \geq 19$ and every $b = 1, \dots, 48$, with no undetermined cell in any of the 48 sweeps. Independently re-verified at $K = 14$ by the combinatorial criterion of Section 6 ($K=14$ `python kwin.py ...`) on the necklaces $0^{19}1^b$ for $b = 1, 2, 3, 4$ (42/68, 47/72, 56/89, 44/71 states/edges), $0^{19}10^{20}11$ (18/44) and $0^{19}1110^{19}101$ (32/65): every recurrent component a single cycle.

Every word run in more than one implementation received the same verdict in each.

11 What is not proved here

1. Theorem 3.1 is cited, with its proof reproduced. Theorem 7.4's family-level certificate at zero-runs ≥ 19 (uniform in the 1-run lengths) is reproduced by the code shipped here (`cover14.py`, Section 10), so it no longer rests on an unpublished working record;

the six representative necklaces remain as an independent cross-check by the combinatorial criterion of Section 6. The extension to 17 is reproduced word by word only (`s3_threshold17.py`). It is used above only for tails with a run ≥ 19 , which are also covered directly for the computed ranges.

2. The lower bounds in Theorems 8.1, 8.2, 8.4 are the smallest values at which the whole family is certified; individual smaller words (1^70 at $K = 10$, 0^71 at $K = 12$) are certified separately, so the families are not intervals, and “not certified at K ” is not “not excluded”. Since the criterion may be applied to any rotation of a necklace (Lemma 6.1), every uncertified word above was tested in all its rotations.
3. Uncountability of $L(01)$ and its siblings is not proved, and Proposition 9.2 is stated under exactly that hypothesis — not entropy — which suffices for its conclusion but is not necessary for it (Remark 9.5). The distinction is not editorial. The proposition’s sufficient condition, exponential growth of $|L_k|$, is *unavailable* here rather than merely unproved: the measured ratio is 1.094 at $k = 60$ and falling, which is consistent with zero entropy, so counting cannot deliver the hypothesis and uncountability must be reached by some other route. An earlier version of this item read “positive entropy of $L(01)$ and its siblings is measured, not proved”; that was ~~WITHDRAWN~~ as contradicting the exact counts recorded above and as misnaming the hypothesis of the proposition it cites. The second objection stands; the first does not, since the stream measurement of Remark 9.6 shows the counts describe the row-0 transient: positive entropy of $L(01)$ is now **MEASURED** (on random right halves, not on the counts, which the row-0 transient dominates), the zero-entropy expectation is withdrawn, and uncountability is still not proved.
4. Nothing here bears on the single seed beyond what holds for every number-like configuration, and nothing here excludes any word without a long run.

12 Note added: the constant fibres and the period-two horizon

Condrey [91] (arXiv, September 2026) classifies the two constant-trace fibres exactly: for a prescribed right half the unique left half with identically zero centre is an alternating tail selected by the first one of the right half (a prefix-OR of the right half), and with identically one centre it is one universal checkerboard; every nonzero member of either fibre has infinitely many ones, so no nonzero finite configuration has an eventually constant column, and for support radius w the sharp maximal constant prefix has length $w + 2$ with 2^w or $2^w - 1$ extremisers by parity of w . Theorem 8.6 above gives the eventual-constancy exclusion for the wider class of number-like configurations by the two-column theorem, and its one-case is attributed by [91] to Jen [12, Thm. 7a], a citation we have not been able to verify at source and which should be treated as second-hand; the explicit fibres and the sharp horizons are Condrey’s and are not contained here. His Lemma 1 is Proposition 9.1 in its finite form.

Condrey defines $H(p, w)$, the sharp maximal length of an eventually p -periodic centre prefix over rows of support radius w , notes $H(2, w) \geq w$ from left permutivity, and leaves period two open, which is the target of Part I. We computed the period-two horizon exactly by exhaustion over all 2^{2w+1} rows (**EXACT**; prefix lengths, both phases of 01 counted):

w	1	2	3	4	5	6	7	8	9	10
alternating prefix length	7	7	7	7	9	10	10	15	17	17
extremisers	1	1	1	2	7	11	67	3	1	2

so $H(2, w) = 6, 6, 6, 6, 8, 9, 9, 14, 16, 16$ in his index; the same exhaustion reproduces his constant-case table exactly (lengths 3, 3, 5, 5, 7, 7, 9 and 2, 4, 4, 6, 6, 8, 8 with counts $2^w - 1$ and 2^w). The period-two horizon runs about six to seven ahead of w with an irregular staircase and no sign of a closed law; the one-sided version of the same quantity (left support bounded, right half free) is the kill instrument of Section 22, whose excess over w was measured to grow like $\log w$.

Part II

The Duhamel class and its decimation

13 The Duhamel formula and the 2-kernel

Write $U_t(z) = \sum_x u_x^t z^x \in \mathbb{F}_2[z^{\pm 1}]$ for row t , $\Lambda(z) = z + 1 + z^{-1}$ for the symbol of Rule 150, and $Q_t(z) = \sum_x u_x^t u_{x+1}^t z^x$ for the census of adjacent pairs. Let $T_m(z) = \Lambda(z)^m$ and $T_m(x) = [z^x]T_m(z)$; T_m is the row m of Rule 150 from the seed, and $T_m(x) = T_m(-x)$.

Lemma 13.1 (KNOWN, elementary). $T_{2m}(z) = T_m(z^2)$ and $T_{2m+1}(z) = \Lambda(z) T_m(z^2)$. Moreover $T_m(0) = 1$ for every $m \geq 0$.

Proof. In characteristic 2, $\Lambda(z)^2 = \Lambda(z^2)$, which gives the first two identities. For the third, $T_m(0) = [z^m](1 + z + z^2)^m = \sum_b \binom{m}{2b} \binom{2b}{b}$ (choose the b factors contributing z^2 and the $m - 2b$ contributing z), and $\binom{2b}{b}$ is even for $b \geq 1$ (the binary addition $b + b$ carries), so $T_m(0) \equiv \binom{m}{0} = 1$. $\square$

Theorem 13.2 (Duhamel formula; PROVED). For all $t \geq 0$,

$$U_{t+1}(z) = \Lambda(z) U_t(z) + Q_t(z), \quad \text{hence} \quad U_t = \Lambda^t U_0 + \sum_{s < t} \Lambda^{t-1-s} Q_s,$$

and, reading the constant term with $\delta_x = [x = 0]$ and $q_x^s = u_x^s u_{x+1}^s$,

$$c_t = \langle T_t, \delta \rangle + \sum_{s < t} \langle T_{t-1-s}, q^s \rangle = 1 + \sum_{s < t} \sum_x T_{t-1-s}(x) u_x^s u_{x+1}^s, \quad \langle a, b \rangle = \sum_x a_x b_x.$$

In the notation of Section 15, $c = c[\delta, q]$.

Proof. The rule is $u_x^{t+1} = u_{x-1}^t + u_x^t + u_{x+1}^t + u_x^t u_{x+1}^t$; the three linear terms are ΛU_t and the quadratic term is Q_t . Iterate. The constant term of $\Lambda^m F$ is $\sum_x T_m(-x) F_x = \langle T_m, F \rangle$ by symmetry of T_m , and $\langle T_t, \delta \rangle = T_t(0) = 1$ by Lemma 13.1. Verified numerically for $t \leq 200$ (EXACT). $\square$

Rule 30 is thus Rule 150 driven by the source q ; the same identity with the Rule 90 symbol $z + z^{-1}$ and the source $u_x^s(1 + u_{x+1}^s)$ (the right-hand ends of black runs) gives c_t as the parity of those sites on the Pascal support (PROVED, ledger 10.68). Both are the same statement in different linear bases.

Cartier operators, the 2-kernel and automaticity (Known)

For $G(y) = \sum_t g_t y^t \in \mathbb{F}_2[[y]]$ put $\Lambda_r G = \sum_t g_{2t+r} y^t$ ($r = 0, 1$). The 2-kernel of a sequence g is $\mathcal{K}(g) = \{(g_{2^k m+r})_{m \geq 0} : k \geq 0, 0 \leq r < 2^k\}$; its elements are the coefficient sequences of $\Lambda_{r_{k-1}} \cdots \Lambda_{r_0} G$ over binary words. A sequence is 2-automatic iff its 2-kernel is finite (Eilenberg [69]), iff its generating series is algebraic over $\mathbb{F}_2(y)$ (Christol [44]). An eventually periodic sequence is 2-automatic. Hence **an infinite 2-kernel of c implies P1**; this is the automaticity route, and it is strictly stronger than P1.

Proposition 13.3 (the uniform-measure two-point function lives on one ray; PROVED). *Let F be any left-permutive radius-one rule and let μ be the uniform Bernoulli measure. Then the two-point function $C(x, t) = \mathbb{E}_\mu[(-1)^{u_0^0 + u_x^t}] - \mathbb{E}_\mu[(-1)^{u_0^0}] \mathbb{E}_\mu[(-1)^{u_x^t}]$ vanishes except on the single ray $x = t$. In particular $C(0, t) = 0$ for every $t \geq 1$: in the measure in which all of this machinery operates, the centre column is exactly uncorrelated with the seed.*

Remark 13.4 (the balance itself is KNOWN). The balance $\Pr_\mu(u_0^t = 1) = \frac{1}{2}$ for every $t \geq 1$ under every left-permutive rule is not the special case $C(0, t) = 0$ but a different statement: it is the one-point identity $\mathbb{E}_\mu[(-1)^{u_0^t}] = 0$, while $C(0, t) = 0$ is the two-point identity $\mathbb{E}_\mu[(-1)^{u_0^0 + u_0^t}] = 0$. Both follow from $u_0^t = u_{-t}^0 \oplus g_t(u_{-t+1}^0, \dots, u_t^0)$ with u_{-t}^0 uniform and independent of the pair (u_0^0, g_t) . For Rule 30 the balance is classical: the rule is surjective [14, Thm. 6.6], so by the balance property [14, Thm. 5.4] it preserves μ (Section 26) and u_0^t has the law of u_0^0 . It is also stated as Theorem 3 of Chan-López and Martín-Ruiz [90], but the XOR lemma their proof rests on, that $X \oplus Y$ is a fair bit for every Y even when Y depends on X , is false (take $Y = X$); nothing in the balance is new here. What Proposition 13.3 adds is the full two-point function. Its proof uses only that a radius-one rule bijective in its left argument is $l \oplus h(c, r)$ over $\mathbb{F}_2$, which is all that “strong” left permutivity (defined in the proof of Theorem 14.7) means at radius one.

Proof. Two lines from strong left permutivity: for $x < t$ the cell u_x^t is, for each fixed assignment of the other coordinates in its light cone, an exclusive-or with the leftmost coordinate, so summing over that coordinate cancels the expectation; for $x > t$ the two cells are independent. $\square$

Remark 13.5 (why the solvable-circuit analogy must not be made). The exactly solvable circuit literature has correlations supported only on the light cone, computed by a transfer operator in the space direction, and the resemblance to Corollary 15.5 is a trap. Their confinement is measure-theoretic, following from preservation of the maximal-entropy measure in both directions together with tracelessness of the observable; ours is an identity over $\mathbb{F}_2$, valid orbit by orbit. Proposition 13.3 is the decisive separator: theirs is a statement in the uniform Bernoulli measure, and in that measure our centre column carries no signal at all. The class is also empty of informative members — the elementary rules satisfying the classical hypothesis are the shifts, whose correlation is frozen, and the linear rules, whose output vanishes — and the values of C on the ray $x = t$ for Rule 30 satisfy no linear recurrence of order at most 5 over $\mathbb{Q}$, where a dual-unitary qubit circuit forces order at most 3.

One item in that literature transfers as a target rather than a theorem, and we record why it does not close either. Influence-solvability in the sense of Hübner and Kim [61] is characterised by necessary and sufficient local conditions (their §III), is stated there for reversible gates with an extension to non-reversible ones that they call straightforward (§II.B), and its criterion is a condition on the width-two trace measure; the deciding question is whether that trace subshift is sofic. Computing its exact language to horizon 11 — 38 616 realisable traces of 4 194 304 initial windows — and counting distinct sets of length- m continuations over contexts of length k gives

4, 7, 8, 11, 14, 16, 18, 19 at $m = 3$, then 4, 8, 11, 18, 28, 37, 47 and 4, 10, 16, 31, 54, 89 at $m = 4$ and $m = 5$. The flattening of the shallow rows is not evidence of soficity and cannot be: for fixed m the count is non-decreasing in k and bounded by 2^{2^m} , so it saturates in *every* subshift, sofic or not, and here the $m = 3$ row is still rising at the deepest context available. What the probe yields is a rising sequence of *lower* bounds on the number of follower sets — 7 at $m = 2$, and at least 19, 47 and 89 at $m = 3, 4, 5$ — and never an upper bound, while soficity requires finiteness. It is therefore the obstruction of Remark 14.5 in another guise.

Remark 13.6 (the same object over $\mathbb{Z}_2$, and why the two literatures never meet). Part II works in $\mathbb{F}_2[[y]]$, where Christol’s theorem lives. The same recursion has a second natural home, and the comparison explains a gap that is otherwise puzzling.

Read a configuration as a 2-adic integer and pass to the co-moving frame. Rule 30 becomes $H(a) = a \oplus (2a \vee 4a)$, whose n -th output digit depends only on digits $0, \dots, n$ of the input — a 1-Lipschitz map in the sense of [50] — and the centre column is the *diagonal* read $c(t) = \text{digit}_t(H^t(1))$, which we verified against simulation for $t < 3000$. The map is measure-preserving, being a bijection mod 2^n for every $n \leq 16$, and left-permutivity is why: the opposite orientation $G(a) = 4a \oplus (2a \vee a)$ is 1-Lipschitz but not measure-preserving, its image mod 2^{16} having 18 869 of 65 536 elements.

Two things follow, and the first corrects a reflex. One is tempted to dismiss ergodic statements here because the seed is a single point of measure zero; for a 1-Lipschitz map on $\mathbb{Z}_2$ that is wrong, since ergodicity is equivalent to transitivity mod 2^n for every n , a statement about every orbit. The route fails for sharper reasons: H is not ergodic at all, as $H(0) = 0$ makes $2^n\mathbb{Z}_2$ invariant; and ergodicity would control each *fixed* digit, that is each fixed diagonal, never digit_t at time t . The diagonal read is the obstruction, as it is throughout this paper.

The comparison is the useful part. Rule 90 is $(1 + y^2) \cdot a$, natural in $\mathbb{F}_2[[y]]$ and in $\mathbb{Z}_2$ alike, having neither carries nor an OR; Rule 30’s OR is natural only over $\mathbb{Z}_2$, while Christol’s theorem needs characteristic 2 and no carries. Each literature holds half of our object and neither holds it whole, which is why no result crosses between them. For completeness, the 2-adic complexity of the centre column is finite exactly when the sequence is eventually periodic, so it restates Problem 1.

14 Certified bounds

A *Mahler relation of depth n and degree D* for $G = \sum c_t y^t$ is $\sum_{i=0}^n p_i(y)G(y^{2^i}) = 0$ with $\deg p_i \leq D$, not all zero; G is algebraic iff such a relation exists for some (n, D) . Its existence is the vanishing of a null vector of an explicit $\mathbb{F}_2$ -matrix built from c , and full rank of a square submatrix on a stated set of rows is a certificate of non-existence for that (n, D) .

Theorem 14.1 (EXACT). *For the centre column of Rule 30:*

1. *no Mahler relation of depth $n \leq 12$ and degree $D \leq 8000$ exists ([81, Thm. 9], where each determinant is computed twice, by a row-streaming elimination and by an elimination of the transposed matrix; re-certified by independent code to $D \leq 4000$ at $n \leq 12$, with certificates `ag2_cert_n<n>_D<D>.txt` each re-verified from its file by a transposed elimination, the box (12, 8000) itself not being re-run; controls Thue–Morse $(n, D) = (2, 4)$, paperfolding $(2, 4)$, rational and Rule 90/150 columns all recovered at their known (n, D));*
2. *the truncated kernel elements are pairwise distinct wherever they have been compared. From n computed terms the maximal window at level k is $W_k = 2^{n-k}$ simultaneously for every*

residue (PROVED), so the census is a clean (k, W) grid; on that grid every computable cell with $W \geq 32$ is injective, and all non-injectivity sits in the columns $W \leq 16$, where it is forced by pigeonhole and is therefore a truncation artefact rather than a kernel coincidence. At $n = 21$ the map $(k, r) \mapsto \text{element}$ is injective on every pair with $\max(k, k') \leq 15$, which is 2,147,385,345 comparisons at window length at least 64, and the 65 536 residues of level 16 are pairwise distinct on 32 terms; of the 6,442,352,640 pairs whose higher level is 16 exactly one is unresolved, against an independence expectation of 1.5, and the closest separated pair agrees on 31 terms and differs at the 32nd. The sharpest quantity the prefix supports is not the count of a single level but the largest antichain in the prefix trie, two elements being provably distinct exactly when neither prefix extends the other. That quantity has a name: it is Shallit's automaticity function, whose second definition [71, §1] is the maximum number of pairwise n -dissimilar sequences in the n -truncated k -kernel, with dissimilarity meaning that neither is a prefix of the other, and it comes with the same Dilworth optimality remark. So the statement to make is not a bound but an exact value: $A_2^c(2^{20} - 1) = 80\,660$ and $A_2^c(2^{21} - 1) = 138\,672$, computed by the algorithm of [71], hence at least 138 672 states for a least-significant-digit-first automaton and at least 18 for a most-significant-digit-first one. The order is maximal: $A_k^s(n) = O(n/\log_k n)$ for every sequence, and our value sits at $1.389 n/\log_2 n$, so no computation on a prefix can improve it by more than a constant factor. The figure 1,048,486 of [81] is $2^{20} - 1 - 89$, that document's own census at 2^{24} terms with 89 residues dropped, a count its method predicts at 96 under independence; the method is sound but needs 2^{24} terms, roughly 46 hours here, and is not re-derived.

Remark 14.2 (the census is capped; PROVED). By item (2), from 2^ν terms a level- k kernel element is seen through a window of length $2^{\nu-k}$, and injectivity is certified only at $W \geq 32$; so the census can certify at most $\sum_{k \leq \nu-5} 2^k = 2^{\nu-4} - 1$ elements from 2^ν terms, 131 071 at $\nu = 21$ — about $n/16$. The antichain bound 138 672 of item (2) is a different object (pairwise incomparable truncations, certified by prefix comparison rather than by window) and is not subject to this cap.

Remark 14.3 (what Christol's theorem gives effectively, and why the conversions are needed). Since Part II converts kernel bounds into exclusions on an algebraic presentation, it matters how much the classical theorem gives by itself. Reading the original [44]: the equivalence between 2-automaticity and algebraicity is the only clause needing p prime, the equivalence with the finite k -kernel is Eilenberg's [69] and is base-agnostic, and $q = p^s$ -automatic coincides with p -automatic. The effective content, however, is *doubly exponential* — their §7 bounds the kernel by $q^{s(N+1)}$ with Ore height at most $2hq^d$ — so running the theorem backwards from an algebraic presentation to a state count gives only a doubly logarithmic exclusion. That is why the sharper conversions of Bridy and of Adamczewski and Yassawi are necessary rather than merely convenient, and it also locates the binding constraint: the bounds $hd \geq 18$, $(h+1)d \geq 17$ and $h + d + g \geq 19$ are limited by our census of 138 672, not by anything in Christol. One attribution to correct while we are here: the degree bound $q^m - 1$ that we took from Bridy is already in [44, §6].

The same paper's later sections supply a concrete exclusion we can state. For a generalised Morse sequence $\mu(u)$ the minimal homogeneous Mahler relation has order 2 and degree $2^\ell + 2$, which at $\ell = 1$ is $(2, 4)$ and reproduces the Thue–Morse relation we use as a control. Theorem 14.1(1) therefore already excludes $c = \mu(u)$ for all 527 345 non-trivial generalised words of length at most 12 — a named published family that contains both Thue–Morse and Rudin–

Shapiro. It also exposes a weakness in our control set, which we record rather than repair here: the non-rational controls we ran reach only degree 82 — Thue–Morse and paperfolding sit at $(n, D) = (2, 4)$, and the off-centre Rule 90 and Rule 150 columns at offset k , leading zeros stripped, need depth 2 and degree $D = 4k$ and $D = 4k + 2$ respectively (EXACT at $k = 1, 2, 3, 5, 7, 14, 20$, each relation verified on 10^4 terms) — while the scan claims to reach degree 8000, and [44] supplies an infinite depth-two ladder at degrees 4, 6, 10, 18, $\dots$, 4098 with closed-form minimal polynomials, which is the right family to test against.

Remark 14.4 (a route, with the hypothesis verified as far as we can compute). The same paper supplies the only technique the literature has for proving an automaticity lower bound on an *explicit* sequence, and it converts our census into a hypothesis rather than an endpoint. Shallit’s separation lemma [71, Lemmas 3, 12] needs, for every level i and every pair of distinct residues $a \neq b$ below 2^i , a witness $m \leq M(2^i)$ with $c(2^i m + a) \neq c(2^i m + b)$; if M is polynomially bounded the sequence is not 2-automatic, and hence by Christol’s theorem G is transcendental over $\mathbb{F}_2(y)$ and c is not eventually periodic. Our census *is* that hypothesis, verified to $i = 16$ with the constant witness bound $M = 32$, far inside the polynomial regime; with $M = O(\log r)$ the same proof reaches the universal ceiling $\Omega(n/\log n)$. Two calibrations keep this honest. The witness in every published application comes from an infinite arithmetic theorem, never from data — the examples run through the primes, the squarefree numbers, the Thue–Morse and Sturmian words — and the best bound proved for any explicit non-automatic sequence is $n^{2/5}$, some 320 times below what we measure. And a decision procedure does exist for one class: Krawczyk and Müllner [74, Thm. B] decide automaticity for nonperiodic uniformly recurrent substitutive sequences by a finite eigenvector check on a return substitution. One correction to how we first read that route: the missing input is *not* simply “a substitution for c ”. Christol, Kamae, Mendès France and Rauzy’s substitution clause [44] is for *uniform* substitutions and is Cobham’s theorem, so exhibiting a uniform substitution would make c automatic — the opposite of what we want. Any useful substitution here must be non-uniform, and then it is not Cobham that applies.

Remark 14.5 (what the census does and does not evidence). Controls fix the honest reading. The same census returns exactly 2 on Thue–Morse and exactly 4 on the paperfolding sequence, so the implementation is sound; but a periodic sequence of period 12,345, whose 2-kernel is finite, scores 47,086, which is larger than any figure above, and two independent coin sequences score within 0.24% of Rule 30’s value. A kernel census therefore measures how much counting room a prefix allows, not aperiodicity, and a large count is not evidence for an infinite kernel. The limitation is structural rather than a matter of taking more terms, and it has a one-line witness: the sequence that agrees with c on the first 2^{20} terms and is zero thereafter is ultimately periodic, hence 2-automatic with a finite 2-kernel — computed at 107 047 classes — and it agrees with c on every term any census of that prefix can read. Since every finite word extends to an automatic sequence, no measured value, growth rate, or proximity to the universal ceiling can imply non-automaticity. The right name for this is elementary and we use it rather than a grander one: the ultimately periodic sequences are dense in the Cantor topology on $\{0, 1\}^{\mathbb{N}}$, since $w \cdot 0^\infty$ lies in the cylinder of w . It is not a profinite statement — the profinite topology on the free monoid is discrete, and profinite closeness and agreement on a prefix are incomparable notions. This is KNOWN, under the name *diversity*: Shallit [71] proves that almost all binary sequences are maximally diverse, warns that “it is not so easy to prove that any individual sequence has the maximally diverse property”, and closes by saying he knows no explicit example of a diverse sequence with diversity measure $O(\log n)$ — which is exactly the wall met here, stated in 1996.

The one part that does appear to be unrecorded is the quantitative control: a periodic sequence can score high because for odd period p one has $|\mathcal{K}| \leq p \cdot \text{ord}_p(2)$, which accounts for the 47 086 of the period-12 345 control. Moreover the certified family is capped at $2^{\nu-4} - 1$ elements from 2^ν terms, about $n/16$ (Remark 14.2), so no census of a prefix can produce the one-parameter family of distinct kernel elements that the criterion of Allouche, Shallit and Yassawi requires. The census route to P1 is closed; what survives is the exclusion of low-complexity algebraicity by the Mahler certificates.

Source: [81] and the independent re-run `agent_certificate.md` (2026-09-16). The positive control set is audited: `ag2_controls.py` and its log record the box shapes and recover Thue–Morse at $(n, D) = (2, 4)$, paperfolding at $(2, 4)$, the rational cases, and the Rule 90 and Rule 150 off-centre columns at their known depth-2 relations, each relation verified on 10^4 terms, so the search does exercise depth ≥ 2 on non-rational algebraic series; the apparent false negative for Rule 150 at offset 20 is a bound of the scan, not a failure of the instrument. These remain bounds, not a proof of non-automaticity.

Proposition 14.6 (the centre column is one designated ANF coefficient; PROVED). *Let F_t be the centre cell at time t as a Boolean function of the initial row on the window $[-t, t]$, and write x_j for the variable at position j . Then*

- (a) $F_t = x_{-t} \oplus g_t(x_{-t+1}, \dots, x_t)$: *flipping the leftmost cell always flips the centre, so the coefficient of the monomial x_{-t} is 1 for every t .*
- (b) *The Walsh spectrum $W_t(a) = \sum_x (-1)^{F_t(x) + a \cdot x}$ vanishes identically for every mask a with $a_{-t} = 0$. Half the spectrum is zero, and the entire first-order structure is the single integer $W_t(e_{-t})$.*
- (c) $c(t) = F_t(\delta_0)$ *is exactly the ANF coefficient of the single monomial x_0 .*
- (d) *Adding that one monomial, $F_t \mapsto F_t + x_0$, flips $c(t)$ while permuting the Walsh spectrum by $W(a) \mapsto W(a \oplus e_0)$, hence leaving the Walsh multiset, the algebraic degree, and every invariant defined from them unchanged.*

Proof. (a) Rule 30 is left-permutive, and the flip of the leftmost cell propagates along the left edge of the light cone to the centre in exactly t steps. (b) Pair x with $x + e_{-t}$: by (a) the exponent $F_t(x)$ changes while $a \cdot x$ does not when $a_{-t} = 0$, so the two terms cancel. (c) $F_t(0) = 0$, so the constant term of the ANF vanishes, and $F_t(\delta_0)$ is the sum of the coefficients of the monomials supported in $\{0\}$, which is the coefficient of x_0 alone. (d) Since $x_0 = e_0 \cdot x$, the exponent $F_t + x_0 + a \cdot x$ equals $F_t + (a \oplus e_0) \cdot x$. Verified exactly for $t \leq 11$ in (a)–(c) and $t \leq 7$ in (d), with the ANF computed to $t = 15$, where F_{15} has 344 592 268 monomials. $\square$

Theorem 14.7 (the degree law for the centre cell; PROVED). *Let F_t be as in Proposition 14.6. For every $t \geq 2$, $\deg F_t = 2t - 1$. For $t \geq 3$ the top-degree part of F_t is the single monomial $\prod_{j=-t+2}^t x_j$ — every light-cone variable except the two leftmost — and its coefficient is 1.*

Proof. Three ingredients. (i) *Strong left permutivity.* In the ANF of any cell the leftmost light-cone variable occurs only as the bare singleton: by induction, the new cell’s leftmost variable lies in L ’s cone alone, $L = x \oplus$ (terms free of x), and C , R and CR never see it. (ii) *Upper bound and uniqueness,* by induction from $\deg F_2 = 3$. With $F_{t+1} = L \oplus C \oplus R \oplus CR$ and $\deg L, C, R \leq 2t - 1$, only CR can exceed $2t - 1$. Every monomial of degree ≥ 2 omits the new leftmost variable x_{-t-1} by (i), so $\deg \leq 2t + 2$. A monomial of degree $2t + 2$ or $2t + 1$ that contains x_{-t} must take it from C , whose factor is then the singleton x_{-t} by (i), leaving R to supply $2t + 1$ or $2t$ variables against $\deg R \leq 2t - 1$. Hence $\deg F_{t+1} \leq 2t + 1$, and

the only candidate of that degree omits x_{-t-1} and x_{-t} : the product of the remaining $2t + 1$ variables. (iii) *Non-vanishing*. By Möbius inversion the candidate's coefficient is the parity of $\#\{x : x_{-t} = x_{-t+1} = 0, F_t(x) = 1\}$, which has the parity of $R_t/2$ for $R_t := \#\{x : x_{-t} = x_{-t+1} = 0, \partial F_t / \partial x_t = 1\}$. Since $\partial(l \oplus (c \vee r)) / \partial r = 1 \oplus c$, the centre depends on x_t exactly when the chain of cells $a(k, t - k - 1)$, $k = 0, \dots, t - 1$, down the right edge of the cone all vanish. Each of those cells is permutive in its own leftmost variable, the odd-position cell x_{t-2k-1} , so the system is triangular: the odd positions are determined and the even ones free, and x_{-t+1} becomes a function ψ_t of the $t - 1$ free interior cells (x_t enters nowhere), with $R_t = 2^t - 2 \text{wt}(\psi_t)$. Thus $R_t \equiv 2 \pmod{4}$ iff $\bigoplus_x \psi_t(x) = 1$. Put $b(k, j) = a(k, t - 1 - k - j)$, the cell j steps left of the gate diagonal in row k (so $b(0, j) = x_{t-1-j}$ and $b(0, 2t - 2) = x_{-t+1}$); the rule reads $b(k + 1, j) = b(k, j + 2) \oplus (b(k, j + 1) \vee b(k, j))$, permutive in $b(k, j + 2)$, and the gates are $b(k, 0)$. Solve sideways: put $B(k, 0) = 0$, $B(k, 1) = c_k$ and $B(k, j + 2) = B(k + 1, j) \oplus (B(k, j + 1) \vee B(k, j))$ (two adjacent columns fix the space-time of a permutive automaton; read in the coordinates a , each step is Meier and Staffelbach's completion backwards [82, eq. (4)]). Then $c \mapsto \text{row } 0$ is a bijection from $\mathbb{F}_2^{t-1}$ onto the row-0 words with all gates closed, $B(k, j)$ depends only on $c_k, \dots, c_{k+\lfloor (j-1)/2 \rfloor}$, and $G_m := B(0, 2m - 1) = c_{m-1} \oplus g_m$ with g_m free of c_{m-1} . Write $F_m := B(0, 2m)$ and $T_m(X) := \bigoplus_{c \in \mathbb{F}_2^m} X(c)$, so that $\bigoplus \psi_t = T_{t-1}(F_{t-1})$ (a cube sum is invariant under a bijection of the cube). Expanding the OR, $F_m = B(1, 2m - 2) \oplus G_m \oplus F_{m-1} \oplus G_m F_{m-1}$. The first three terms have $T_m = 0$: $B(1, 2m - 2)$ misses c_0 , G_m is the bare c_{m-1} plus a term missing c_{m-1} , and F_{m-1} misses c_{m-1} ; and $T_m((c_{m-1} \oplus g_m)F_{m-1}) = T_{m-1}(F_{m-1})$ because $gh \oplus (1 \oplus g)h = h$. Hence $T_m(F_m) = T_{m-1}(F_{m-1}) = \dots = T_1(F_1) = T_1(c_0) = 1$, so $\bigoplus \psi_t = 1$ for every $t \geq 2$, $R_t \equiv 2 \pmod{4}$ and the coefficient is 1. A second, independent proof runs the same telescoping in the original variables through a two-to-one map onto the $(t - 1)$ -instance. Both are reproduced by `s3_verify_topcoef.py` and `s3_gate_fast.py`; the census R_t is exact to $t = 22$ ($R_{22} = 2066214$), and a brute force of the original definition agrees to $t = 12$. On provenance: we have not found the law in print, and its ingredients are classical (completion backwards, and the top ANF coefficient as a cube sum). It is a different statement from the Fibonacci degree growth recorded in [90], which concerns a univariate polynomial encoding the support of a single-seed row and has no free input variables, whereas F_t is a function of all $2t + 1$ light-cone cells and its degree is linear in t . $\square$

Proposition 14.8 (the right-edge influence law; PROVED). *For uniform $x \in \mathbb{F}_2^{2t+1}$ let $\text{Inf}_j = \Pr[F_t(x) \neq F_t(x \oplus e_j)]$ be the influence of the light-cone variable x_j on the centre cell $F_t = a(t, 0)$. Then $\text{Inf}_{-t} = 1$ and $\text{Inf}_t = \text{Inf}_{t-1} = 2^{-t}$ for every $t \geq 1$. (EXACT alongside, $t \leq 12$: $\text{Inf}_{t-2} = (2t - 1)2^{-t}$.) This is the exact right-edge form of the sensitivity profile that Chan-López and Martín-Ruiz measure by Monte Carlo for $t \leq 20$ [90, §8.2], whose “flat left profile $\approx \frac{1}{2}$ ” is not exact. The endpoint $j = -t$ has value 1; the next two cells have $\text{Inf}_{-t+1} = \text{Inf}_{-t+2} = \frac{1}{2}$ exactly at every odd t (PROVED, ledger 10.142c2: for odd t the derivative in x_{-t+1} is permutive in x_{-t+2}) and at no even t computed (EXACT, $t \leq 12$ for x_{-t+1} and $t \leq 10$ for x_{-t+2} ; for instance $\text{Inf}_{-t+1} = 5/16$ at $t = 4$); and the remaining left cells $x_{-t+3}, \dots, x_{-1}$ take values between 0.467 and 0.523 at $t = 11$ (EXACT).*

Proof. $\text{Inf}_{-t} = 1$ is left permutivity. For the right edge let $G_j = a(j, t - j - 1)$ and $P_j = a(j, t - j)$ be the two rightmost cells of row j of the cone and $H_j = a(j, t - j - 2)$ the next. (a) *Independence*. The map from row j to (row $j + 1, G_j, P_j$) is a bijection, since left permutivity recovers row j from its two rightmost cells and row $j + 1$ one cell at a time from the right; so if row j is uniform then row $j + 1$ is uniform and independent of (G_j, P_j) , and by induction the pairs (G_j, P_j) , $j < t$,

are independent fair bits, with $H_j = P_{j+1} \oplus (G_j \vee P_j)$. (b) *Damage*. A flip of x_t or of x_{t-1} changes in row j only G_j and P_j , by $(\delta G_j, \delta P_j)$ say, and the rule gives

$$\delta G_{j+1} = \delta G_j (1 \oplus H_j), \quad \delta P_{j+1} = (G_j \vee P_j) \oplus ((G_j \oplus \delta G_j) \vee (P_j \oplus \delta P_j)),$$

with $\text{Inf} = \Pr[\delta P_t = 1]$. For x_t the damage starts at $(0, 1)$, δG stays 0 and $\delta P_{j+1} = \delta P_j(1 \oplus G_j)$: the derivative $\partial F_t / \partial x_t$ is $\prod_{k < t} (1 \oplus G_k)$, the gate chain of Theorem 14.7, and its probability is 2^{-t} by (a). For x_{t-1} the damage starts at $(1, 0)$, and the chain on $(\delta G_j, \delta P_j, P_j)$ driven by the fresh bits G_j and P_{j+1} has, after $j \geq 1$ steps, the law $\Pr[(1, 1, P)] = 4^{-j}/2$ and $\Pr[(0, 1, P)] = (2^j - 1)/(2 \cdot 4^j)$ for each P , $\Pr[(1, 0, 1)] = (2^j - 1)/4^j$, $\Pr[(1, 0, 0)] = 0$, the remaining mass dead; the induction is the two displayed formulas read case by case. Hence $\Pr[\delta P_j = 1] = 2^{-j}$. The chain reproduces the brute-force influences for $t \leq 12$ and gives 2^{-t} exactly to $t = 60$ (`s3_influence.py`); the value was known by enumeration to $t = 11$ (ledger 10.141b), and the phenomenon — the right sensitivity decays because $l \oplus (c \vee r)$ is free of r when $c = 1$ — is in [90, §8.2]. $\square$

Remark 14.9 (which rules carry the parity, and the same monomial as the census). The argument in (iii) applies verbatim to every left-permutive rule $a' = l \oplus h(c, r)$ with $h = h_{00} \oplus h_{10}c \oplus h_{01}r \oplus h_{11}cr$, and gives $\bigoplus \psi_t = h_{10}h_{11}^{t-2}$, so $R_t \equiv 2 \pmod{4}$ exactly when $h_{10} = 1$ and either $h_{11} = 1$ or $t = 2$: Rules 30, 75, 180 and 225 at every t , and Rules 60, 105, 150 and 195 at $t = 2$ only. PROVED by two independent derivations and EXACT by brute force of the original count over all sixteen such rules for $t \leq 8$ (`s3_verify_general.py`).

The step from R_t to the top coefficient is Rule 30's, and it does not generalise: it runs through $\partial(l \oplus (c \vee r)) / \partial r = 1 \oplus c$, whereas the general gate derivative is $h_{01} \oplus h_{11}c$. The coefficient of $\prod_{j=-t+2}^t x_j$ in F_t is in fact $h_{11}(1 \oplus h_{10} \oplus h_{01})$, independent of t and of h_{00} , so the rules reaching the maximal degree with that monomial are 30, 120, 135 and 225 — not the same four, and Rules 60, 105, 150 and 195 do not reach it even at $t = 2$. EXACT, all sixteen rules and $t \leq 8$ (`s3_crossrule_check.py`); no proof of the general formula is claimed here, only of Rule 30's (Theorem 14.7). An earlier version of this remark carried the R_t list over to the top coefficient and read “the top coefficient is 1 at every t for exactly Rules 30, 75, 180 and 225”; that sentence is WITHDRAWN, being false at four rules.

The cleanest instance is Rule 30 = $150 \oplus cr$: the single monomial cr switches on both the parity and, by Remark 15.12, the single-seed aperiodicity that Rule 150 lacks. That is the same quadratic coupling of the centre to a neighbour which the census found necessary — and, the census is explicit, far from sufficient — so this is one monomial appearing in two results, not an explanation of the second by the first.

Remark 14.10 (why this closes the Boolean-invariant route exactly). Proposition 14.6 turns Problem 1 into a statement about one designated coefficient of an explicit Boolean function, which sounds like progress and is in fact an obstruction, because (d) says that coefficient is invisible to the standard invariants. Degree, nonlinearity, correlation immunity, the rank of the quadratic part, and the dimension of the linear space are all functions of the Walsh multiset or of the monomial support above degree one, and adding x_0 changes $c(t)$ while fixing every one of them. So no such invariant can decide the centre column, however precisely we compute it — and we can compute a good deal, and one thing we can now prove: by Theorem 14.7 the degree is exactly $2t - 1$, near-maximal in $2t + 1$ variables, with a single top monomial for $t \geq 3$, and the linear space is $\{0, e_{-t}\}$ (EXACT to $t = 9$; at $t = 2$ it has dimension 2). One correction to how an earlier draft described the open half of that law. The divisibility $W_t(e_{-t}) \equiv 0 \pmod{8}$ is not it.

Since $F_t = x_{-t} \oplus g_t$ one has $W_t(e_{-t}) = 2^{2t+1} - 4 \text{wt}(g_t)$, so the divisibility says exactly that $\text{wt}(g_t)$ is even, i.e. that the full monomial of g_t is absent — which is the *upper* bound $\deg \leq 2t - 1$, the half that was already proved. The lower bound is the non-vanishing of one designated coefficient, and $W_t(e_{-t})$ does not see it even modulo 16: the residues run 8, 0, 0, 0, 8, 0, 0, 8 at $t = 2, \dots, 9$ while the coefficient is 1 throughout. This obstruction is worth distinguishing from the Rule 90 obstruction of Proposition 26.1, which says a hypothesis is too weak to separate two rules. Here the hypothesis class is blind to the quantity itself, on one rule. For the record, the low-degree layers of the ANF are not independent information either: the degree-one layer is the mirrored single-seed row, the degree-two layer is the pairwise superposition defect and the degree-three layer the triple defect, verified on all 615 pairs and 1 035 triples with no mismatch. And the bias bound available for a quadratic form over $\mathbb{F}_2$, which is governed by its rank, does not apply, its hypothesis $\deg = 2$ failing here by a factor that grows to 97 over the range computed.

Remark 14.11 (the Hankel rank: one object behind two censuses, and why it cannot discriminate). The rational-series view makes our two censuses one object and supplies the only bound we have about *weighted* machines. The right Hankel matrix is not the classical $[c_{i+j}]$ — finite rank there means a linear recurrence, which over $\mathbb{F}_2$ is eventual periodicity, so that object restates Problem 1 — but the matrix indexed by base-2 digit words, whose columns *are* the elements of the 2-kernel. Hence $\log_2 |\mathcal{K}(c)| \leq \text{rank} \leq |\mathcal{K}(c)|$, which is why the two censuses report 1024 and 138 672 on the same sequence: they are the logarithm and the cardinality of one set. By the Fliess–Schützenberger theorem a recognisable series has a minimal linear representation of dimension its Hankel rank, so every $\mathbb{F}_2$ -weighted automaton computing c from its base-2 digits has at least 1024 states — every other bound in this Part concerns deterministic machines.

It discriminates nothing, on three counts. Only *boundedness* of the rank characterises anything, so Remark 14.5 applies verbatim. The domination $R_f \leq A_f$ makes the rank strictly weaker than the antichain census, by a factor of 135 on our data. And the coin control is a theorem: a random series over $\mathbb{F}_2$ has expected rank at least $2^{n/2}$, and the split matrix at $n = 8, \dots, 18$ gives Rule 30 as 16, 31, 64, 127, 256, 510 against a coin's 15, 30, 63, 127, 255, 511 on the same prefix, saturating that ceiling and indistinguishable at every n .

That is worth stating as the general fact it is. Every instrument in this paper separates Rule 30 from Rule 90 without difficulty, and not one separates it from a fair coin. The obstruction has been described as the firewall rule satisfying our hypotheses; the sharper statement is that our sequence is indistinguishable from a coin to every quantity we can compute, and Rule 90 is the most convenient witness of that.

Remark 14.12 (k -regularity is not a weaker hypothesis, but one derived statistic does separate). It is natural to hope that k -regularity, where the k -kernel need only span a finitely generated module rather than be finite, is a weaker hypothesis that our data could refute outright. For c it is not weaker at all, and for two independent reasons: a finitely generated module over $\mathbb{F}_2$ is finite, and a k -regular sequence taking finitely many values is k -automatic [73, p. 441]. The same applies to the summatory function of c . So nothing is gained on the centre column itself.

For the *unbounded* statistics of the orbit the notions do differ, and there the module rank is a computable invariant that behaves differently on Rule 30 than on the additive rules — which is more than the census does. Certified lower bounds on the rank of the 2-kernel module are 512 for the row weight, 351 for the diagonal preperiods τ_d and 294 for the periods p_d , against 1 for Rule 90 and 2 for Rule 150. That this separates Rule 30 from the rules that satisfy more of our hypotheses than it does is the one feature no other instrument in this Part has. We record it as an instrument rather than a result: the relevant published measure is Shallit's rationality

measure, whose Theorem 9(b) gives $R_f \leq A_f$, so the rank census is strictly weaker than the antichain census and caps at $\sqrt{2N}$ against $N/\log_2 N$. And the decision procedures available — Krenn and Shallit’s via Jacob’s boundedness algorithm, and Krawczyk and Müllner’s [74] — all consume a presentation as input, which is exactly what these rank bounds say we do not have.

Proposition 14.13 (what the certificates exclude; PROVED from KNOWN bounds). *Write $\text{comp}_2(c)$ for the number of states of the minimal 2-automaton generating c , which equals $|\mathcal{K}(c)|$ [73, Cor. 4.1.9, Thm. 6.6.2]. Adamczewski and Yassawi [70] record $\text{comp}_q(a) \leq q^{d(2H+1)}$ for a series algebraic of degree d and height $H = \max_i \deg A_i$. Contrapositively, the certificate of Theorem 14.1(1) converts into an exclusion rather than a mere bound: for every $d \leq 12$, the series G is not algebraic of degree d with height at most $8000/2^{d+1}$. In particular G is not quadratic of height at most 1000, nor of degree 4 with height at most 250. In the other direction, run backwards against the census of Theorem 14.1(2), Bridy’s bound [75] $\text{comp}_q \leq (1 + o(1))q^{h+d+g-1}$ gives $h + d + g \geq 19$, the exact form of Rowland, Stipulanti and Yassawi removes the asymptotic caveat and gives $hd \geq 18$, and [70, eq. 5] gives $(h + 1)d \geq 17$, for the height, degree and genus of any algebraic presentation of G . Over $\mathbb{F}_2$ a Mahler relation is an Ore relation exactly, the Frobenius being trivial on coefficients.*

15 Decimation of the Duhamel class

The Duhamel formula expresses c_t as a linear functional of the initial row and of the quadratic source $q_x^s = u_x^s u_{x+1}^s$. In this section we treat that functional on an arbitrary pair (initial row, source field), show that the class of such expressions is closed under the Cartier operators, and give a closed form for the functional. The result is an exact description of the 2-kernel of c as the image of an orbit, and a reduction of the algebraicity question to a series built from the chiral part of the source alone.

15.1 The class $c[g, f]$

For a row $g: \mathbb{Z} \rightarrow \mathbb{F}_2$ and a source field $f = (f_x^s)_{s \geq 0, x \in \mathbb{Z}}$, both of finite support in x for each s , define

$$c[g, f]_t = \langle T_t, g \rangle + \sum_{s < t} \langle T_{t-1-s}, f^s \rangle, \quad \langle a, b \rangle = \sum_x a_x b_x \in \mathbb{F}_2.$$

Theorem 13.2 states that $c = c[\delta, q]$ with $\delta_x = [x = 0]$ and $q_x^s = u_x^s u_{x+1}^s$. Write $(\Lambda h)_x = h_{x-1} + h_x + h_{x+1}$ for the action of Λ on rows, and Eh for the even-site restriction $(Eh)_x = h_{2x}$.

Theorem 15.1 (exact decimation; PROVED). *For every (g, f) and every $t \geq 0$,*

$$c[g, f]_{2t} = c[D_0(g, f)]_t, \quad c[g, f]_{2t+1} = c[D_1(g, f)]_t,$$

where

$$\begin{aligned} D_0(g, f) &= \left(Eg, \left(E(f^{2\sigma+1} + \Lambda f^{2\sigma}) \right)_{\sigma \geq 0} \right), \\ D_1(g, f) &= \left(E(\Lambda g + f^0), \left(E(f^{2\sigma+2} + \Lambda f^{2\sigma+1}) \right)_{\sigma \geq 0} \right). \end{aligned}$$

Proof. By Lemma 13.1, $T_{2m}(z) = T_m(z^2)$ and $T_{2m+1}(z) = \Lambda(z) T_m(z^2)$. A pairing with $T_m(z^2)$ samples the even sites, $\langle T_m(z^2), h \rangle = \sum_x T_m(x) h_{2x} = \langle T_m, Eh \rangle$, and a pairing with $\Lambda(z) T_m(z^2)$ samples Λh on the even sites, using the symmetry of Λ : $\langle \Lambda T_m(z^2), h \rangle = \langle T_m, E\Lambda h \rangle$.

For $c[g, f]_{2t}$ the initial term is $\langle T_{2t}, g \rangle = \langle T_t, E g \rangle$. In the source sum split s by parity: $s = 2\sigma$ gives $T_{2t-1-2\sigma} = T_{2(t-1-\sigma)+1} = \Lambda T_{t-1-\sigma}(z^2)$, hence $\langle T_{t-1-\sigma}, E \Lambda f^{2\sigma} \rangle$; and $s = 2\sigma + 1$ gives $T_{2(t-1-\sigma)} = T_{t-1-\sigma}(z^2)$, hence $\langle T_{t-1-\sigma}, E f^{2\sigma+1} \rangle$. Collecting terms with the same σ gives $c[D_0(g, f)]_t$.

For $c[g, f]_{2t+1}$ the initial term is $\langle T_{2t+1}, g \rangle = \langle T_t, E \Lambda g \rangle$. The $s = 0$ source term is $\langle T_{2t}, f^0 \rangle = \langle T_t, E f^0 \rangle$ and joins the initial row. For $s \geq 1$ write $s = 2\sigma + 1$ ($T_{2(t-1-\sigma)+1}$, giving $E \Lambda f^{2\sigma+1}$) and $s = 2\sigma + 2$ ($T_{2(t-1-\sigma)}$, giving $E f^{2\sigma+2}$). Collecting gives $c[D_1(g, f)]_t$. $\square$

Corollary 15.2 (PROVED). *Let $L(g, f) = (c[g, f]_t)_{t \geq 0}$, a linear map into $\mathbb{F}_2^{\mathbb{N}}$, and let $\mathcal{O} = \{D_w(\delta, q) : w \in \{0, 1\}^*\}$ be the orbit of (δ, q) under the words in D_0, D_1 . Then*

$$\mathcal{K}(c) = L(\mathcal{O}).$$

In particular $\mathcal{K}(c)$ is finite if and only if $\mathcal{O}$ is finite modulo $\ker L$.

Proof. The kernel element $(c_{2^k m+r})_m$ is $\Lambda_{r_{k-1}} \cdots \Lambda_{r_0} G$ with $r = \sum r_i 2^i$, and Theorem 15.1 says $\Lambda_r L(g, f) = L(D_r(g, f))$. $\square$

15.2 Closed form of the functional

Theorem 15.3 (closed form; the resolvent is KNOWN, Litow–Dumas [76]). *Let $\alpha \in y \mathbb{F}_2[[y]]$ be the unique solution of*

$$y \alpha^2 + (1 + y) \alpha + y = 0, \quad \text{equivalently} \quad y = \frac{\alpha}{1 + \alpha + \alpha^2}.$$

Then for every (g, f) ,

$$\sum_{t \geq 0} c[g, f]_t y^t = \frac{1}{1 + y} \left(\sum_x g_x \alpha^{|x|} + y \sum_{s, x} f_x^s y^s \alpha^{|x|} \right).$$

Proof. Let $\hat{g}(z) = \sum_x g_x z^x$ and $\hat{f}^s(z) = \sum_x f_x^s z^x$. Summing the definition of $c[g, f]$ against y^t and using $T_m = \Lambda^m$,

$$\sum_t c[g, f]_t y^t = [z^0] \left(\sum_t y^t \Lambda^t \right) \left(\hat{g} + y \sum_s y^s \hat{f}^s \right) = [z^0] \theta \cdot \left(\hat{g} + y \sum_s y^s \hat{f}^s \right),$$

where $\theta = (1 + y \Lambda)^{-1} \in \mathbb{F}_2[z^{\pm 1}][[y]]$ is the unique solution of $y \theta_{x-1} + (1 + y) \theta_x + y \theta_{x+1} = [x = 0]$. The ansatz $\theta_x = C \alpha^{|x|}$ satisfies the equation for $x \neq 0$ because $C \alpha^{|x|-1} (y + (1 + y) \alpha + y \alpha^2) = 0$, and at $x = 0$ gives $C (2y \alpha + 1 + y) = C (1 + y)$ in characteristic 2, so $C = 1/(1 + y)$. Since $\alpha = y + O(y^2)$, $\alpha^{|x|} = O(y^{|x|})$ and the ansatz lies in the ring, hence is the solution. Pairing θ with $\hat{g} + y \sum_s y^s \hat{f}^s$ and taking $[z^0]$ gives the formula. $\square$

Remark 15.4 (provenance of Theorem 15.3). The resolvent is not new. Litow and Dumas [76] write the generating function of an additive cellular automaton as $M(x, z) = C_0(z)/(1 - xQ(z))$, which is $\theta = (1 + y \Lambda)^{-1}$ in the present notation, and evaluate its coefficients as a sum over the roots of the denominator; for $Q = z + 1 + z^{-1}$ that evaluation reduces in characteristic 2 to $\alpha^{|x|}/(1 + y)$, normalisation included. Their worked Motzkin example moreover exhibits the quadratic $y \alpha^2 + (1 + y) \alpha + y = 0$ and hence the element α itself, in print in 1993. What is added here is only the characteristic-2 proof by the ansatz together with the jump condition

at $x = 0$, and the extension of the functional from an initial row to a pair (initial row, source field), which is what makes Theorem 15.1 and Corollary 15.5 available. The decomposition of Rule 30 as Rule 150 plus the single term cr is likewise KNOWN and stated in Wolfram’s own prize announcement [68]: all of the complexity is attributed there to that one nonlinear term, “for without that term, one would have rule 150”. The reading of Corollary 15.5 as saying that the centre column is fed by the asymmetry of the rule is also in Chan-López and Martín-Ruiz [90].

For Rule 30 itself ($g = \delta$, $f = q$), rewriting in the variable α with $1 + y = (1 + \alpha)^2 / (1 + \alpha + \alpha^2)$ gives

$$G = \frac{1 + \alpha + \alpha^2}{(1 + \alpha)^2} + \frac{\alpha}{(1 + \alpha)^2} H, \quad H = \sum_{s,x} q_x^s \frac{\alpha^{s+|x|}}{(1 + \alpha + \alpha^2)^s}.$$

Since $\mathbb{F}_2(\alpha)/\mathbb{F}_2(y)$ is a finite extension, G is algebraic over $\mathbb{F}_2(y)$ if and only if H is algebraic over $\mathbb{F}_2(\alpha)$.

15.3 Only the chiral part of the source is visible

The weight $\alpha^{|x|}$ in Theorem 15.3 does not distinguish x from $-x$. Hence $c[g, f]$ depends on f only through the centre line f_0^s and the *chiral part*

$$\chi_x^s = f_x^s + f_{-x}^s, \quad x > 0,$$

and likewise on g only through g_0 and $g_x + g_{-x}$.

Corollary 15.5 (PROVED). $\ker L = \{(g, f) : \hat{g}(\alpha) + y \hat{f}(y, \alpha) = 0\}$ with $\hat{g}(\alpha) = \sum_x g_x \alpha^{|x|}$ and $\hat{f}(y, \alpha) = \sum_{s,x} f_x^s y^s \alpha^{|x|}$. In particular every mirror-symmetric pair ($g_x = g_{-x}$, $f_x^s = f_{-x}^s$) with $g_0 = 0$ and $f_0^s = 0$ for all s lies in $\ker L$.

Remark 15.6. The source of Rule 30, $q_x^s = u_x^s u_{x+1}^s$, is not mirror-symmetric: the rule is left-permutive and its orbit from the single cell is not symmetric, so q is not killed by $\ker L$. For a mirror-symmetric rule started from the single cell the field is symmetric, $\chi \equiv 0$, and the centre column depends on the source only through its own centre line f_0^s . This is the algebraic form of the observation that the centre column of Rule 30 is fed entirely by the asymmetry of the rule.

Remark 15.7 (the whole family of single-monomial perturbations; EXACT). Rule 30 and Rule 22 are not isolated cases. Rule 150 has algebraic normal form $l + c + r$, and adding one monomial gives exactly five rules; Corollary 15.5 classifies all of them, with no exceptions.

added monomial	none	1	lc	cr	lr	lcr
rule	150	105	86	30	54	22
symmetric in $l \leftrightarrow r$	yes	yes	no	no	yes	yes
orbit mirror-symmetric	yes	yes	no	no	yes	yes
chiral weight of the source	0	0	16 784	16 784	0	0
centre density to $t = 2^{13}$	1.0000	0.5000	0.5010	0.5010	0.5000	0.0002
centre eventual period	1	2	none	none	4	1

The unperturbed base case is the sharpest entry and was missing from earlier statements of this table: the centre column of Rule 150 itself is *identically* 1. From a single seed, row t is $(1 + z + z^2)^t$ over $\mathbb{F}_2$, so the centre bit is the central trinomial coefficient, which is always odd;

the corresponding statement for Rule 90 is that the centre bit is $\binom{t}{t/2} \bmod 2$, and $\binom{2m}{m} = 2\binom{2m-1}{m-1}$ is even for every $m \geq 1$, so that column is $1, 0, 0, \dots$. Both additive parents therefore have a constant centre column, and the whole contrast of this Part is between a constant sequence and one with no detected period, across a single monomial. The three symmetric perturbations give a mirror-symmetric orbit, hence a mirror-symmetric source, hence chiral weight exactly zero, and in each case the centre column is eventually periodic — with period 2, 4 and 1 respectively. The density row is a measurement at a stated depth and nothing more; in particular Rule 22's centre column contains exactly two ones and then vanishes, so its density is 0 rather than the small positive number a short run suggests, and the 0.5010 for Rules 30 and 86 is a finite-depth figure that drifts (it reads 0.5469 at $t = 2^6$ and 0.4810 at $t = 10^3$). The two chiral perturbations are Rule 30 and its mirror Rule 86, they share the same chiral weight as they must, and neither centre column has any period up to 64. So within this family, being hard is exactly the same as having a chiral perturbation, which is the content of Corollary 15.5 read as a classification rather than as a remark. Rule 54 is a second exactly solvable nonlinear control for the machinery of this Part, genuinely quadratic and with a period-4 centre column, and Rule 105 a third.

Remark 15.8 (the equivalence does not leave this family; EXACT). The hedge “within this family” in the previous remark is load-bearing, and the control is cheap enough that it should be stated rather than assumed. Taking all 128 elementary rules with a quiescent background ($000 \mapsto 0$) and running the centre column to $t = 2^{12}$, exactly three have no eventual period with preperiod ≤ 64 and period ≤ 1024 : Rules 30 and 86, and Rule 126. Restricting to the quiescent rules understates the set, and the full census is given in Remark 15.12; the conclusion drawn here is unaffected, because the counterexamples below are already inside the quiescent half. Ninety-four chiral rules are eventually periodic, and one mirror-symmetric rule is not, so “chiral” and “hard” are equivalent on the five rules of Remark 15.7 and on nothing larger. Two entries deserve naming. Rule 110 is chiral, Turing-universal [34], and its centre column from a single seed is constant 1: universality of the rule says nothing about the column. The same separation is what closes the rank route of Remark 23.9: communication rank measures the prediction function over all inputs and not the column, and Rules 22 and 54, whose single-seed columns are eventually 0 and of period 4, have rank over $\mathbb{F}_2$ of the same order as Rule 30's or larger at every t computed. And 110 of the 128 columns are constant within two steps, so the single-seed centre column is a very thin probe — it is not that most elementary rules are simple, but that most of them say nothing from this initial condition. What survives of the classification is therefore a statement about the Rule 150 neighbourhood, not a mechanism; the remark below says why that neighbourhood is nonetheless the right place to look.

Remark 15.9 (confinement: the period exponent is not a property of the rule, the surjectivity defect is; EXACT and MEASURED). Confine a rule to a ring of width N . Every rule then cycles, by pigeonhole, so periodicity under confinement discriminates nothing and the content can only be in how the period scales. We report this experiment together with its own refutation, because the refutation is the useful part.

Measuring the single-seed orbit for $N = 10..22$ over all 256 rules, each tagged by mirror symmetry *and* by affineness so the two are not confounded, the mean slope of $\log_2(\text{period})$ against N is 0.198 and 0.171 for the symmetric and chiral affine rules and 0.071 and 0.122 for the symmetric and chiral nonlinear ones: overlapping ranges, and a sign that reverses between the rows, so chirality does not act in a consistent direction. On that window the class $\{45, 75, 89, 101\}$ sits at 0.95, the class of Rule 30 at 0.45 — the two classical laws, a random permutation of 2^N states giving cycles $\sim 2^N$ and a random map $\sim 2^{N/2}$ — and everything else below 0.28. Those

exponents are stable under changing the initial state, with spread 0.07–0.22 against a gap of 0.51, and exactly 0 for the affine rules as a linear map requires.

They are nevertheless *not measurements of anything*, and seed-stability is what misled us: the instability is in N , not in the seed. Extending the same computation to $N = 36$ for Rule 30 and $N = 52$ for the rules whose periods stay small refutes the window fit outright. The $N \leq 22$ fit for Rule 30 under-predicts its period at $N = 25$ by a factor 46.6; the fit for Rule 45 over-predicts at $N = 26$ by a factor 977. The fitted exponent is a function of the window and not of the rule: for Rule 30 it reads 0.447, 0.718, 0.329, 0.587, 0.511, 0.059 and -0.463 on the windows 10..22, 10..26, 22..36, 10..36, 20..36, 26..36 and 30..36, a range of more than one unit that changes sign. Even the *ordering* reverses — Rule 30’s period exceeds Rule 45’s at $N = 12, 16, 24, 26$ and is smaller by a factor 745 at $N = 19$. The decisive figure is the raw series: Rule 30’s period runs 18 480 630 at $N = 35$, then 2 844 at $N = 36$, then past a cap of 6×10^7 at $N = 37$, so consecutive widths differ by more than four orders of magnitude. No exponent survives that. (The $N = 36$ value is exactly the $N = 18$ value because the orbit does fall into the spatially 18-periodic invariant subspace, at $t = 193\,613$; Rule 30’s ring map is not injective, so such a collapse is available, and the subspace is shift-invariant hence preserved. It is the *only* width in 12..40 where this happens, so it explains this one point and not the scatter: neither $N = 35$ nor $N = 37$ involves a collapse.) By contrast the rules whose periods stay small are stable out to $N = 52$: Rule 54 has period 4 at every width, exponent exactly 0 over 31 widths, and Rules 90, 150 and 126 sit at 0.17–0.18. The instability is specific to the rules the statistic appeared to single out. We therefore withdraw the exponent and record the route as closed: the ring-period scaling of the single-seed orbit is not a property of the rule at any width we can reach.

What survives is the quantity the exponents pointed at, and it survives because it involves no fitting. Writing $\delta_N = 1 - |\text{im}|/2^N$ for the surjectivity defect of the width- N ring map: Rule 45 has $\delta_N = 2^{-N/2}$ at every even N and $\delta_N = 0$ at every odd N — it is surjective on every odd ring — PROVED in Proposition 15.10; Rule 30’s defect decays with the exponent $\log_2 \lambda - 1 = -0.2382$, $\lambda = 1.6956$ the real root of $x^3 - x^2 - 2$, an algebraic constant and not a fit (the same proposition; the window fits -0.2366 , -0.2376 and -0.2382 on $N = 8..20$, $8..28$ and $20..28$ are stable, unlike the period exponent above); Rule 110’s image collapses ($+0.061$, MEASURED on $N = 8, 10, \dots, 20$); Rule 90 is exactly four-to-one at every even width and two-to-one at every odd one; and Rule 150 is four-to-one exactly at the multiples of three and bijective otherwise, since $1 + z + z^2 \mid z^N - 1$ iff $3 \mid N$. Rules 45, 30 and 110 are all chiral while spanning that whole range.

The sharp form of the conclusion is an identity in the algebraic normal form, and it needs none of the retracted material. Rule 30 = 150 + cr and Rule 45 = 90 + 1 + cr carry the *same* chiral monomial over different additive parents, and differ from each other by the single term $1 + c$; their defects differ not merely in exponent but in kind: Rule 45’s is a parity phenomenon in N , exactly $2^{-N/2}$ at even N and absent at odd N , while Rule 30’s is a genuine algebraic decay at every N with no parity structure, its Garden-of-Eden counts 1, 1, 3, 5, 6, 12, 22, 33, 57, 101, 166, 280 for $N = 1, \dots, 12$ containing no power of two past $N = 2$. So the chiral monomial does not by itself determine the behaviour even across the two additive parents this Part is built on. This leaves Corollary 15.5 untouched, which is a proved statement about the chiral part of the *source* being the only part visible at the centre; what it constrains is the reading of Remark 15.7 as a mechanism, which is now hedged for a second independent reason.

Proposition 15.10 (ring surjectivity defects from the transfer monoid; PROVED). *For an el-*

elementary rule f let A_b , $b \in \{0,1\}$, be the Boolean 4×4 matrix on pairs of adjacent cells with $(A_b)_{(p,q),(q,r)} = [f(p,q,r) = b]$. A configuration y on the ring $\mathbb{Z}/N$ has a preimage iff the Boolean product $A_{y_1} \cdots A_{y_N}$ has a nonzero diagonal entry, so $|\text{im}_N|$ is the number of length- N words accepted by the automaton whose states are the finite monoid generated by A_0, A_1 , and $|\text{im}_N|$ satisfies a linear recurrence of order at most the size of that monoid, which is 8, 12, 21, 38, 41, 75 and 2375 for Rules 90, 150, 45, 18, 30, 110 and 22. Consequently:

- (a) Rule 45 has exactly $2^{N/2}$ Garden-of-Eden configurations on every even ring and none on any odd ring: it is surjective on every odd ring.
- (b) Rule 150 is four-to-one when $3 \mid N$ and bijective otherwise; Rule 90 is four-to-one at even N and two-to-one at odd N .
- (c) Rule 30's Garden-of-Eden count $s_N = 2^N - |\text{im}_N|$ satisfies the recurrence of $Q(x) = (x-1)^7 x^{11} (x^2+x+1)^2 (x^3-x^2-2)^5$, the characteristic polynomial of its automaton with the factor $(x-2)^4$ removed, and $\lim_N N^{-1} \log_2 s_N = \log_2 \lambda$ with $\lambda = 1.695620\dots$ the real root of $x^3 - x^2 - 2$; its defect exponent is therefore $\log_2 \lambda - 1 = -0.238186$, an algebraic constant and not a fit.

Proof. The automaton is the transfer-matrix description of preimages on a ring, and the recurrence is Cayley–Hamilton for its transition matrix M , since $|\text{im}_N| = uM^Nv$ with u the identity state and v the accepting states. (a) $2^N - |\text{im}_N|$ satisfies a recurrence of order ≤ 22 and $2^{N/2}[2 \mid N]$ one of order 2, that of $x^2 - 2$; the two agree for $N = 1, \dots, 60$ (`s3_ring_defect.py`, the automaton counts cross-checked against brute force for $N \leq 14$), which exceeds the 24 consecutive values that force agreement at every later N . (b) is linear algebra over $\mathbb{F}_2$: the kernel of $x \mapsto x_{i-1} + x_i + x_{i+1}$ on $\mathbb{Z}/N$ has dimension $\deg \gcd(1 + z + z^2, z^N - 1)$, which is 2 iff $3 \mid N$, and that of $x \mapsto x_{i-1} + x_{i+1}$ has dimension $\deg \gcd(1 + z^2, z^N - 1)$, which is 2 or 1 by the parity of N . (c) Let R be the characteristic polynomial of the 41×41 matrix M , computed exactly, and $R = (x-2)^4 Q$ with $Q(2) \neq 0$. The sequence $Q(\sigma)s$, σ the shift, lies in the four-dimensional solution space of $(x-2)^4$ and vanishes at $N = 0, \dots, 43$ (`s3_ring_defect2.py`), hence identically; so the eigenvalue-2 component of s is zero and s satisfies Q . The roots of Q have moduli 0, 1, λ and $(2/\lambda)^{1/2} < \lambda$, and s does not satisfy $Q/(x^3 - x^2 - 2)^5$, so the λ -component is nonzero and $s_N = p(N)\lambda^N(1 + o(1))$ for a nonzero polynomial p . Numerically $s_{N+1}/s_N = 1.69562077$ from $N = 41$ on, so p is constant to the precision available. $\square$

Remark 15.11 (the provable case in this family is the symmetric one; EXACT). Rule 126 is $150 + lc + cr + lr$, the *full symmetric* quadratic, and it is mirror-symmetric. Its centre column from a single seed is 1 exactly at $t = 2^k - 1$, verified for every $t \leq 2^{14}$, so it is the Fredholm–Rueppel sequence [94] as far as computed ($t \leq 2^{14}$, EXACT); if the identity holds for all t — the single-seed pattern of Rule 126 is Rule 90's with each cell thickened inward, which we have checked but not written out — then the column is not eventually periodic, since the gaps double. The inversion is worth stating plainly. Within the single-monomial neighbourhood of Rule 150, the one case whose non-periodicity admits a proof is the symmetric one, and the chiral perturbation is precisely where the proof is unavailable. Nothing here suggests that the chiral monomial *causes* non-periodicity; what the family shows is that it is the perturbation which removes every handle we have, which is a statement about our methods and not about the rule.

Remark 15.12 (the full aperiodic census, and the exact necessary condition; EXACT). Running all 256 rules rather than the quiescent half — for a non-quiescent rule the background blinks, but the centre column is still a well-defined sequence and periodicity is still the right question — exactly fifteen have no eventual period at preperiod ≤ 64 , period ≤ 1024 and $t \leq 2^{12}$, and two

of them are artefacts of the preperiod bound: Rules 137 and 193, a mirror pair, are eventually periodic with period 240 and preperiod 2851 (EXACT to $t = 20\,000$). The other thirteen show no eventual period at preperiod ≤ 4000 , period ≤ 1024 and $t < 8000$:

30, 45, 75, 86, 89, 101, 126, 129, 135, 149, 161, 167, 181.

The set is consistent with the one symmetry that preserves the experiment: the mirror fixes a symmetric seed, and every one of the thirteen has its mirror in the set, giving eight mirror classes. The complement does *not* preserve it, since it sends a single 1 on a 0 background to a single 0 on a 1 background, and indeed three of the thirteen (161, 167 and 181) have a complement outside the set. Six mirror classes are complement-closed as well, and they group into three families: Rule 30's {30, 86, 135, 149}, Rule 45's {45, 75, 89, 101}, and Rule 126's {126, 129}. Those three are the rules whose centre column is aperiodic from a single seed on either background.

The census yields one sharp necessary condition, and it is a condition on a single pair of monomials. *Every* one of the thirteen has cr or lc in its algebraic normal form — a quadratic term coupling the centre to one neighbour — and no affine rule appears at all. The condition is specific to that pair rather than to nonlinearity: of the 32 rules carrying lr but neither cr nor lc , none is aperiodic, and of the 32 carrying lcr but neither, none is aperiodic. So a quadratic gate on the two *outer* cells, or a cubic on all three, never suffices; the coupling must reach the centre. This is the precise form of the informal statement that the chiral monomial is what breaks periodicity, and it also explains why Rule 54 ($150 + lr$) and Rule 22 ($150 + lcr$) are the tame members of Remark 15.7 while 30 and 86 are not.

It is necessary and far from sufficient: 192 rules carry cr or lc and only thirteen are aperiodic, and no Boolean expression of at most three literals over the monomial indicators, the light-cone growth flags and the permutivity flags — conjunctive or disjunctive — selects the thirteen exactly. One tempting half of the informal statement is definitely false: light-cone growth is not necessary, since Rules 45, 101, 129 and 161 send both 001 and 100 to 0. What the census supplies is therefore a clean necessary condition and no sufficient one, which is the honest position: the gate is required, and what selects thirteen rules out of the 192 that have it is not visible at the level of the normal form.

Remark 15.13 (what the literature has solved in this family, and what it has not). The classification of Remark 15.7 lines up exactly with the state of the published art, and the alignment is not a coincidence. Of the five rules, the three symmetric ones have all been solved or rigorously treated: Rule 105 is affine and falls to [64]; Rule 54 is exactly solvable in its reversible two-step form, with a complete quasiparticle catalogue [59]; and Rule 22 has a rigorous finite-seed theory [29] as well as the closed form of [90] that we verified in Remark 15.17. Neither chiral one has been solved, and they are Rule 30 and its mirror. The same split appears one rung down, on Rule 90: the *symmetrised* chiral pair $90 + lc + cr$ is Rule 18, the rule whose defect is the one nonlinear elementary defect with a proved statistical theorem [52], and $90 + lc + cr + lcr$ is Rule 146. So the literature has solved every symmetric member of both families and the symmetrised chiral pair of one of them, and has nothing on a single chiral monomial. The structural reason is visible in the list of techniques: each of them uses a reflection symmetry, a sublattice or parity invariant, or a conserved particle number, and one chiral monomial breaks all three at once.

There is a historical remark that belongs here, because the record turns out to agree with the classification. Rule 30's first appearance in print is, as far as we can tell, a single unlabelled panel in Fig. 7 of [86, p. 608] — we checked the panel against the true orbit — and it is neither

named in the caption nor discussed nor measured anywhere in that paper. The reason is stated on its p. 603: the analysis is restricted to the 32 rules called “legal”, and Rule 30 is excluded because it is not symmetric under reflection, $110 \mapsto 0$ against $011 \mapsto 1$. The body then records the expectation that the excluded rules “introduce no qualitatively new features”. So the exact property that removed Rule 30 from the first systematic survey of these automata is the property that this remark identifies as the discriminator between the tractable members of its family and the two intractable ones.

The classification by reflection symmetry itself is published, and we should use its vocabulary rather than invent one. Wuensche and Lesser’s atlas sorts the elementary rules as *symmetric*, *semi-symmetric* and *full-asymmetric* under reflection, tabulated in [43, §4.3], and the lists agree with the table above at every entry: 150, 105, 54, 22 and 90 are symmetric, and Rule 30 is semi-symmetric. The term of art is therefore *asymmetric*, not *chiral*, and we keep the word *chiral* only for the algebraic object of Corollary 15.5, the mirror-antisymmetric part of the source. There is also independent corroboration from a different direction: Wuensche’s four *chain rules*, reached through the Z -parameter and not through any symmetry consideration, are 30, 45, 106 and 154, every one semi-symmetric and not one symmetric. What is ours here is therefore narrower than the observation that asymmetry matters: it is the single-monomial decomposition of the family, the exact cancellation of Corollary 15.5, and the correlation with what the literature has managed to solve.

There is a second reason Rule 18 is the wrong precedent for Rule 30, and it is quantitative. The technique behind every rigorous Rule 18 result is that the nonlinear rule coincides with a linear one on an invariant set — the sublattice or invariant-set linearization of Grassberger, Jen and, in its modern form, Rupe and Crutchfield [53]. For Rule 30 that set is as small as it can be.

Proposition 15.14 (no nontrivial additive subdynamics; EXACT). *Since $30 = 150 + cr$, Rule 30 and Rule 150 agree on a configuration exactly when it contains no factor 11, so the set on which the two rules agree for all time is the largest Rule-30-invariant subshift inside the golden-mean shift. That set is*

$$\{0^{\mathbb{Z}}, (01)^{\mathbb{Z}}, (10)^{\mathbb{Z}}\},$$

three points, of zero entropy. Moreover the ordered background actually present in the seed’s orbit is not one of them.

Computation. The argument is a window exhaustion, with a cyclic sweep as corroboration. Note first that a cyclic sweep alone would not suffice: the set in question is closed and invariant, and a nonempty subshift need not contain a periodic point, so excluding periodic points leaves Sturmian possibilities open. What is needed is a bound on the language.

Take all 2^{23} words of length 23 and iterate, at each step testing only the determined region, which loses one cell at each end per step, for the factor 11. The first test alone, at $t = 0$, leaves the 11-free words, $75\,025 = F_{25}$ of them; the second, at $t = 1$, leaves six; and those same six survive all eleven steps. They are 0^{23} , the two checkerboards, and the three words whose only ones sit at positions 0 and 22, the two cells the determined region never covers at any positive time. In particular every length-21 factor of a configuration in the set is 0^{21} or a checkerboard, which forces the set to be contained in the three configurations listed; and each of the three is indeed fixed by Rule 30 and 11-free, the checkerboards because every cell there has exactly one neighbouring one. Hence equality.

The cyclic sweep agrees and extends the range: for every period $p \leq 29$, taking all 2^p configurations, keeping the 11-free ones (1149851 at $p = 29$) and iterating 400 steps while discarding any configuration whose image contains 11, leaves exactly one word for odd p and three for even $p = 0^p$ and, when $2 \mid p$, the two checkerboards — and nothing else at any period. For the last clause of the statement, the 40 cells at the light-cone edge are the same word 110010001110011110001111010001111101000 at $t = 200$ and at $t = 400$, and it contains 11 nine times, so the quadratic term fires throughout the ordered region. $\square$

Remark 15.15 (the contrast with Rule 90, and why the precedents are on the other side). The reason the linearization industry has been productive on Rule 90 and not on Rule 150 is that on Rule 90 the perturbation is switched off. From the single seed, Rule 90's orbit is supported on $x \equiv t \pmod{2}$, so no two adjacent ones ever occur and no monomial in $\{lc, cr, lcr\}$ ever fires: rules 18, 26, 82, 146, 154, 210 and 218 are therefore *identical* to Rule 90 from a seed, which we verified over 40 steps, with zero adjacent-one pairs in those rows. On the Rule 150 side the count of adjacent-one pairs in the same 40 rows is 208, and rules 30, 86, 54 and 22 all diverge from Rule 150 at $t = 2$, Rule 150's first row being 111, which fires every monomial at once. So the whole class of results that make a nonlinear rule tractable by exhibiting it as a linear rule on an invariant set lives on the side where the perturbation is off, and Proposition 15.14 says Rule 30 is not merely on the other side but as far from that side as the golden-mean shift allows. This raises rather than lowers the standing of Theorem 15.3: it is not a restriction of the dynamics to where the rule is linear, it is a genuine variation-of-constants formula with a live source.

Remark 15.16 (the literature slot this Part occupies; KNOWN framing). The decomposition used throughout this Part has a name in the cellular-automaton literature and an explicit open problem attached to it. Fukś [54, 55] sets up exactly our situation: write the local rule as $f = g + b$ with g a rule whose iterate is known in closed form and b a perturbation, expect a solution of the form $[F^n(x)]_j = [G^n(x)]_j + P(x, n, j)$ by analogy with variation of constants, and then states flatly that “we do not know any general method for finding $P(x, n, j)$ ” [54, §2]. He carries the programme out for $156 = 140 + b$ with b supported on a single word, and his appendix lists solution formulae for more than sixty elementary rules — but always with a tame g . Our Theorem 15.3 and Corollary 15.5 are an identification of that P for $g = \text{Rule 150}$, which is chaotic, and the honest statement of novelty is therefore narrow: the decomposition and the question are KNOWN and named, and what is ours is the closed form for a chaotic g , together with the chirality obstruction that says which part of P is visible at the centre. His verification technique — guess the perturbation term, write it as a density polynomial, induct on n — is also the natural template for checking any refinement of ours.

Where the topological classification actually places this rule is settled in print, and it is worth recording because it also ends the enquiry. Schüle and Stoop give the complete Kůrka table for all 88 minimal elementary rules [36]: the positively expansive class is exactly $\{90, 105, 150\}$, and Rule 30 sits one rung below, sensitive but not positively expansive. Their Proposition 13 is the elementary-rule form of the equivalence between positive expansivity and bipermutivity, and their Proposition 3 notes that blocking words of length at most 4 suffice for elementary rules — the finite search that the undecidability of the classification in general forbids in the large.

Two consequences. First, no rule is excluded from solvability *by* its topological class. The criterion sometimes attached to that hierarchy is sufficiency and not necessity: Fukś writes that the initial value problem is solvable for almost all rules possessing some equicontinuity property, while noting that simpler methods usually suffice in those cases [54, §2]. Rule 90 is positively expansive and solvable in closed form, so it is out of scope there rather than a counterexample.

What Fuk s does name as the obstruction is defects performing a pseudo-random walk, and that does not reach us either, since the mechanism assumes a constant defect population while ours grows by Theorem 17.8.

Second, and this is why we stop here: across that whole table Rule 30 and Rule 90 agree on sensitivity, surjectivity, transitivity and Devaney chaos, and differ on exactly one label, positive expansivity. The hierarchy carries one bit of information about the distinction we care about, and that bit is bipermutivity, which we already use throughout. There is one place where the separation runs our way instead: an affine rule cannot be globally universal, with Rule 90 named, while Rule 30’s status is open.

Remark 15.17 (a worked nonlinear control: Rule 22). Rule 22 has algebraic normal form $a + b + c + abc$ and Rule 30 has $a + b + c + bc$, so both are Rule 150 plus a single monomial: the symmetric cubic in one case, the chiral quadratic in the other. The Duhamel identity therefore applies verbatim to Rule 22 with the cubic source $q_x^s = u_{x-1}^s u_x^s u_{x+1}^s$, and Rule 22 is exactly solvable from the single seed: Chan-L pez and Mart n-Ruiz [90] give the closed form $Q_{2j}(x) = (x^2 + x^{-2})^j$, $Q_{2j+1}(x) = (x^{-1} + 1 + x)Q_{2j}(x)$ for the row polynomials, which we have verified against simulation for every $t < 129$, together with the support count $2^{s_2(j)}$ on even rows, s_2 the binary digit sum. Since the constant term of $(x^2 + x^{-2})^j$ is $\binom{j}{j/2} \bmod 2$, which vanishes for $j \geq 1$, the centre column of Rule 22 is 1, 1, 0, 0,

This is a complete test of the machinery above, and it isolates what makes Rule 30 hard. EXACT: the Duhamel sum reproduces the Rule 22 centre column for all $t < 200$; the cubic source is supported exactly on the odd rows and carries substantial weight, 1251 ones for $t < 200$, yet contributes nothing beyond $t = 1$. Corollary 15.5 says why. Rule 22 is symmetric, so its single-seed orbit and hence its source are mirror-symmetric, and the chiral part χ vanishes identically (EXACT: chiral weight 0 for $t < 200$, against 7666 for Rule 30 over the same range). By Corollary 15.5 the centre column then depends on the source only through its own centre line, which here is the single bit $f_0^s = [s = 1]$, and Theorem 15.3 gives the answer in one line:

$$\sum_t c_t y^t = \frac{1}{1+y} (\hat{g}(\alpha) + y \hat{f}(y, \alpha)) = \frac{1+y^2}{1+y} = 1+y,$$

so $c_0 = c_1 = 1$ and $c_t = 0$ thereafter, which is the closed form’s consequence recovered without the closed form. The contrast is the content: a nonlinear rule with a source of weight 1251 can be completely invisible to its own centre column, and what makes Rule 30’s centre column resistant is not the size of its source but the 7666 units of *chiral* weight the mirror-symmetric kernel can see. A symmetric rule is therefore the right reference object for Rule 30 precisely because it is the $\chi \equiv 0$ case of Corollary 15.5.

Remark 15.18 (verification). Theorem 15.1 was checked for the seed and source q for all $t < 60$ in both parities, and Theorem 15.3 reproduces c_t for all $t < 160$, with α computed to order 160 as the fixed point of $\alpha = y(1 + \alpha + \alpha^2)$ (independent code, 2026-09-16).

15.4 What remains

The question “does the quadratic source decimate?” has two readings. Whether the orbit $\mathcal{O}$ itself is finite is the weak one: $\ker L$ is large, and infinitely many distinct sources can give the same kernel element. Whether $\mathcal{O}$ is finite modulo $\ker L$ is, by Corollary 15.2, exactly the finiteness of $\mathcal{K}(c)$. Equivalently, by Theorem 15.3 and Corollary 15.5, G is algebraic over $\mathbb{F}_2(y)$ if and only if the chiral series H is algebraic over $\mathbb{F}_2(\alpha)$. The certified bounds of Section 14 are

consistent with $\mathcal{K}(c)$ being infinite but cannot establish it. The action of the Cartier operators on the chiral data is not what is missing: Λ and E commute with the mirror $x \mapsto -x$ and preserve the rows with $h_x = h_{-x}$ and $h_0 = 0$, which are exactly the rows with zero centre and zero chiral part, so D_0 and D_1 descend to the half-line data $(g_0, \chi_g; f_0^s, \chi^s)$ (PROVED), and Section 16 writes the descended maps out and checks them (Proposition 16.1). What remains is to exhibit an invariant of that recursion that cannot take only finitely many values. We do not claim such an invariant here.

Remark 15.19 (the control the growth measurements were missing; EXACT). Every other claim in this paper is checked against Rule 90, and the interface language was not. It should have been, because the comparison is the sharpest separation we have between our rule and the linear ones, and it needs no fitting.

For Rule 90 and Rule 150 the interface language is the *full* free monoid: $|L_k| = 2^k$ exactly, verified to $k = 40$ by two independent methods, so the entropy is exactly one bit per zero-row. For Rule 30 it is 2, 3, 5, 8, 12, 17, 25, 36, $\dots$, reaching $|L_{60}| = 103\,220$. Two estimators of the entropy have to be kept apart here, by the same standing rule that corrected the Rule 150 growth exponent: the block estimate $k^{-1} \log_2 |L_k|$ reads 0.2776 at $k = 60$, but it converges from above and logarithmically slowly, whereas the local slope $\log_2(|L_{k+1}|/|L_k|)$ reads 0.1295 at $k = 60$ (0.1535 at $k = 46$), a factor 2.1 apart on the same data. Both fail, and the local slope is the one to quote. Neither carries the separation, which needs no estimator at all: Rule 90's entropy is exactly one bit, clearing both by more than a factor of three. The separation is visible without any asymptotics at all: for a linear rule $|L_k|$ is a power of two at every depth, and Rule 30's is a power of two at only two of the twenty depths we list, with $|L_{60}| = 103\,220$ lying strictly between 2^{16} and 2^{17} . That is an integrality obstruction at essentially every depth, and it is exact.

The corollary is worth stating because it inverts the natural reading. Rule 90's fibre algebra *is* the free algebra on two generators; Rule 30's contains no free subalgebra at all. The rule with the intermediate growth is the constrained one, not the rich one — our language is smaller than the linear rules', not larger, and the difficulty of Rule 30 does not show up here as extra freedom.

16 The chiral recursion in practice

Proposition 16.1 (EXACT). *Let the state be $(g_0, \chi_g; f_0^s, \chi^s)$ with $\chi^s(x) = f_x^s + f_{-x}^s$, $x \geq 1$. The maps D_0, D_1 of Theorem 15.1 act on this state by $\chi'^\sigma = E(\chi^{2\sigma+1} + \Lambda\chi^{2\sigma})$, $f_0'^\sigma = f_0^{2\sigma+1} + f_0^{2\sigma} + \chi^{2\sigma}(1)$ for D_0 , and the shifted analogues for D_1 (with g' absorbing f^0), where $(\Lambda\chi)(x) = \chi(x-1) + \chi(x) + \chi(x+1)$ with $\chi(0) = 0$ and $(E\chi)(x) = \chi(2x)$. Evaluating Theorem 15.3 on this state reproduces c_t for $t < 64$ and every kernel element $(c_{r+j2^k})_j$ for all 126 words of length ≤ 6 (script `s3_chiral.py`). For depths 1 to 9 the orbit is injective on the chiral windows (rows < 16 , $|x| \leq 16$): all 2^k words give distinct windows (depth 10 exhausts the 16-row window and is uninformative). For every word to depth 8 the chiral data has a nonzero entry within its first four rows.*

Remark 16.2 (what this rules out). The zero-run criterion is the elementary observation that if for every M there are k and $r < 2^k$ with $c_{r+j2^k} = 0$ for all $j < M$ and $c_{r+j2^k} = 1$ for some $j \geq M$, then these progressions are nonzero kernel elements of unbounded y -adic valuation, so $\mathcal{K}(c)$ is infinite and c is neither 2-automatic nor eventually periodic (PROVED; the clause $j \geq M$

is needed, since an identically zero progression is the zero kernel element). The leading zero run of a kernel element (its y -adic valuation) is the only invariant of the orbit that is visible in the data, and it behaves like a coin across residues: over the 4096 residues $r < 2^{12}$ at level $k = 12$, read on the first 2^{20} terms of c , $\Pr[\text{run} \geq m]$ matches 2^{-m} to sampling accuracy for every $m \leq 12$ (MEASURED), so the criterion’s hypothesis holds at the coin’s rate as far as it can be tested. Since the chiral data never vanishes near the top, long zero runs arise from cancellation inside the α -weighted sum of nonzero data, not from vanishing data. Any invariant that proves the kernel infinite must therefore be a class function of the series G_w itself, not a support or valuation of the chiral arrays. None is known. OPEN.

Proposition 16.3 (no small linear lift; EXACT). *There is no radius-one linear cellular automaton over $\mathbb{Z}/4$, $\mathbb{Z}/8$, $\mathbb{F}_4$, $\mathbb{F}_8$ or $\mathbb{F}_2^2$ (with 2×2 matrix coefficients), no cellwise recolouring to $\{0, 1\}$ and no lift of the seed, that reproduces the Rule 30 space-time diagram for 40 rows (`s3_linearlift.py`; 19,840 lifts). KNOWN obstruction: a linear automaton over a finite abelian group from a finite seed has an automatic space-time diagram [77], and a cellwise coding preserves automaticity, so any such lift would make c automatic, with an automaton of at least the 138 672 states forced by the census above. The implication is in fact an equivalence: over $\mathbb{F}_q$ in characteristic p , a sequence is p -automatic if and only if it is a column of a linear cellular automaton with memory from eventually periodic initial conditions (Rowland–Yassawi [93], KNOWN), so the lift search is a search for automaticity in other coordinates. Nesme [72] extends [77] to commutative monoids and, more to the point here, to k -automatic initial configurations, so the conclusion strengthens from “no lift of the seed” to “no k -automatic lift of the seed”.*

Part III

The front of the diagonal defects

Source: `front_section.tex` (referee dialogue, 2026-09-16/17), labels prefixed `fr`;; Section 18 writes out the unbounded-periods theorem in the diagonal frame; the theorem itself is KNOWN (Gravner–Griffeath [67], Proposition 3.3), as recorded in the ledger on 2026-09-17.

17 The front of the diagonal defects

Throughout, arithmetic is in $\mathbb{F}_2$, u_x^t is the Rule 30 evolution

$$u_x^{t+1} = u_{x-1}^t + u_x^t + u_{x+1}^t + u_x^t u_{x+1}^t, \quad u_x^0 = [x = 0],$$

and $c_t = u_0^t$ is the centre column. For a finitely supported $f: \mathbb{N} \rightarrow \mathbb{F}_2$ put $\deg f = \max\{t : f(t) = 1\}$ and $\deg 0 = -1$. For $a \in \mathbb{Z}$, $(a)_+ = \max(a, 0)$.

17.1 Diagonals, background and defect

Definition 17.1 (Diagonals). For $d \in \mathbb{Z}$ and $t \geq 0$ let $D_d(t) = u_{d-t}^t$. Thus D_0 is the left edge of the light cone, and D_d passes through the centre cell $x = 0$ at time $t = d$, so $D_d(d) = c_d$.

Lemma 17.2 (PROVED). (a) $D_d(t+1) = D_{d-2}(t) + D_{d-1}(t) + D_d(t) + D_{d-1}(t)D_d(t) = D_{d-2}(t) + (D_{d-1}(t) \vee D_d(t))$.

(b) (Light cone) $D_d(t) = 0$ whenever $d > 2t$; $D_d \equiv 0$ for $d < 0$; $D_d(0) = [d = 0]$.

(c) $D_0 \equiv 1$ and $D_1(t) = [t \geq 1]$.

Proof. (a) Put $x = d - (t + 1)$ in the rule; the neighbours $x - 1, x, x + 1$ are the cells $(d - 2) - t, (d - 1) - t, d - t$. (b) $u_x^t = 0$ for $|x| > t$ by induction on t , and $d - t > t \iff d > 2t$; $d < 0$ gives $d - t < -t$. (c) From (a) and (b): $D_0(t + 1) = D_0(t)$, $D_0(0) = 1$; $D_1(t + 1) = 1 + D_1(t) + D_1(t) = 1$. $\square$

Proposition 17.3 (Eventual periodicity and reset formula; PROVED). *Let $d \geq 0$.*

(a) (Reset formula) *If $s < t$, $D_d(s) = 1$ and $D_d(u) = 0$ for $s < u < t$, then $D_{d+1}(t) = 1 + \sum_{u=s}^{t-1} D_{d-1}(u)$. If $D_d(u) = 0$ for all $u < t$, then $D_{d+1}(t) = D_{d+1}(0) + \sum_{u<t} D_{d-1}(u)$.*

(b) *Every D_d is eventually periodic, with minimal eventual period p_d a power of 2 (eventual periodicity of the diagonals, with period a power of 2, is stated by Rowland [66, §5] as a consequence of Jen [12, Thm. 4], which we have not been able to verify at source, so the attribution is second-hand; the proof below does not use it, and Jen's 1990 paper does not treat diagonals). Let $\tau_d = \min\{t \geq 0 : D_d(u + p_d) = D_d(u) \ \forall u \geq t\}$, $T = \max(\tau_{d-1}, \tau_d)$, $q = \max(p_{d-1}, p_d)$.*

(c) *If D_d is not eventually 0, then $p_{d+1} \mid q$ and $\tau_{d+1} \leq T + p_d$.*

(d) *If D_d is eventually 0, put $w = \sum_{u=T}^{T+p_{d-1}-1} D_{d-1}(u)$. Then $p_{d+1} \mid p_{d-1}$ if $w = 0$, $p_{d+1} = 2p_{d-1}$ if $w = 1$, and $\tau_{d+1} \leq T$.*

Proof. (a) By Lemma 17.2(a), $D_{d+1}(u + 1) = D_{d-1}(u) + 1$ if $D_d(u) = 1$ and $D_{d+1}(u + 1) = D_{d-1}(u) + D_{d+1}(u)$ if $D_d(u) = 0$; iterate from $u = s$. (b)–(d) by induction on d , starting from Lemma 17.2(c) ($p_0 = p_1 = 1$). (c) Let $s_0 = \min\{s \geq T : D_d(s) = 1\} \leq T + p_d - 1$. For $t \geq s_0 + 1$ let $s(t)$ be the last 1 of D_d before t ; then $s(t) \geq T$, $s(t + q) = s(t) + q$ and, as $p_{d-1} \mid q$, $\sum_{u=s(t)+q}^{t+q-1} D_{d-1}(u) = \sum_{u=s(t)}^{t-1} D_{d-1}(u)$. By (a), $D_{d+1}(t + q) = D_{d+1}(t)$ for $t \geq s_0 + 1$. (d) For $t \geq T$, $D_{d+1}(t + 1) = D_{d+1}(t) + D_{d-1}(t)$, so $D_{d+1}(t + p_{d-1}) = D_{d+1}(t) + w$; if $w = 1$ the minimal period divides $2p_{d-1}$ and not p_{d-1} , and is a power of 2, hence equals $2p_{d-1}$. $\square$

Remark 17.4 (status of the reset formula, and of the transient). We have not found the reset formula, or any statement eliminating the recursion in the diagonal index, in print; Brunnbauer, who computes this system furthest, says of his own recursion that “the recursion over n cannot be easily removed”. In the frozen case ours collapses to the parity criterion for period doubling of Wolfram, Rowland and Gravner–Griffeath, so it is a strict generalisation of a published statement.

One structural fact does have a published explanation: the periods are powers of two because the orbit closure is an odometer. Coven, Pivato and Yassawi prove that a left-permutive rule with no memory and local rule $t_0 + \psi(t_1, \dots, t_r)$ over $\mathbb{Z}/p$, started from a right-fixed tail with infinite orbit, has orbit closure topologically conjugate to the p -adic odometer with every scale factor p [21, Thm. 4]. Rule 30 as $\sigma \circ F$ satisfies every hypothesis, and the infinite-orbit condition is Theorem 18.3; instantiating their scale data returns our own doubling positions. It applies verbatim to Rule 90, so it explains a shared phenomenon rather than separating the rules, and *where* the doublings occur is not explained by it — their index is defined as the doubling position, with no bound on it anywhere.

Of the *transient* there is no theory at all. [67] contains no occurrence of “transient” or “preperiod” in its body, Rowland fixes a common starting time and bounds only the period, and the preperiods exist as data in OEIS A363346. The one paper whose title promises otherwise, Jen’s on transience and dislocations, maps certain nonlinear rules onto linear template systems and bounds collision times within that correspondence, which is valid only while the *number* of dislocations stays constant; it fails for us twice, since Proposition 15.14 denies any nontrivial set on which Rule 30 agrees with a linear rule, and Theorem 17.10 makes our defect grow. It is therefore the third instance of one mechanism, with the Rule 18 linearisation and the Rule 22 defect calculus.

The shape of the gap can be said exactly. For nonlinear rules, every published mechanism we have found for bounding an infinite-lattice preperiod runs through a *blocking word*, which exists precisely when the rule has an equicontinuity point — precisely when the transient is inert. Rule 30 is sensitive and has no blocking word. Linear rules are handled otherwise, by algebra: for Rule 90 from the seed, Kummer’s theorem gives $\tau_d = 0$ on every diagonal (Remark 17.32), although Rule 90 is permutive, hence sensitive, and has no blocking word either. A nonlinear alternative is sometimes credited to Li and Nordahl for Rule 110 [23], and does not survive reading: their paper contains no theorem, its transient is a finite-lattice quantity fitted as $T_{\text{ave}} \sim N^{1.08}$, and its glider decay is a least-squares fit from uncorrelated random initial states. So for nonlinear rules the field’s only preperiod tool works only where there is nothing to explain, and the general results in the area — finite-lattice hardness, and undecidability of ever reaching a periodic regime — quantify over rules and say nothing about one fixed rule, exactly as Remark 23.2 distinguishes for Problem 3.

Definition 17.5 (Background and defect). r_d is the unique p_d -periodic sequence on $t \geq 0$ with $r_d(t) = D_d(t)$ for $t \geq \tau_d$; the *defect* is $\Delta_d = D_d + r_d$. For $d < 0$: $D_d = r_d = \Delta_d = 0$, $p_d = 1$, $\tau_d = 0$. Put $P_d = \max(p_{d-1}, p_d, p_{d+1})$, a common period of r_{d-1}, r_d, r_{d+1} .

Lemma 17.6 (PROVED). (a) Δ_d has finite support and $\deg \Delta_d = \tau_d - 1$.

(b) (Background equation) $r_d(t+1) = r_{d-2}(t) + r_{d-1}(t) + r_d(t) + r_{d-1}(t)r_d(t)$ for all $t \geq 0$.

(c) (Defect equation) For $d \geq 0$ and all $t \geq 0$,

$$\Delta_{d+1}(t+1) = h_{d+1}(t) + (1 + D_d(t))\Delta_{d+1}(t), \quad h_{d+1} := \Delta_{d-1} + (1 + r_{d+1})\Delta_d,$$

$$\text{and } \Delta_{d+1}(0) = r_{d+1}(0).$$

(d) There is no $d \geq 0$ with $r_d \equiv r_{d+1} \equiv 0$.

Proof. (a) If $\tau_d \geq 1$ and $\Delta_d(\tau_d - 1) = 0$ then $D_d(\tau_d - 1) = r_d(\tau_d - 1) = r_d(\tau_d - 1 + p_d) = D_d(\tau_d - 1 + p_d)$, contradicting minimality of τ_d . (b) The identity holds for $t \geq \max(\tau_{d-2}, \tau_{d-1}, \tau_d)$ by Lemma 17.2(a); both sides are periodic with a common period (powers of 2), so it holds for all t . (c) Subtract (b) at $d+1$ from Lemma 17.2(a) at $d+1$: $\Delta_{d+1}(t+1) = \Delta_{d-1} + \Delta_d + (1 + D_d)D_{d+1} + (1 + r_d)r_{d+1}$, and $(1 + D_d)D_{d+1} + (1 + r_d)r_{d+1} = (1 + D_d)\Delta_{d+1} + r_{d+1}\Delta_d$. $D_{d+1}(0) = 0$ for $d \geq 0$. (d) If $r_d \equiv r_{d+1} \equiv 0$, (b) at $d+1$ gives $r_{d-1} \equiv 0$; descending gives $r_0 \equiv r_1 \equiv 0$, contradicting Lemma 17.2(c) ($r_0 \equiv r_1 \equiv 1$). $\square$

Definition 17.7 (Front quantities). For $d \geq 0$:

$$\begin{aligned} M_d &= \max(\deg \Delta_{d-1}, \deg \Delta_d), & S_d &= M_d - d, & H_{d+1} &= \deg h_{d+1}, \\ \kappa_{d+1} &= M_d - H_{d+1}, & \beta_{d+1} &= \Delta_{d+1}(M_d + 1). \end{aligned}$$

$M_d + 1 = \max(\tau_{d-1}, \tau_d)$ is the *front*; S_d the *slack*; $\kappa_{d+1} \geq 0$ (since $h_{d+1}(t) = 0$ for $t > M_d$) the *cancellation depth*. If $\deg \Delta_{d-1} \leq \deg \Delta_d$ put $\mu_{d+1} = (\deg \Delta_{d-2} - H_{d+1})_+$, otherwise $\mu_{d+1} = 0$.

17.2 Exact degree formula

Theorem 17.8 (A: exact degree formula; PROVED). *Let $d \geq 0$, $H = H_{d+1}$, $\pi = \Delta_{d+1}(H + 1)$ and $\rho = \min\{t \geq H + 1 : D_d(t) = 1\} \in \mathbb{N} \cup \{\infty\}$.*

- (a) $\Delta_{d+1}(t) = \sum_{u=s(t)}^{t-1} h_{d+1}(u)$, where $s(t) = \max\{u < t : D_d(u) = 1\}$; if there is no such u , $\Delta_{d+1}(t) = r_{d+1}(0) + \sum_{u < t} h_{d+1}(u)$. In particular π is the parity of h_{d+1} over $[s(H+1), H]$, plus $r_{d+1}(0)$ when D_d has no 1 below $H + 1$.
- (b) If $\pi = 0$ then $\deg \Delta_{d+1} = H$, and $D_d(H) = 0$ when $H \geq 0$.
- (c) If $\pi = 1$ then $\rho < \infty$ and $\deg \Delta_{d+1} = \rho$.
- (d) $h_{d+1}(t) = \Delta_{d+1}(t + 1) + (1 + D_d(t))\Delta_{d+1}(t)$; hence $H_{d+1} \leq \deg \Delta_{d+1}$, with equality iff $\Delta_{d+1} = 0$ or $D_d(\deg \Delta_{d+1}) = 0$.

Proof. (a) Iterate Lemma 17.6(c): at a 1 of D_d the accumulator is overwritten by $h_{d+1}(s)$, between 1s it adds h_{d+1} . (b),(c) For $t \geq H + 1$, $\Delta_{d+1}(t + 1) = (1 + D_d(t))\Delta_{d+1}(t)$. If $\pi = 1$, $\Delta_{d+1} = 1$ on $[H + 1, \rho]$ and $\Delta_{d+1}(\rho + 1) = 0$; $\rho = \infty$ contradicts Lemma 17.6(a). If $\pi = 0$, $\Delta_{d+1} \equiv 0$ on $[H + 1, \infty)$; if $H \geq 0$, $0 = \Delta_{d+1}(H + 1) = 1 + (1 + D_d(H))\Delta_{d+1}(H)$ forces $\Delta_{d+1}(H) = 1$, $D_d(H) = 0$. (d) Rearrange Lemma 17.6(c); at $t_0 = \deg \Delta_{d+1}$ it gives $h_{d+1}(t_0) = 1 + D_d(t_0)$. $\square$

Corollary 17.9 (Modes; PROVED). $\beta_{d+1} = 1 \iff (\pi = 1 \text{ and } \rho > M_d)$. If $\beta_{d+1} = 0$, exactly one of: (E) $\pi = 0$ and $\deg \Delta_{d+1} = M_d - \kappa_{d+1}$; (R) $\pi = 1$, $\rho \leq M_d$ and $\deg \Delta_{d+1} = \rho \in [M_d - \kappa_{d+1} + 1, M_d]$. In all cases $\deg \Delta_{d+1} \geq M_d - \kappa_{d+1}$. If $\deg \Delta_{d-1} > \deg \Delta_d$ then $\kappa_{d+1} = 0$.

Proof. $h_{d+1} = 0$ on $(H, M_d]$, so $\Delta_{d+1}(M_d + 1) = \pi \prod_{H < u \leq M_d} (1 + D_d(u)) = \pi[\rho > M_d]$. The rest is Theorem 17.8(b),(c). If $\deg \Delta_{d-1} > \deg \Delta_d$ then $h_{d+1}(M_d) = \Delta_{d-1}(M_d) = 1$. $\square$

17.3 The front is monotone

Theorem 17.10 (B; PROVED). *Let $d \geq 0$.*

- (a) $M_d \leq M_{d+1}$.
- (b) If $\beta_{d+1} = 0$ then $M_{d+1} = M_d$.
- (c) If $\beta_{d+1} = 1$ then $r_d \not\equiv 0$ and $\deg \Delta_{d+1} = M_{d+1} = \min\{t \geq M_d + 1 : r_d(t) = 1\}$; hence the jump $J_{d+1} := M_{d+1} - M_d \in [1, p_d]$.
- (d) If $r_d \equiv 0$ then $\beta_{d+1} = 0$.

Proof. (a) By Lemma 17.6(c), $\Delta_{d-1}(t) = \Delta_{d+1}(t + 1) + (1 + r_{d+1}(t))\Delta_d(t) + (1 + D_d(t))\Delta_{d+1}(t)$, which vanishes for $t > M_{d+1}$; so $\deg \Delta_{d-1} \leq M_{d+1}$ and $M_d \leq M_{d+1}$. (b) $\Delta_{d+1}(M_d + 1) = 0$ and $h_{d+1} = 0$ beyond M_d give $\deg \Delta_{d+1} \leq M_d$, so $M_{d+1} \leq M_d$. (c) For $t \geq M_d + 1$: $h_{d+1}(t) = 0$ and $D_d(t) = r_d(t)$, so $\Delta_{d+1}(t + 1) = (1 + r_d(t))\Delta_{d+1}(t)$ with $\Delta_{d+1}(M_d + 1) = 1$. Finite support forces a 1 of r_d in $[M_d + 1, M_d + p_d]$, and $\deg \Delta_{d+1} > M_d \geq \deg \Delta_d$. (d) Contrapositive of (c). $\square$

17.4 The exact failure condition

Theorem 17.11 (C; PROVED). *Let $d \geq 0$.*

(a) $\tau_{d+1} \leq d \iff (\kappa_{d+1} \geq S_d + 1 \text{ and } \Delta_{d+1}(d) = 0)$.

(b) *If $\tau_d \geq d$ and $\beta_{d+1} = 1$ then $\tau_{d+1} \geq d + 1$.*

(c) *If $\kappa_{d+1} \leq S_d$ then $\tau_{d+1} \geq d + 1$.*

Proof. (a) $\tau_{d+1} \leq d$ iff $\Delta_{d+1}(t) = 0$ for all $t \geq d$. By Theorem 17.8(d) this implies $h_{d+1}(t) = 0$ for $t \geq d$; conversely, if $\Delta_{d+1}(d) = 0$ and $h_{d+1} = 0$ on $[d, \infty)$, Lemma 17.6(c) propagates $\Delta_{d+1}(t) = 0$. Finally $h_{d+1} = 0$ on $[d, \infty)$ iff $H_{d+1} \leq d - 1$ iff $\kappa_{d+1} \geq M_d - d + 1$. (b) $M_d \geq \deg \Delta_d = \tau_d - 1 \geq d - 1$, and by Theorem 17.10(c) $\tau_{d+1} = M_{d+1} + 1 \geq M_d + 2$. (c) Immediate from (a). $\square$

17.5 The cancellation window

Lemma 17.12 (F: window slaving; PROVED). *Let $d \geq 1$ with $\deg \Delta_{d-1} \leq \deg \Delta_d =: b$, and $W = (H_{d+1}, b]$. Put $\omega_d = 1 + r_{d-1} + r_d(1 + r_{d+1})$, a P_d -periodic sequence. For $t \in W$:*

$$(F1) \quad \Delta_{d-1}(t) = (1 + r_{d+1}(t)) \Delta_d(t), \quad (F2) \quad \Delta_d(t+1) = \Delta_{d-2}(t) + \omega_d(t) \Delta_d(t).$$

Proof. (F1) is $h_{d+1}(t) = 0$. Lemma 17.6(c) at diagonal d reads $\Delta_d(t+1) = \Delta_{d-2}(t) + (1 + r_d)\Delta_{d-1}(t) + (1 + r_{d-1} + \Delta_{d-1}(t))\Delta_d(t)$. Insert (F1) and use $\Delta_d^2 = \Delta_d$: the coefficient of $\Delta_d(t)$ becomes $(1 + r_d)(1 + r_{d+1}) + 1 + r_{d-1} + (1 + r_{d+1}) = 1 + r_{d-1} + r_d(1 + r_{d+1})$. $\square$

Lemma 17.13 (G; PROVED). *Under the hypotheses of Lemma 17.12 let $m = \deg \Delta_{d-2}$ and $x = \max(m, H_{d+1})$. Then $x \leq b$, and if $x < b$: $\Delta_d = 1$ on $(x, b]$, $\omega_d = 1$ on $[x+1, b-1]$, $\omega_d(b) = 0$, and $\Delta_{d-1} = 1 + r_{d+1}$ on $(x, b]$. Consequently $b - x \leq P_d$ and, for every $d \geq 1$,*

$$\kappa_{d+1} \leq P_d + \mu_{d+1}.$$

Proof. $m \leq M_{d-1} \leq M_d = b$ by Theorem 17.10(a), and $H_{d+1} \leq M_d$. For $t \in (x, b] \subseteq W$, $\Delta_{d-2}(t) = 0$, so (F2) gives $\Delta_d(t+1) = \omega_d(t)\Delta_d(t)$. At $t = b$: $\Delta_d(b) = 1$, $\Delta_d(b+1) = 0$, so $\omega_d(b) = 0$. Descending from $t = b - 1$ to $x + 1$, $\Delta_d(t+1) = 1$ forces $\omega_d(t) = \Delta_d(t) = 1$. The last claim is (F1). Since ω_d is P_d -periodic and $\omega_d(b) = 0$, a run of 1s of ω_d has length at most $P_d - 1$, so $b - 1 - x \leq P_d - 1$. Then $\kappa_{d+1} = (b - x) + (x - H_{d+1}) \leq P_d + \mu_{d+1}$. If $\deg \Delta_{d-1} > \deg \Delta_d$, $\kappa_{d+1} = 0$ by Corollary 17.9. $\square$

Remark 17.14 (Level two; identity PROVED, bound OPEN). For $t, t+1 \in W$, substituting (F1), (F2) into Lemma 17.6(c) at diagonal $d - 1$ gives, with $y = \Delta_d$,

$$\Delta_{d-3}(t) = \xi(t)y(t+1) + \zeta(t)y(t) + (1 + r_{d+1}(t))y(t)y(t+1),$$

$\xi(t) = r_{d+1}(t+1) + r_{d-1}(t)$, $\zeta(t) = (1 + r_{d-2}(t))(1 + r_{d+1}(t)) + \omega_d(t)(r_{d-1}(t) + r_{d+1}(t))$. Where $\Delta_{d-3}(t) = 0$ this is a transition relation on $(y(t), t \bmod P)$. Since $\Delta \equiv 0$ solves Lemma 17.6(c), every back-solved relation is homogeneous in y and the transition $0 \rightarrow 0$ is always admissible: no level of the telescoping yields a finite bound on μ_{d+1} by itself. A bound on μ_{d+1} is OPEN.

17.6 Plateaus, stall runs and the row reading

Definition 17.15 (Jumps and stall runs). Let $d_1 < d_2 < \dots$ enumerate $\{d \geq 1 : \beta_d = 1\}$. For each i : the front $\sigma_i = M_{d_i} + 1$; the stall run $k_i = d_{i+1} - d_i - 1 \in \mathbb{N} \cup \{\infty\}$ (∞ if d_i is the last jump); the jump $J_i = M_{d_i} - M_{d_i-1}$. The stall steps of plateau i are the d with $d_i \leq d \leq d_{i+1} - 2$ (those with $\beta_{d+1} = 0$). Put $P_i = \max\{P_d : d_i \leq d \leq d_{i+1} - 1\}$ and $\mu_i = \max\{\mu_{d+1} : d_i \leq d \leq d_{i+1} - 2\}$ ($\mu_i = 0$ if $k_i = 0$).

Lemma 17.16 (Row reading; PROVED). *For each i :*

- (a) $\sigma_i = \tau_{d_i}$ and $M_j = \sigma_i - 1$ for $d_i \leq j < d_{i+1}$;
- (b) $\Delta_j(\sigma_i) = 0$ for $d_i \leq j < d_{i+1}$, and $\Delta_{d_{i+1}}(\sigma_i) = 1$ if $k_i < \infty$;
- (c) $k_i = \min\{j > d_i : \Delta_j(\sigma_i) = 1\} - d_i - 1$, i.e. in cells $x = j - \sigma_i$ of row σ_i : $k_i + 1$ is the distance from the front cell $d_i - \sigma_i$ to the first cell $x > d_i - \sigma_i$ with $u_x^{\sigma_i} \neq r_{x+\sigma_i}(\sigma_i)$.

Proof. (a) Theorem 17.10(c) gives $\deg \Delta_{d_i} = M_{d_i}$; Theorem 17.10(b) keeps M_j constant on the plateau. (b) $\deg \Delta_j \leq M_j = \sigma_i - 1$; and $1 = \beta_{d_{i+1}} = \Delta_{d_{i+1}}(M_{d_{i+1}-1} + 1) = \Delta_{d_{i+1}}(\sigma_i)$. (c) Combine (b) with $D_j(\sigma_i) = u_{j-\sigma_i}^{\sigma_i}$. $\square$

Lemma 17.17 (Lattice path; PROVED). *For each i : $S_d = \sigma_i - 1 - d$ for $d_i \leq d < d_{i+1}$; $\sigma_{i+1} = \sigma_i + J_{i+1}$ and $S_{d_{i+1}} = S_{d_i} - k_i + J_{i+1} - 1$ if $k_i < \infty$, with $J_{i+1} = \min\{t \geq \sigma_i : r_{d_{i+1}-1}(t) = 1\} - \sigma_i + 1 \in [1, p_{d_{i+1}-1}]$; and if $1 \leq k_i < \infty$, $\min\{S_d : d_i \leq d \leq d_{i+1} - 2\} = \sigma_i - d_i - k_i$.*

Proof. Lemma 17.16(a) and Theorem 17.10(c) at $d = d_{i+1} - 1$; the minimum is at $d = d_i + k_i - 1$. $\square$

17.7 Reduction of Conjecture H to stall runs

Conjecture 17.18 (H; OPEN). $\tau_d \geq d$ for every $d \geq 20$.

EXACT: the conjecture holds for $20 \leq d \leq 4 \times 10^6$ (τ_d per diagonal, with a certificate; independent codes agree to 2×10^6). The threshold 20 is sharp: $\tau_d < d$ exactly for $d \in \{3, \dots, 15, 17, 19\}$ (E1, and the same set to 4×10^6 by E10). Theorem 17.20 reduces the conjecture to two inequalities on the front; their measured margins are in Remark 17.21.

Remark 17.19 (the column as the sum of two diagonals; PROVED, EXACT and MEASURED parts as marked). The centre column is the diagonal of the frame, $c(t) = D_t(t)$, so with $D_t = r_t + \Delta_t$,

$$c(t) = b_B(t) \oplus \Delta_t(t), \quad b_B(t) := r_t(t),$$

the background column of Remark 19.3 XOR the defect of diagonal t at its own row (PROVED, an identity; ledger 10.234d A). Since $\tau_t > t$ for every $20 \leq t \leq 8 \times 10^5$ (EXACT, ledger 10.242d A14, strengthening E7's $\tau_d \geq d$; $\tau_{19} \leq 19$), on that range the second summand is a live transient bit, about $0.32t$ rows below the front of its diagonal; its persistence is, up to one row, Conjecture H (ledger 10.242d A14). On rows 1 000–1 130 000 the bit $\Delta_t(t)$ has density 0.50015 (coin standard deviation 0.00047), geometric runs, and conditional frequencies within 2.1 standard deviations of $\frac{1}{2}$ against b_B , against the wall's phase modulo 16, 32 and 64, against the local context of r_t , against the period, and, to 8×10^5 , against the state of the edge at diagonal t — move or wait, depth below the front, mode (MEASURED; ledger 10.234d B, 10.242d A15). Nothing the left half's order knows is seen to leak into the column's bits at this length. The same entry

argues that the edge speed equals the leftward damage speed of a single flipped cell (0.2429 against Wolfram's 0.2428, Remark 17.24); the argument uses the universality of the front, which is MEASURED (Theorem 18.9 and the discussion following it), so that lemma is conditional on it and is not a theorem (ledger 10.234d C).

Theorem 17.20 (H*; PROVED). *Let $d_{i_0} \geq 20$ be a jump diagonal and assume (Base) $\tau_d \geq d$ for $20 \leq d \leq d_{i_0}$.*

- (U) *If $k_i \leq \sigma_i - d_i - P_i - \mu_i$ for every $i \geq i_0$ with $k_i \geq 1$, then $\tau_d \geq d$ for all $d \geq 20$.*
- (C) *If $\kappa_{d+1} \leq P_d - 1$ for all $d \geq d_{i_0}$, and $k_i \leq \sigma_i - d_i - P_i + 1$ for every $i \geq i_0$ with $k_i \geq 1$, then $\tau_d \geq d$ for all $d \geq 20$.*

Proof. Induction on $d \geq d_{i_0}$; assume $\tau_d \geq d$. If $\beta_{d+1} = 1$, Theorem 17.11(b) gives $\tau_{d+1} \geq d + 1$. If $\beta_{d+1} = 0$, then $d + 1$ lies strictly inside a plateau $i \geq i_0$, so $k_i \geq 1$ and d is a stall step; the hypothesis excludes $k_i = \infty$. By Lemma 17.17, $S_d \geq \sigma_i - d_i - k_i$. Under (U) this is $\geq P_i + \mu_i \geq P_d + \mu_{d+1} \geq \kappa_{d+1}$ (Lemma 17.13); under (C) it is $\geq P_i - 1 \geq P_d - 1 \geq \kappa_{d+1}$. Theorem 17.11(c) gives $\tau_{d+1} \geq d + 1$. $\square$

Remark 17.21 (the two halves of (C), and their measured margins; EXACT and MEASURED parts as marked, used in no proof). Condition (C) has two halves. The first, $\kappa_{d+1} \leq P_d - 1$, is EXACT on $[21, 8 \times 10^5]$ (E2, E7; ledger 10.156c2). The second asks $k_i \leq \sigma_i - d_i - P_i + 1$, which on the seed's data allows stall runs of about $0.32 d_i - P_i$, far above the $P_i - 1$ one would read off if the two halves were conflated (ledger 10.242d A13). The coin-driven edge machine of Remark 18.10 gives the tail of the stall-run distribution to 10^{-10} per cycle over 4×10^{10} diagonals: geometric with ratio 0.433 at both periods, the longest stalls seen (each run's start-up stall of 43–62 excluded) being 24 at period 32 and 26 at period 64, against the empirical $k_i \leq P_i - 1$ of E3, that is 31 and 63, which is EXACT to 8×10^5 and not a proven bound; extrapolated, a stall of 32 once per 2.7×10^{12} diagonals and one of 64 once per 10^{24} (MEASURED; ledger 10.233d H). That is a margin and not a proof; the machine can say nothing about a stall that the universal front forbids or allows beyond what its coin sees, and nothing in this remark enters Theorem 17.20.

Corollary 17.22 (Light cone and necessity; PROVED). *Let $1 \leq k_i < \infty$.*

- (a) $k_i \leq 2\sigma_i - d_i + 2P_i - 1$.
- (b) *If $\tau_j \geq j$ for all $j \in (d_i, d_{i+1})$, then $k_i \leq \sigma_i - d_i$.*

Proof. (a) Let $j^* = d_i + k_i$. By Lemma 17.6(d) some $j \in \{j^* - 1, j^*\} \subseteq [d_i, d_{i+1})$ has $r_j \neq 0$, so $r_j(t) = 1$ for some $t \in [\sigma_i, \sigma_i + p_j - 1]$. As $\deg \Delta_j \leq \sigma_i - 1$ (Lemma 17.16), $D_j(t) = 1$, hence $j \leq 2t$ by Lemma 17.2(b). Thus $d_i + k_i - 1 \leq 2\sigma_i + 2P_i - 2$. (b) $\tau_{j^*} = \deg \Delta_{j^*} + 1 \leq \sigma_i$ and $\tau_{j^*} \geq j^* = d_i + k_i$. $\square$

Remark 17.23 (the shape of this argument is a recognised technique, and the literature prices it). Two published proofs have exactly the shape of Theorem 17.20, and it is worth saying which, because it converts our saturation computations from an ad hoc device into an instance of a method.

The first is Baetens and Gravner [26, Thm. 4.2]. They model defect accumulation along a trajectory as a branching walk and prove that if there is a finite set M with $|M| \geq 2$ and a time t_M such that a single defect at the origin has spread over all of M by time t_M for arbitrary

initial configuration, then the defect dynamics is expansive with an explicit lower bound on the Lyapunov profile. Rule 30 is one of the four rules their theorem handles, but only on its rightward, permutive side: they apply it with $M = \{1, 3\}$ and $t_M = 3$, so the bound lives on velocities $\alpha \in [\frac{1}{3}, 1]$, and the left edge -0.41 of their defect shape for Rule 30 is an empirical estimate. We have found nothing in print that bounds Rule 30's leftward damage, defect or front speed from below, and the left side is where our front lies. Two features matter for us: it is one of the very few theorems in this literature that applies to Rule 30 with *no* hypothesis on an initial measure, and its proof is a finite verification over all backgrounds in a window followed by an induction — which is precisely the form of the hypotheses (U) and (C) above and of the saturation computations that support them.

The second is Gravner and Griffeath's treatment of Rule 22 [29], the symmetric sibling of our rule. Their Lemma 3.3 defines an irregular block, gives an explicit succession rule for how the irregular blocks evolve, and concludes that their number is nonincreasing in time; Theorem 1 then follows. That is the template a proof of Conjecture H would follow: a finite list of defect types, a closed succession rule, a monotone count, all deterministic and all along one orbit. It does not transfer as a theorem, and the two reasons are worth recording because they are exactly the difficulty. Their argument needs the nonlinear rule to *be* Rule 90 on the background, which Proposition 15.14 excludes for Rule 30; and their defects shrink by one cell from each end per step whereas ours grow, by Theorem 17.10, so the monotone quantity runs the wrong way and the list of defect types is not finite.

Finally, the general theory says why no borrowed theorem will close the gap. Pivato's kinematic classification [27] sorts a defect between two subshifts L and R by their entropies. When both vanish the defect is a finite automaton; when both are positive it is Turing equivalent; our case is $h(L) = 0 < h(R)$, the asymmetric pushdown regime, in which he shows the analogous questions — including whether the defect eventually stops moving — are equivalent to the halting problem. His framework also assumes throughout that defects stay bounded in width, and he singles out the unbounded case as probably undecidable in general, citing [28]. Our Δ_d is exactly that case, and Theorem 17.8 is an exact formula for its growth, which is more than the framework has; but the price is that any proof of Conjecture H must be specific to Rule 30, and this is the citation for that.

Remark 17.24 (what precision the settling constant actually has; MEASURED). Every number in this paper that is keyed to τ_d inherits an uncertainty we had been quoting too optimistically, and the correction matters for the remark that follows, so we state it here.

Two facts about the sequence τ_d . First, least-squares error bars on it are meaningless. Fitting $\tau_d = \rho d + c$ on disjoint blocks of the exact data to $d = 90\,000$ gives block-to-block standard deviations of 0.0158 at block width 10^4 and 0.0077 at width 3×10^4 , against within-block standard errors of about 10^{-4} : the residuals are strongly autocorrelated, because τ_d moves in plateaus, and a regression bar understates the spread by roughly two orders of magnitude. Over $d \in [8000, 90\,000]$ the smoothed ratio τ_d/d is flat at 1.335 to within ± 0.001 , with no drift visible inside that range at all.

Second, the drift is real but lives further out, beyond 90 000, and on the exact data to 4×10^6 none is visible past about 10^6 . The two published constants quoted below are both from [88], but the difference-pattern constant 0.2428 has an earlier, two-digit version: Wolfram's CRYPTO '85 extended abstract [87] gives the left Lyapunov exponent of this rule as 0.25, with 1 on the right, and Meier and Staffelbach [82, p. 191] report that a bit change propagates to the left with speed roughly $\frac{1}{4}$, citing Wolfram's 1986 paper in *Adv. Appl. Math.*, which we have not

seen. Neither gives the range measured. We read Wolfram’s 1983 survey [86] page by page and it contains no measurement of this boundary speed, nor any quantity convertible to one — its only difference-pattern data, Fig. 20, is for Rules 90 and 126 over some 220 steps, and the quantity plotted there is a width times an interior density, which for the symmetric Rule 126 simply recovers the light cone at density $\frac{1}{2}$. So no third range is available from the literature against which to test the drift. Our own exact values of M_d/d , now per diagonal to $d = 4 \times 10^6$ (with a certificate; independent codes agree to 2×10^6), are 1.336660 at $d = 5 \times 10^4$, then 1.334278, 1.332570, 1.329870, 1.326183, 1.324145, 1.321118, 1.320079 at $d = 9 \times 10^4, 10^5, 2 \times 10^5, 3 \times 10^5, 4 \times 10^5, 6 \times 10^5, 8 \times 10^5$, and 1.320763 at $d = 4 \times 10^6$, above the value at 8×10^5 : the ratio is not monotone. The bin medians of M_d/d from the word-granular frame of E9 fall from 1.3370 on the bin at $d \approx 9.5 \times 10^4$ to 1.3202 on $[7 \times 10^5, 10^6]$ and then sit at 1.3209–1.3212 on the six bins covering $[10^6, 4 \times 10^6]$ (spread 0.0003), and the per-diagonal medians on those six bins reproduce them to ± 0.0001 ; the linear fit over $d \geq 3 \times 10^6$ is $1.32039d + 1830$, and the local slope shows no trend from 10^5 to 4×10^6 (mean 1.32046 over blocks of 10^5 diagonals, block standard deviation 0.005, drift 0.00005 ± 0.00073 per 10^6). So the bin medians are locked to four digits on $[10^6, 4 \times 10^6]$; whether the ratio converges is OPEN, and these data give no bound on the limit: the fitted slope 1.32039 ($v = 0.24265$) is a local rate of advance and not a bound. Two earlier versions of this remark gave upper bounds, 1.3204 and then, under continued monotone decrease, 1.3209 (front speed at most 0.2429); both are WITHDRAWN, the second because the ratio is not monotone. Extrapolation does not do better, and that can be shown rather than assumed. Fitting bin medians on $d \leq 2 \times 10^5$ to $\rho_\infty + Cd^{-\alpha}$, to $\rho_\infty + C/d$, to $\rho_\infty + C/\log d$, and by Aitken acceleration on the dyadic subsequence, returns 1.25–1.34, 1.332–1.338, 1.29–1.41 and 1.340 respectively; the two tight estimates (1.332–1.338 and 1.340) overshoot the bin median 1.3209 measured at $d = 4 \times 10^6$, and the two loose ones merely contain it. That measurement is a held-out test point which no method predicts, so the approach to the limit is of none of those forms over the range we hold. This is unsurprising once the input is examined: the bin medians are not monotone there, rising to 1.3395 at $d = 32\,768$ before falling, and extrapolating a sequence that has just turned over is exactly the case in which extrapolation misleads. The two published constants are 0.2428 [88, p. 949] and 0.252 [88, p. 871]. The locked bin value $1 - 1/1.3209 = 0.2429$ agrees with the first to three digits and is inconsistent with the second at 3.7%; an earlier version of this remark spoke of three numbers agreeing to three digits, which is WITHDRAWN, and the agreement with 0.2428 is a coincidence of ranges until a limit is proved to exist. What the data establish is only that the local slope gives a speed near 0.2427 on $[10^5, 4 \times 10^6]$; that the limiting speed is below $\frac{1}{4}$ rests on the conditional bracket of Remark 17.29, and nothing finer is known. Two numerical coincidences in the older literature should not be mistaken for our quantities: the decay constant $\lambda \approx \frac{4}{3}$ of [86] governs a triangle density and is unrelated to the settling constant near $\frac{4}{3}$ that we refuted as an asymptote, and the “effective density” 0.25 there is not the mean-field 0.25 of [88, p. 949].

Remark 17.25 (the front is a min-plus first-passage law, and not a finite one). The front has an exact tropical form, which names the difficulty precisely and then supplies a negative result.

Theorem 17.10 can be written as a first-passage law in the min-plus sense: the front advances from M_d to the next position at which the background r_d permits it,

$$M_{d+1} = M_d + \beta_{d+1} \cdot \text{gap}_{r_d}(M_d \bmod p_d),$$

an identity we verified on all 90 001 diagonals, with every one of the 49 121 jumps lying in $[1, p_d]$ and mean increment 1.334289 — the slope *on that range*, which is the figure to compare

against $M_{90\,000}/90\,000 = 1.334278$ and not against 1.324145 at $d = 4 \times 10^5$, since the ratio falls further beyond that range (Remark 17.24). The min-plus part — the gap function of a periodic background — is exactly the kind of object max-plus spectral theory handles. The gate β_{d+1} is not: by Theorem 17.8(a) it is a parity of h_{d+1} on an interval and is not a function of the background at all. That single factor is what makes the front a cocycle rather than a matrix power, and it is the whole difficulty of this Part in one line. In the same vocabulary, Lemma 17.17 with Theorem 17.20 is a Lindley reflected walk against the barrier $P_d + \mu_{d+1}$.

Now the negative result, and it is the strongest unconditional statement we have about the front. For an irreducible max-plus matrix, the Cyclicity Theorem [45, Thms. 3.109, 3.112] gives $A^{t+\gamma} = \lambda^\gamma \otimes A^t$ for all large t , so every entry is eventually affine plus periodic and its deviation from the line $\lambda t + c$ is *bounded*, at any dimension. Measure that deviation for our front: the Chebyshev band half-width $W(D) = \min_a \frac{1}{2} (\max_{d \leq D} (M_d - ad) - \min_{d \leq D} (M_d - ad))$ takes the values 20.6, 29.8, 56.7, 109.0, 175.4, 216.7 at $D = 10^3, 3 \times 10^3, 10^4, 3 \times 10^4, 6 \times 10^4, 9 \times 10^4$, a fitted growth of $D^{0.54}$ over these six points (0.63 over the last three), while the three natural bounded controls are flat across the same two orders of magnitude: a Beatty sequence of irrational slope gives 0.50 throughout, a rational slope 0.33, and an actual irreducible max-plus matrix power 3.00. A random walk with increments in $\{1, 2\}$ gives 6.5, 29.4, 98.2, so our growth is of that order. So the front is not the max-plus power of any fixed finite irreducible matrix; the growth of W is MEASURED, and the implication from unbounded W is PROVED.

Two consequences for the dichotomy stated in the next remark. First, it is incomplete as we stated it: a Beatty sequence with irrational slope has discrepancy below 1, so irrationality by itself does not produce a growing band, and there is a third branch — irrational slope with *unbounded* discrepancy — which is the branch the growing band points to (MEASURED). Second, our own appeal to the growth of the transfer matrix was the wrong mechanism. Blondel, Gaubert and Tsitsiklis [46] show that at *fixed* finite dimension the max-plus Lyapunov exponent is already NP-hard to approximate and that deciding whether it vanishes is undecidable, so irrationality does not require an unbounded system: the dividing line is autonomy, not dimension. Our 2^k growth is a strengthening of the obstruction, not its source. Consistently with that, the jump increments do not grow with the period, their mean being flat at 2.44 across $p = 4, \dots, 32$.

Remark 17.26 (the front speed: a dichotomy, and what each branch costs). Proved front velocities for cellular automata are rational with one class of exception, and the exception sits on our side of the question. Pivato's Theorem 2.1 [27] explains the rule: a defect of bounded width between σ -periodic backgrounds has a finite internal state driven by periodic input, so its average velocity is rational by construction. The concrete method agrees — Gravner and Griffeath's (a, τ) -blockers [67, Props. 5.1, 5.2] return $2, 1, \frac{4}{3}, \frac{3}{4}, \frac{1}{2}$ for the range-two edge automata, each by a finite verification, and their remark that the blocker bound is independent of the rest of the environment makes it the one published method built for an ordered solution competing against a chaotic one.

The exception is the multiplication automaton: multiply by p in base pq , a radius-one rule on a full shift, whose digit string from a quiescent background grows by $\log_{pq} p$ per step, irrational. This is Kopra's Theorem 5.5 [19], and Hochman realises every non-negative real as a Lyapunov exponent on a subshift. Note which side it falls on: that automaton's exponent as a supremum over configurations is rational, while the exponent *along the single orbit* is irrational, and our front is the single-orbit quantity. The right general statement is a published question rather than a theorem: Cyr, Franks and Kra ask whether any automaton on a full shift has an irrational Lyapunov exponent [20, Question 5.7].

Blockers do not apply to Rule 30 as it stands, since Theorem 18.3 denies it a periodic solution at the light-cone edge. That obstruction has a published way around it: Proposition 3.3 forbids only *one-sided* solutions, and Gravner and Liu define a weak robust periodic solution — doubly periodic, preserved on the left half-line, advancing into any environment — with a decidable criterion and the bound $v \geq a/T_a$ [22]. That is the object we had described informally as the ordered texture invading the chaotic bulk, and it separates the rules: Rules 90 and 150 have none, their fronts retreating because they are bipermutive, while Rule 30’s checkerboard advances against every environment in a sample of 912.

So the dichotomy, and its price. A universal (a, τ) -blocker certifies a/τ and costs 2^a , both bounds quantifying over all environments of length a , so the numerator of the true speed is the price of the certificate. At the interval 0.242653 ± 0.00017 used when this was computed (an error bar that Remark 17.24 does not support; no bound on the limit is now claimed), no rational of numerator below 25 lies in it, the first admissible fraction being $25/103$; the published velocities were certified at $a \leq 4$. And the price rises as the measurement sharpens — $33/136$ at a third of the error bar, $223/919$ at a hundredth. A small numerator is the only case the published method reaches, and it is the case the measurement excluded at that error bar. Given Remark 17.24 the number 25 is conditional on an error bar we do not trust; what is unconditional is that any interval tight enough to identify the speed excludes every numerator a blocker search could feasibly reach.

Remark 17.27 (the existence of the front limit: one route dead, one live; MEASURED). The existence of the limit, as opposed to its value, is cheaper, and one route is dead while another is live. Dead: the increment $a(d, e) = M_{d+e} - M_d$ is a telescoping difference, hence an exact cocycle, verified on 20 000 random triples with no violation, so every two-parameter subadditivity hypothesis — Kingman’s or Liggett’s — holds with equality and yields nothing, and Fekete applied to it is a tautology. The only non-vacuous superstructure quantifies over restart states, $G^+(e) = \sup_d (M_{d+e} - M_d)$, which is subadditive; but $G^+(1) = \sup_d J_d$ and the maximal jump saturates the diagonal period, the ratio $\max J/p$ being $1.00, 1.00, 1.00, 0.94$ at $p = 1, 4, 8, 16$, so with periods unbounded that constant is $+\infty$. Note that this is the same sup-over-environments structure a blocker quantifies, so the blocker route dies here a second time, independently of its cost. Kingman cannot be repaired for a deterministic array either: its stationarity hypothesis becomes equality of increments, forcing J_d constant.

Live, and sharper than we first posed it. The de Bruijn–Erdős theorem [48], in Füredi and Ruzsa’s form [49], needs only near-diagonal almost-subadditivity: if $M_{d+e} \leq M_d + M_e + f(d+e)$ for all $d \leq e \leq 2d$ with $\sum_n f(n)/n^2 < \infty$, then $\lim M_d/d$ exists. It quantifies over no restart states and needs no randomness, monotonicity or stationarity, which is why it survives where the routes above fail. On the exact orbit the inequality holds with f constant: the maximum of $M_{d+e} - M_d - M_e$ over the 13 333 533 333 band pairs with $d+e \leq 400\,000$ is 326, attained at $(d, e) = (10\,391, 16\,064)$, and it is the same at each of 32 000, 90 000, 120 000, 150 000, 180 000, 200 000 and 400 000 with the same maximising pair throughout; τ is exact to $d = 400\,000$, all 400 001 diagonals settled, agreeing with the independent $d \leq 90\,000$ computation at all 89 999 diagonals settled in both.

A running maximum that never moves is also what one would see if the extremal region were simply never revisited, so what is reported below is the maximum restricted to each window of $d+e$, which is the only form in which the figure carries information. The extremum is localised in $d+e \in [25\,000, 50\,000]$ and is never approached again: over the ten billion band pairs with $d+e \geq 200\,000$ — three times the entire range available before this computation — the largest

deficit is 94. The decline is also not mysterious. M_d/d falls across the computed range, through 1.336660, 1.332570, 1.329870, 1.326183, 1.324145 at $d = 5 \times 10^4, 10^5, 2 \times 10^5, 3 \times 10^5, 4 \times 10^5$, and a falling ratio favours $M_{d+e} < M_d + M_e$ at large scales, though it forces nothing pointwise, the ratio not being monotone (below, and Remark 17.24); what the constant measures is the small-scale roughness of δ , and the window table is the evidence that the roughness does not grow.

Examining where that maximum lives improves the target. First, the deficit is exactly the deviation combination: writing $\delta_d = M_d - \rho d$, we have $M_{d+e} - M_d - M_e = \delta_{d+e} - \delta_d - \delta_e$, and at the extremal pair this reads $384.3 - (-48.7) - 107.0 = 326$. So the inequality is a statement about the *roughness* of δ and not its size — a smooth concave δ gives a negative deficit, and only local excursions give a positive one. That the front is smoother than generic noise of the same amplitude can be checked: a synthetic sequence with our slope and a random walk of matched amplitude gives band maximum 629, and one with our measured exponent 1126, in both cases with the maximising pair drifting to the far end as the range grows, where ours is fixed.

Second, and this is the useful part, the positive deficits are confined. Taken by window of $d + e$ the maximum runs

$d + e/10^3$	20–30	30–40	40–55	55–70	70–90	90–110	110–130	130–150	150–175	175–200
max	+326	+285	+26	+227	+233	–54	–40	–323	–503	–429

A negative maximum means every band pair in that window satisfies $M_{d+e} < M_d + M_e$ outright. On the range available when that table was computed, $d + e \leq 200\,000$, every window above 90 000 was negative, and we concluded that the front is plainly subadditive on the band beyond that point with a margin that grows, so that the target could be stated without an error term as $M_{d+e} \leq M_d + M_e$ for $d + e \geq N_0$. Doubling the range refutes both halves of that, and the refutation is recorded here because the fallback it triggers was registered in advance. Continuing the same windows, now 20 000 wide, to $d + e = 390\,000$:

$d + e/10^3$	190–210	210–230	230–250	250–270	270–290	290–310	310–330	370–390
max	–167	–93	+94	–40	–571	–770	–872	–1374

The positive deficits *do* recur: exactly one window above 90 000 is positive, at $d+e \in [230\,000, 250\,000]$ with maximum +94 attained at (101 905, 146 203). So plain subadditivity does not hold on the tail $d + e \geq 90\,000$, and the margin does not grow monotonically; quoting the smallest N_0 that works on the range computed so far would repeat the error the doubling just exposed. What survives is the almost-subadditive form with f constant, which is what [49] needs and which the recurrence at +94 does not threaten, since 94 is far below the global 326. That is now the target, and the clean form is withdrawn. Two cautions, and the second can now be quantified. The crossover sits at 87 866, the last doubling position in the seed’s lineage (E8), so the positive region is the one preceding it, and we do not know whether that is causal. The doublings do not stop there, so positive deficits may return at the next one — but the next one is much further away than “beyond our range” suggests. Reading the doubling positions off the exact periods, the first diagonal carrying period p occurs at

p	2	4	8	16	32
first d	3	8	29	400	87 867
$\log_2(\text{first } d)$	1.585	3.000	4.858	8.644	16.423

and the successive differences of that last row are 1.415, 1.858, 3.786, 7.779, whose ratios are 1.31, 2.04, 2.06. The three usable ratios are consistent with a *doubly exponential* onset in $\log_2 p$

after a short transient, and extrapolating the two stable ratios puts the first period-64 diagonal near $2^{32.4} \approx 5 \times 10^9$. This is MEASURED and rests on three usable ratios of which two agree to one per cent, so it is a reading rather than a law; but it says that no computation at any scale we can contemplate will reach the next doubling, and that the whole computed range lives inside a single period regime. The positive deficit we do find above 90 000, at $d + e \in [230\,000, 250\,000]$, is accordingly *not* at a doubling, so whatever produces it is not the doubling structure. In the same range, 399 251 of the 400 001 diagonals carry period 16 or 32, and the mean of M_d/d continues to fall *within* the period-32 class alone — 1.3321, 1.3295, 1.3268, 1.3250 over $d \in [88\,000, 150\,000)$, $[150\,000, 250\,000)$, $[250\,000, 320\,000)$, $[320\,000, 400\,000]$ — but this does not separate the drift from the period: M_d/d is a running average over all diagonals and carries the range below 87 867 with it, while the local slope shows no trend from 10^5 to 4×10^6 (Remark 17.24). Proving either form of the inequality is the open problem we would most like solved.

Exact monotonicity of the ratio would suffice and is false. If $\rho_n = M_n/n$ were non-increasing then $\rho_{d+e} \leq \min(\rho_d, \rho_e)$ and hence $M_{d+e} = (d+e)\rho_{d+e} \leq d\rho_d + e\rho_e = M_d + M_e$ immediately. But $\rho_{n+1} - \rho_n = (J_n - \rho_n)/(n+1)$ exactly, where $J_n = M_{n+1} - M_n$, so ρ rises whenever a jump exceeds the running ratio, that is whenever $J_n \geq 2$ — which happens at 39.9% of steps, and at every scale we measured, from 41.7% in $[2^{10}, 2^{11})$ to 39.8% in $[2^{17}, 2 \times 10^5]$. What does decay is the size of each rise, bounded by $(p_n - \rho_n)/(n+1)$ and measured falling from 1.27×10^{-2} to 1.01×10^{-4} across those scales, in proportion to $1/n$ as that bound predicts.

One further route is priced rather than closed. Baetens and Gravner's transfer-matrix theory [26, Thms. 6.1, 6.2] gives the asymptotic shape and Lyapunov profile of a defect on a doubly periodic background as a Legendre transform, with one state per phase of the background — so the matrix has size proportional to the period, which is 2^k in our regime index. That is the same wall our own transfer computation hit at $K = 18$: the literature's best deterministic front theorem and our computational ceiling are one ceiling.

Remark 17.28 (the target reduced to one index; PROVED reduction, MEASURED hypothesis). The band inequality quantifies over pairs. It can be replaced by a statement about a single index, and the exchange also shows that the constant form of f we have been pursuing is stronger than needed.

Let G be the least concave majorant of M on $[0, N]$ and $h = M - G \leq 0$. For concave G with $G(0) = c$, concavity at x and at y against the endpoints 0 and $x + y$ gives $G(x) + G(y) \geq c + G(x + y)$; here $c = M_0 = -1$. Hence

$$M_{d+e} - M_d - M_e = [G_{d+e} - G_d - G_e] + [h_{d+e} - h_d - h_e] \leq 1 + |h_d| + |h_e| \leq 1 + 2 \max_{m \leq d+e} |h_m|,$$

using $h \leq 0$ throughout. So $f(n) = 1 + 2 \max_{m \leq n} |h_m|$ is admissible, and since $\sum n^{\alpha-2}$ converges for every $\alpha < 1$:

If M stays within $O(N^\alpha)$ of its least concave majorant for some $\alpha < 1$, then $\lim M_d/d$ exists.

The reduction is PROVED. The hypothesis is MEASURED: recomputing the majorant on $[0, N]$ for $N = 12\,500, 25\,000, 50\,000, 10^5, 2 \times 10^5, 4 \times 10^5$ gives $\max |h| = 115.9, 190.7, 241.7, 367.5, 557.6, 762.8$. So $\max |h|$ grows across the computed range (MEASURED), and a constant f does not come from this route — but the ratio per doubling is 1.646, 1.267, 1.520, 1.517, 1.368, every one below the value 2 that $\alpha = 1$ would require. That is the point of this formulation: the hypothesis is

a *threshold* question rather than an exponent estimate, so it does not rest on fitting α — an exercise withdrawn elsewhere in this paper (Remark 15.9). The largest observed ratio is 1.65 against a threshold of 2, and the growth is consistent with the $D^{0.54}$ Chebyshev band above, which is the same quantity measured against a line instead of against the majorant.

Two limits, stated so the reduction is not oversold. The bound is lossy in the large: its maximum over band pairs is 1525.7, at (176 617, 176 617), against a measured band maximum of 326. It is nevertheless tight where it matters — the twelve largest deficits all lie in one neighbourhood near (10 390, 16 064) and sit 38 to 41 below their bound. And the reduction proves nothing on its own: it converts a statement quantified over 13 billion pairs into a statement about one sequence, which is progress in form and not in content. The exact-monotonicity route above is the $\alpha = 0$ case of the same picture, and it is false; this asks only for $\alpha < 1$.

Three caveats belong with this, and the first prevents the reduction from being read as more than it is. *The $o(d)$ form of the hypothesis is equivalent to the conclusion, not merely sufficient for it.* If $M_d/d \rightarrow L$, fix $\varepsilon > 0$, choose D with $M_d \leq (L + \varepsilon)d$ for $d \geq D$ and put $C = \max_{d \leq D} M_d$; then $A(t) = (L + \varepsilon)t + C$ is a concave majorant of M , so $G \leq A$ and $\limsup G_d/d \leq L + \varepsilon$ for every ε , while $G \geq M$ gives the reverse. Hence $\lim G_d/d = L$ and $h_d/d \rightarrow 0$. So a hypothesis of the form $|h_d| = o(d)$ is a restatement and cannot be a target; what is stated above, $|h_d| = O(N^\alpha)$ with $\alpha < 1$, is strictly stronger than the conclusion and is a genuine one. Second, the measured quantity is weaker than it looks: $\max |h|$ on $[0, N]$ is *non-decreasing in N by construction*, since for $N < N'$ the restriction of $G^{(N')}$ to $[0, N]$ is a concave majorant there and so dominates $G^{(N)}$; and the object in the theorem is the deficit below the full-horizon majorant $G^{(\infty)} \geq G^{(N)}$, so every finite- N figure is a *lower* bound on a quantity that must be bounded above. Third, $\sup_d M_d/d < \infty$ is MEASURED and not proved — the only proved recursion is $\tau_{d+1} \leq \max(\tau_{d-1}, \tau_d) + p_d$ with p_d unbounded — and without it $G \equiv +\infty$ and the majorant has no object. The measured value is $\sup_{d \leq 8 \times 10^5} M_d/d = 1.4588$, at $d = 85$.

Remark 17.29 (a criterion strictly between the two; PROVED reduction, MEASURED hypothesis). The gap left by the previous remark — weaker than $O(N^\alpha)$, stronger than the tautology — is filled by a dyadic criterion on the slope itself. Put $s_d = M_d/d$ and

$$W_k = \max\{s_b - s_a : 2^k \leq a < b \leq 2^{k+1}\},$$

the largest *upward* move of the slope inside one dyadic window. Then $\sum_k W_k < \infty$ implies $\lim M_d/d$ exists, with $\lim s_a \leq s_a + \sum_{i \geq k} W_i$ for every $a \geq 2^k$: given $2^j \leq a < b \leq 2^{k+1}$, insert the intervening powers of two, note each consecutive pair lies inside a single window, and sum, so $\limsup s - \liminf s \leq \sum_{i \geq j} W_i \rightarrow 0$. It is weaker than the $O(N^\alpha)$ hypothesis, since $|h| = O(d^\alpha)$ gives $W(D) = O(D^{\alpha-1})$, summable exactly when $\alpha < 1$; and stronger than the conclusion, since convergence does not imply summability.

Measured on the exact front to $d = 8 \times 10^5$:

k	10	11	12	13	14	15	16	17	18
W_k	5.13e-2	2.91e-2	2.31e-2	1.80e-2	8.47e-3	3.77e-3	5.28e-3	3.18e-3	8.25e-4

with partial sums reaching 0.1431 and a fit $W_k \sim 2^{-0.666k}$; extrapolating the tail beyond $k = 18$ gives 0.0023 and the conditional bracket $\lim M_d/d \leq 1.3201 + 0.0023 = 1.3224$. Two features recommend this over $\max |h|$. The windows are *disjoint*, so the nine numbers are independent measurements rather than a nested sequence, and correspondingly W_k is *not* monotone — $k = 16$

exceeds $k = 15$. And it is the threshold, summability, that is being claimed: three estimators of the same underlying deviation now read 0.54 from $\max |h|$, 0.666 from W_k and 0.54–0.63 from the Chebyshev band, by window, and that spread is exactly why the exponent itself is not quoted.

Remark 17.30 (two routes closed, with their citations; KNOWN theorems, MEASURED inputs). Unique ergodicity is unavailable, and not for a technical reason. Exactness of the cocycle makes the question Birkhoff’s rather than Kingman’s, and Oxtoby’s theorem gives uniform convergence at *every* point of a uniquely ergodic system — but three independent obstructions apply. The observable is not continuous, since τ_d is a preperiod and no finite window determines it; it is not bounded, the largest jump to $d = 8 \times 10^5$ being 19 at $d = 746\,573$, and $\max_{d \leq D} J_d \sim \log_2 D$ because the jump tail halves; and the system is not uniquely ergodic in any case, because the defect-free pair is itself an orbit in the closure carrying an invariant measure on which the front observable vanishes identically. The structure that makes the front well defined — a periodic background — is what supplies the second measure.

Second, the max-plus exclusion can be strengthened at no cost. Kohlberg [37] shows a nonexpansive piecewise-linear self-map of $\mathbb{R}^m$ has a unique invariant half-line, so $f^n(x) - n\alpha$ is *bounded*; any Lipschitz readout of such a model therefore has a bounded band. Our band grows through the computed range (MEASURED); if it is unbounded, this upgrades the exclusion from “no autonomous finite max-plus model with ultimate period at most 44 000” to *no finite-dimensional sup-norm-nonexpansive piecewise-affine model of the front at any dimension*, conditionally on that unboundedness.

One identity is free and worth recording. Since M is non-decreasing, $J_d \geq 0$, and $d(s_d - s_{d-1}) \geq -\sup_d M_d/d$, which is Landau’s slowly-decreasing condition; so by the Hardy–Landau Tauberian theorem for $(C, 1)$, $\lim M_d/d$ exists *if and only if* the Cesàro average $D^{-1} \sum_{d \leq D} M_d/d$ converges. The logarithmic average is not enough.

Remark 17.31 (the slope as a series over the jump tail; PROVED identity, MEASURED tail law). There is a second reduction, to a statement about the distribution of a *single* increment. Write $J_n = M_{n+1} - M_n$ and $q_k(N) = \#\{n < N : J_n \geq k\}/N$. For a non-negative integer variable $\mathbb{E}[J] = \sum_{k \geq 1} \Pr[J \geq k]$, and M_N/N is the empirical mean of the jumps, so

$$\frac{M_N}{N} = \sum_{k \geq 1} q_k(N)$$

exactly, at every N — verified numerically to six decimals at $N = 1.25 \times 10^4, \dots, 4 \times 10^5$. By Tannery’s theorem, if the $q_k(N)$ are dominated uniformly in N by a summable sequence and each converges as $N \rightarrow \infty$, then $\lim M_N/N$ exists and equals $\sum_k \lim_N q_k(N)$. So the front speed exists as soon as the *jump distribution* converges with a controlled tail.

The domination is measured and is unusually clean: the tail halves. On the fresh block $n \in [2 \times 10^5, 4 \times 10^5]$, the ratios q_{k+1}/q_k for $k = 2, \dots, 11$ read

$$0.478, 0.498, 0.500, 0.491, 0.487, 0.486, 0.487, 0.495, 0.491, 0.492,$$

i.e. $q_k \asymp 2^{-k}$ across a full decade of k , and the truncation error behaves as a tail of ratio just below $\frac{1}{2}$ requires, $\sum_{k > K} q_k$ lying within 3% below $2q_{K+1}$ at every $1 \leq K \leq 8$. A uniform bound $q_k(N) \leq C2^{-k}$ is therefore the missing hypothesis of the domination half, and it is the part of this route that looks provable rather than merely measured. We note but do *not* claim a

connection to the halving of survivors in Section 22: that is a different object, and sharing the factor 2 is not evidence.

Two features of the distribution deserve recording. It is not monotone — $\Pr[J = 0] = 0.455$, $\Pr[J = 1] = 0.147$, $\Pr[J = 2] = 0.207$, so a jump of 2 is commoner than a jump of 1, which the definition $M_d + 1 = \max(\tau_{d-1}, \tau_d)$ makes unsurprising but which any model of the increment must reproduce. And the convergence half is genuinely open. Measured on disjoint blocks rather than cumulatively, every q_k drifts downward over the first 4×10^5 diagonals, by -0.0027 , -0.0031 , -0.0035 , -0.0048 , -0.0032 , -0.0026 , -0.0014 , -0.0006 for $k = 1, \dots, 8$ between the first block and the last, and no single k carries the drift. That drift is a transient of the first $2\text{--}3 \times 10^5$ diagonals: past them no q_k with $k \leq 8$ shows a detectable trend on disjoint blocks of 10^5 diagonals up to $d = 2 \times 10^6$, and q_1 shows none over the 39 such blocks of $[10^5, 4 \times 10^6]$, with mean 0.5430 and block s.d. 0.0014 (MEASURED). Flat blocks on a finite range are not convergence: a slow continuing drift, of the form $c/\log d$ with small c , would not be resolved by blocks of this size. So this reduction, like the concave one, changes the shape of the problem without weakening it: what must be proved is now the convergence of the density of a local event rather than a property of the front.

One route is closed by this data. If $\delta_d = M_d - \rho d$ carried an explicit-formula structure — a main term plus oscillatory corrections indexed by some spectrum, the shape a 2-adic odometer would produce and the natural guess given that the diagonal periods are powers of two — the corrections would appear as discrete peaks in $\log_2 d$. They do not. The spectrum of δ against $\log_2 d$ over $d \in [10^3, 4 \times 10^5]$ puts 99.81% of its power in the lowest 200 bins, and its largest components sit at 1, 2, 4, 5 times the reciprocal of the window length, which is the signature of a smooth aperiodic wander rather than of any oscillation. There is no log-periodic structure to expand.

What the tail law does buy is a compression. Since q_k is geometric from $k = 2$ with ratio r , the identity closes on three numbers,

$$\rho = q_1 + q_2 + q_2 \frac{r}{1 - r},$$

and on the fresh block $d \in [2 \times 10^5, 4 \times 10^5]$ this reproduces the measured slope to 5×10^{-4} relative — $q_1 = 0.5445$, $q_2 = 0.3970$, $r = 0.4875$ give 1.319136 against 1.318420. It is important not to overstate what that is: $\rho = \sum_k q_k$ is an identity, so the agreement measures how nearly geometric the tail is and does not *derive* ρ from anything. Its value is that it reduces the convergence question from a whole distribution to three sequences: $\lim M_d/d$ exists as soon as q_1, q_2 and r converge. Past the transient $q_1, \dots, q_8$ show no detectable drift to 2×10^6 (MEASURED), which, as above, is not convergence.

The same three numbers separate rules, and stably. Running the construction on every rule — the diagonal frame transfers verbatim, since $D_d(t+1) = f(D_{d-2}, D_{d-1}, D_d)$ is the same f — and recomputing at $D_{\max} = 2000, 6000, 12000$ gives $r = 0.5000, 0.5116, 0.5135$ for Rule 30 against 0.6757, 0.6271, 0.6445 for Rule 110, with Rules 135 and 137 tracking their partners to three decimals at every size. So r is a stable invariant of the rule and not of the window, which the front exponents of Remark 15.9 were not. Two exclusions are needed: Rules 22, 126 and 182 report jumps of order 10^4 against a computation of depth 1.1×10^4 , which are horizon artefacts — Rule 22's diagonals are purely periodic from the seed with periods doubling without bound (Remark 17.32), so past the horizon the detector reports the horizon — and their ratios are meaningless; the mirror rules 86, 149, 101, 169 do not resolve at all, because the frame $D_d(t) = u_{d-t}^t$ tracks the left front specifically and for a mirror rule the corresponding front is

on the other side at the other speed; and Rule 45 has the trivial light-cone front $M_d = 2d + 1$ (Remark 17.32) and resolves nothing for the same reason. That last is a limitation of the frame, not a fact about those rules.

17.8 Computations

All statements in this subsection are finite computations (EXACT) unless labelled otherwise. $D_d(t)$ was computed from Lemma 17.2(a) for $0 \leq d \leq 90\,000$, $0 \leq t < 132\,000$. By Proposition 17.3(c),(d), $\tau_{d+1} \leq \max(\tau_{d-1}, \tau_d) + p_d$ and $p_{d+1} \leq 2 \max(p_{d-1}, p_d)$. In the range, $\max_d \tau_d + 2 \cdot 64 < 132\,000 - 512$ and $p_d \leq 32$. Periods were read on a terminal window of length 512, and preperiods as the last disagreement with the shift p_d . So p_d , τ_d , r_d and Δ_d are determined by data inside the horizon.

- E1** (EXACT) For $1 \leq d \leq 90\,000$: $\tau_d < d$ iff $d \in \{3, \dots, 15, 17, 19\}$; $\min_{20 \leq d \leq 90\,000} (\tau_d - d) = 1$, attained exactly at $d \in \{23, 29\}$; $\tau_d/d \geq 1.267$ for $1000 \leq d \leq 90\,000$ (minimum at $d = 1026$). First occurrences of $p_d = 2, 4, 8, 16, 32$ are at $d = 3, 8, 29, 400, 87\,867$. $r_d \equiv 0$ exactly for $d \in \{2, 7, 28, 399, 53\,207, 58\,286, 87\,866\}$.
- E2** (EXACT) $\kappa_{d+1} \leq P_d - 1$ for $21 \leq d+1 \leq 90\,000$, and two later exact computations (one of them the third frame described in E7) extend it without exception to $21 \leq d+1 \leq 8 \times 10^5$. Below 90 000 the maxima of κ for $P_d = 4, 8, 16, 32$ are 3, 7, 14, 17 and $\mu_{d+1} \leq 11$; to 8×10^5 the maxima are $\kappa_{d+1} = 19$ and $\mu_{d+1} = 15$, both in the regime $P_d = 32$. And $\kappa_{d+1} \leq P_d$ whenever $\mu_{d+1} = 0$, as Lemma 17.13 requires.
- E3** (EXACT) $k_i \leq P_i - 1$ for every plateau with $20 \leq d_i$ and $d_{i+1} < 90\,000$. The maxima for $P_i = 4, 8, 16, 32$ are 1, 5, 11, 6; $\max k_i = 11$, first at $d_i = 4913$.
- E4** (EXACT) Among plateaus with $20 \leq d_i$, $d_{i+1} < 90\,000$ and $k_i \geq 1$, the inequalities of (U) and (C) fail exactly for $d_i \in \{20, 22, 28, 32, 34\}$. For $d_i \geq 36$ they hold, and $\min(\sigma_i - d_i - P_i + 1 - k_i)$ over $d_i \geq D$ equals 2, 28, 258, 3299, 16 812 for $D = 36, 100, 1000, 10^4, 5 \cdot 10^4$.
- E5** (EXACT) $d = 36$ is a jump diagonal, and (Base) of Theorem 17.20 holds with $d_{i_0} = 36$ (E1).
- E6** (EXACT, independent reproduction) E1 was recomputed from the definition of τ_d in a frame using neither a row array nor a windowed readout: iterate the diagonal recursion of Lemma 17.2(a) over d , and for each candidate power of two p record $\max\{u : D_d(u+p) \neq D_d(u)\}$, so that τ_d is that index plus one exactly, a period being accepted only when τ_d falls at least 1024 below the horizon. Over $1 \leq d \leq 90\,000$ with $T = 132\,000$ every diagonal settles and all five statements of E1 are reproduced exactly, including the first period 32 at $d = 87\,867$ and the eventually-zero set $\{2, 7, 28, 399, 53\,207, 58\,286, 87\,866\}$. Their frozen partners, the eventually-*one* diagonals, are $\{0, 1, 4, 9, 30, 401, 53\,209, 58\,288, 87\,868\}$; note $D_7 \equiv 0$ identically, so $\tau_7 = 0$. This frame (`s3_tau_convention.py`) reproduces E1 only. E3, the (C) half of E4 and E5 were reproduced separately from the last-violation array of a second exact per-diagonal run, using no code or data of the paper's frame (49 110 plateaus, 0 discrepancies), and E2 by the two later computations named in E2; the (U) half of E4, which needs μ_i and hence the defect sequences themselves, has not been independently reproduced. Note also that $P_d = \max(p_{d-1}, p_d, p_{d+1})$ is flat across a doubling of p , the increase of P sitting one diagonal earlier: P reaches 32 at $d = 87\,866$ while p reaches 32 at $d = 87\,867$.

- E8** (EXACT, from archived public data) Gravner’s Stage I label tree for this rule at period 16, 2 159 030 nodes to generation 894 234, has been public since 2012; parsing it in full reproduces those counts and identifies the seed’s lineage, whose branch generations are exactly 2, 7, 28, 399, 53 207, 58 286 with leaf 87 866. So the list in E7 is archived data, not a new computation, and the 8 in the published list 2, 8, 28, 399 of [67, §8] is a typo, their own archive giving 7. The tree has 19 branchings and 20 leaves, 16 dying by period doubling, which are their sixteen succession lines; the branching at 72 575 lies on the sibling lineage, not the seed’s. Two consequences for us: Conjecture 18.14 is *true* at $P = 16$, the seed’s exit at 87 866 being the minimum of the sixteen, while remaining refuted at $P = 32$; and Proposition 18.11 survives a much larger test, since across all twenty archived lineages the minimum gap is 5 and the only gaps at most 30 anywhere are one 5 and one 21, with none of the sixteen excluded values occurring. No period-32 tree is archived, so the death generation 65 154 361 quoted in [67, §8] is not reproducible from published files.
- E7** (EXACT) The eventually-zero diagonals below $d = 400\,000$ are exactly 2, 7, 28, 399, 53 207, 58 286, 87 866: the set is unchanged from the paper’s earlier range and there is no eighth such diagonal and no period 64 below 400 000 (bit-packed frame, $T = 608\,192$, 108 s, with p_d read twice on disjoint windows and cross-checked against the exact per-diagonal frame with no mismatch). Their six gaps are 5, 21, 371, 5079, 29 580, 52 808. A previous version of this item quoted a gap histogram with 47 gaps of 2; those figures belong to a different population, the 349 diagonals of $[400, 87\,867)$ with $p_d \mid 8$, and are withdrawn here — for the eventually-zero diagonals a gap of 2 is impossible by Proposition 18.11 below, and the apparent odd-gap deficit in that other population does not recur at matched Poisson power: in the corresponding population of the period-32 regime, the 664 diagonals of $[87\,867, 4 \times 10^7]$ with $p_d < 32$, the gap window $[3, 977]$, whose Poisson expectation matches that of $[3, 10]$ among the 349, holds 5 odd gaps against 5.36 expected (MEASURED). A third implementation of the frame, written from Lemma 17.2(a) alone (rows bit-packed over d , $T = 1\,120\,000$, every diagonal settled at least 63 936 rows below the horizon), reproduces E1, E2, E3, the (C) margins of E4 and E7 and extends them to $d = 8 \times 10^5$: $\kappa_{d+1} \leq P_d - 1$ and $k_i \leq P_i - 1$ hold throughout, the longest stall run is 13, first at $d_i = 184\,979$, the (C) margin $\min(\sigma_i - d_i - P_i + 1 - k_i)$ over $d_i \geq 10^5$ and over $d_i \geq 4 \times 10^5$ is 33 230 and 129 626, and there is no eighth eventually-zero diagonal and no period 64 below 8×10^5 . The hypotheses of (C) are therefore verified for plateaus ending before 8×10^5 (E2, E4 and this extension), and those of (U) only for plateaus ending before 90 000 (E4). Hence the computation reproduces $\tau_d \geq d$ on $[20, 8 \times 10^5]$ through Theorem 17.20(C) and proves nothing beyond; E10 checks $\tau_d \geq d$ directly to 4×10^6 . Conjecture H itself is OPEN, and its target is stated as an open question in print by Wolfram [68]. Gravner and Griffeath’s regularity-expansion machinery cannot reach it: it requires the locked region to be recognised by a finite automaton, which unbounded periods forbid, there is no final Stage I tree for this rule by their own §8, and they diagnose the same failure for the mirror rule in the words “no local method from the present paper applies”.
- E9** (MEASURED, word-granular) The bin medians of Remark 17.24, and its linear fit over $d \geq 3 \times 10^6$, come from a bit-packed frame (`s3_frame_bitpack.py` 4000000 5500000, 9 179 s) holding 64 diagonals per machine word and recording, per word, the *maximum* last violation over its 64 diagonals, de-biased by the constant 84 (validated against the exact per-diagonal frame at $d \leq 9.5 \times 10^4$: offset min 49, max 135, mean 84.8; de-biased

bin medians 1.3370/1.3353 against the exact 1.3371/1.3353). It yields bin medians and the slope robustly and no per-diagonal statement: no bound of the form $\tau_d/d \geq X$ for every $d \leq 4 \times 10^6$ can be read off it, so it does not extend E1's per- d bound; E10 does. Period 32 settles for all 61 129 words of 64 diagonals in $[87\,808, 4 \times 10^6]$, so there is no sixth doubling below 4×10^6 (the lower bound of [88] for it is 2.1×10^9).

E10 (EXACT, with a certificate) τ_d and p_d per diagonal for $0 \leq d \leq 4 \times 10^6$ (`run4Mv2_tau.npy`, a bit-parallel frame restricted to a band of rows). The certificate: the last block of rows computed contains no violation, and a violation at $(t+1, d)$ needs one at (t, d') for some $d' \in \{d-2, d-1, d\}$, so none occurs later. The file agrees with an earlier exact per-diagonal file to 8×10^5 and with an independently written code to 2×10^6 except at $d = 0$ and $d = 7$, which never violate and which this file records as -1 (the other two files give $\tau_0 = \tau_7 = 0$, since $D_0 \equiv 1$ and $D_7 \equiv 0$). $\tau_d < d$ exactly for $d \in \{3, \dots, 15, 17, 19\}$, so $\tau_d \geq d$ on $[20, 4 \times 10^6]$; $\min \tau_d/d$ over $[1000, 4 \times 10^6]$ is still 1.2671, at $d = 1026$, and over $[800\,001, 4 \times 10^6]$ it is 1.3194, at $d = 900\,864$; there is no period 64 below 4×10^6 ; and the largest jump is 24, at $d = 1\,680\,044$.

Remark 17.32 (the front is the one object here that Rule 90 does not have). Throughout this paper the sharpest obstruction to any general argument has been Rule 90, which satisfies more hypotheses than Rule 30 does and fails every conclusion we want (Proposition 26.1, and the width table of Section 26). The front is the exception, and it is worth saying so explicitly because it tells us where to spend effort. For Rule 90 from the single seed, $\tau_d = 0$ for every d : the diagonals are purely periodic from the start, with no transient, which follows from Kummer's theorem on the binomial coefficients and which we verified against direct simulation with no mismatch. So $\Delta_d \equiv 0$, $M_d \equiv -1$, and Rule 90 has no front, no slack and no defect. The same holds for Rules 150, 22 and 106 (EXACT on the computed range; Rule 22's diagonal periods grow without bound). Every statement in this Part is therefore about a structure that the firewall rule does not possess, so no Rule 90 argument can be run against it — unlike our automaticity census and every density statement, where it can. The price is that Rule 90 also cannot serve as a control here: among the rules with a front, the ones we can compute (45 with $M_d = 2d + 1$, and 60 with $M_d = d$) satisfy the band inequality with a non-positive constant precisely because their periods are bounded, so nothing available tests the one feature that matters, $p_d \rightarrow \infty$.

18 The background walk and unbounded periods

Let r_d, p_d, τ_d be as in Definition 17.5, with $r_d = 0$ for $d < 0$. For P a power of 2 let V_P be the set of P -periodic sequences $\mathbb{N} \rightarrow \mathbb{F}_2$, a finite set of size 2^P .

Definition 18.1. For sequences $b, c: \mathbb{N} \rightarrow \mathbb{F}_2$ define $\pi(b, c) = (a, b)$ with

$$a(t) = c(t+1) + b(t) + (1 + b(t))c(t) \quad (t \geq 0).$$

π maps V_P^2 to V_P^2 for every P . Put $v_d = (r_{d-1}, r_d)$ for $d \geq 0$.

Lemma 18.2 (PROVED). $\pi(v_{d+1}) = v_d$ for all $d \geq 0$. Moreover $v_0 = (0, 1)$, $\pi(v_0) = (0, 0)$ and $\pi(0, 0) = (0, 0)$.

Proof. The background equation, Lemma 17.6(b) at index $d+1$, reads $r_{d+1}(t+1) = r_{d-1}(t) + r_d(t) + r_{d+1}(t) + r_d(t)r_{d+1}(t)$ for all $t \geq 0$; solving for $r_{d-1}(t)$ gives $r_{d-1}(t) = r_{d+1}(t+1) + r_d(t) +$

$(1 + r_d(t))r_{d+1}(t)$, i.e. $\pi(r_d, r_{d+1}) = (r_{d-1}, r_d)$. By Lemma 17.2(c), $r_0 \equiv 1$, and $r_{-1} \equiv 0$ by convention, so $v_0 = (0, 1)$. Then $\pi(0, 1) = (1 + 0 + 1, 0) = (0, 0)$ and $\pi(0, 0) = (0, 0)$. $\square$

Theorem 18.3 (unbounded diagonal periods; KNOWN: Gravner–Griffeath [67], Prop. 3.3). *The pairs v_d , $d \geq 0$, are pairwise distinct. Consequently $\sup_d p_d = \infty$: the minimal eventual periods of the left diagonals of Rule 30 from the seed are unbounded.*

Proof. Suppose $v_{d+m} = v_d$ with $m \geq 1$. Applying π^d and using Lemma 18.2 repeatedly, $\pi^d(v_{d+m}) = v_m$ and $\pi^d(v_d) = v_0$, so $v_m = v_0$; then $\pi^m(v_0) = \pi^m(v_m) = v_0$. But $\pi(v_0) = (0, 0)$ is a fixed point of π different from v_0 , so $\pi^m(v_0) = (0, 0) \neq v_0$ for every $m \geq 1$, a contradiction. If the periods were bounded, then, the p_d being powers of 2 (Proposition 17.3(b)), all r_d would lie in V_P for $P = \max_d p_d$, and the infinitely many distinct v_d would lie in the finite set V_P^2 . $\square$

Remark 18.4 (provenance). Gravner–Griffeath [67], Proposition 3.3, proves that a left-permutative edge cellular automaton has no periodic solution, by exactly this argument: left permutativity gives every label a unique predecessor, a repetition on the ancestral line forces the whole ancestral line to carry the zero label, and that contradicts the root label 1. Their labels are our eventual cycles r_d , their root is $r_0 \equiv 1$, and their unique predecessor is our π . Rule 30 is named there as an instance. Jen [13], Proposition 3, is the related statement. The present section is therefore a translation into the diagonal frame, not a new theorem; it is kept because Proposition 18.6 and Theorem 18.9 are stated in that frame.

Remark 18.5. The theorem says nothing about how fast the periods grow. EXACT: the running maximum of p_d first reaches 2, 4, 8, 16, 32 at $d = 3, 8, 29, 400, 87\,867$ — these depths, and a sixth at “2,107,985,255 or more”, are already tabulated in [88, p. 871], and the per-diagonal periods and preperiods are OEIS A363345 and A363346; Brunnbauer’s independently computed table of 410 pairs agrees with ours on every period, with his offset equal to $\tau_d - \lfloor d/2 \rfloor$ throughout, and $r_d \equiv 0$ exactly at $d \in \{2, 7, 28, 399, 53\,207, 58\,286, 87\,866\}$ for $d < 90\,000$ (E1 in Section 17). Rowland’s parity law [66] (the period doubles across a frozen diagonal iff the cycle two diagonals to the left has an odd number of ones) is Proposition 17.3(d) and was verified at all seven frozen diagonals. For $P \leq 16$ the forward tree of $(0, 1)$ under π^{-1} was enumerated exactly: $|F_1| = 3$, $|F_2| = 13$, $|F_4| = 97$, $|F_8| = 3065$, $|F_{16}| = 34\,541\,081$ vertices; the seed’s path exits F_{16} at $d = 87\,866$ along the shortest maximal branch (EXACT; at $P = 32$ it is not the shortest, Remark 18.15). EXACT (ledger 10.231d I, 10.232d B, 10.243d; the companion note [103]; public earlier in [104, 105], as stated below): the label recursion of Lemma 18.2, run forward in C with no simulation of the rule, gives the eighth frozen diagonal of the seed at $d^* = 1,420,878,968$, a fork of even weight (the label r_{d^*-1} has period 32 and weight 20), so its two candidate successors are different walls. On the branch whose label begins `6cbda353` the running maximum reaches 64 at exactly $d = 2,107,985,255$, Wolfram’s number to the digit; on the other branch (`93425cac`) the period does not double below 3.34×10^9 , the next frozen diagonal being 3,340,408,059. This is consistent with reading “or more” as the fork; why it was written is not recorded. These integers are not new here. The eighth frozen diagonal and the first period-64 onsets on its two branches (2,107,985,255 and, on the other side, 9,958,700,505) were, to our knowledge, first made public on 8 September 2026 in two independent code repositories: [104] (commit `94833d2`) and [105] (commit `04f30f6`, from a run of about 30 August 2026). The first gives the frozen diagonal 1,420,878,968 with its two complementary continuations, the onset 2,107,985,255 on one of them, which it matches to [88, p. 871], and the frozen diagonals 3,340,408,059 and 4,989,445,007 on the other. The second gives the branch point 1,420,878,969 with the continuations `93425cac`

and 6cbda353, both onsets, and every branch up to 1.2×10^{10} . We obtained the same integers independently on 24 September 2026 (ledger 10.160c2) and learned of these repositories on 1 October 2026 (ledger 10.165c2). We found the numbers in no paper, preprint or OEIS entry. Neither repository determines the seed's branch: [105] records it as unresolved, and [104] infers it from the match with [88] and lists it as not checked. Beyond the doubling, branch 6cbda353 has no frozen diagonal up to 10^{12} , so its period-64 regime lasts at least to 10^{12} : EXACT on the second machine's arithmetic, with 13 of the run's 8 492 steps of 10^8 diagonals recomputed by two independently written programs and the remaining 8 479 resting on the second machine alone (ledger 10.243d; a step-by-step recheck of the whole log is running). Which branch the seed itself takes is the subject of Remark 18.8.

18.1 The left half as a driven machine, and the universal wall

The cells $x \leq -1$ depend only on cells $x \leq 0$, so the left half of any Rule 30 evolution is Rule 30 on the half-line $x \leq -1$ driven by column 0 as a boundary input, from whatever initial left state it has. Call this the *left machine*; its boundary enters through the OR at $x = -1$ and its output returns to the column through the XOR. Symmetrically the right half is a machine driven by the same column, receiving it through the XOR and returning through the OR. The column is $c_{t+1} = L_t + (c_t \vee R_t)$ with L, R the two outputs. Everything in this Part is a statement about the left machine, and Lemma 18.2 and Theorem 18.3 use nothing about the boundary: aligning diagonals to the trajectory of the leftmost 1 (which moves one cell left per step and is all ones, since $001 \mapsto 1$ and $000 \mapsto 0$), $D_0 \equiv 1$ and $D_{-1} \equiv 0$ hold for every left machine, so its background pairs form a backward orbit of π from $(0, 1)$ and its diagonal periods are unbounded.

Proposition 18.6 (universal background; KNOWN in a stronger form, Gravner–Griffeath [67, §8]). *For a power of two P let F_P be the tree of backward orbits of π from $(0, 1)$ inside V_P^2 . The left machine's background pairs v_d , in edge-aligned indexing, follow a maximal path of F_P for each period regime p until the period doubles. The enumerations $|F_1| = 3$, $|F_2| = 13$, $|F_4| = 97$, $|F_8| = 3065$ (heights 2, 7, 28, 399) show that for $P \leq 8$ the maximal path of F_P is unique up to time translation. Hence, for every boundary sequence and every finite initial left state, the periodic background of the left half agrees with the seed's on every diagonal of edge-aligned index below 400, up to a single time translation, with the same periods and the same doubling positions 3, 8, 29, 400.*

Proof. Lemma 18.2 holds for every left machine after edge alignment, so (v_d) is a backward orbit of π from $(0, 1)$; while the periods stay $\leq P$ it lies in F_P , and by Theorem 18.3 it cannot terminate. Within each regime it is therefore a path of that regime's tree that the orbit never leaves early; that it is *maximal* in the tree needs one more step — the sibling of the orbit's next element, which differs from it by the solution e of $e(0) = 1$, $e(t+1) = (1 + b(t))e(t)$ (both preimages of (a, b) under π obey the same first-order recursion), must not lie in V_P^2 — and that holds whenever $b = r_{d+1} \not\equiv 0$, since then e is finitely supported and nonzero, hence not periodic; at the frozen diagonals, where $b \equiv 0$, we do not check it here: there the statement is KNOWN from Gravner and Griffeath's label tree to $d = 53\,207$ [67, §8] (Remark 18.7) and EXACT to 12 000 by Theorem 18.9, and nothing below rests on more (ledger 10.242d A7). Uniqueness of the maximal path of F_P for $P \leq 8$ is the finite enumeration, and the time translation is the freedom in choosing the phase of a periodic pattern. $\square$

Remark 18.7 (provenance, and the upgrade it supplies). This is KNOWN and the published statement is stronger than the one proved above. Gravner and Griffeath [67, §8], writing about this rule, record that “the first genuine branching occurs for period 16 at generation 53,207” and conclude that “no matter what the seed, there is only one limit cycle on $[0, 53207]$ ”, adding that “all seeds have only 16 distinct limit cycles on the leftmost 65 million cells”; Rowland [66, §5] states the same reduction, that unless the previous column is eventually white the period of a column is independent of the seed. Both cover every seed to $d = 53\,207$, where the proposition above covers $d < 400$ with an enumeration and the accompanying computation reaches 12 000.

Their argument also removes the enumeration from ours. They prove that at a period doubling “the two resulting successors are each other’s rotations by π ” and that “the only label repetition is among the two siblings at a period doubling branching”: their label tree at period π is our F_P , so uniqueness of the maximal path up to time translation holds for *every* P , not merely $P \leq 8$, and the finite enumerations $|F_1| = 3, \dots, |F_8| = 3065$ become illustrations rather than the proof. Their $\pi = 16$ tree, 2,159,030 labels over 894,234 generations, matches our $|F_{16}| = 34\,541\,081$ once labels are counted with their 16 phases, the small deficit being the short-period labels. One bookkeeping note in the other direction: their list of death generations reads 2, 8, 28, 399 where ours reads 2, 7, 28, 399, and 7 is correct here, confirmed four independent ways.

Remark 18.8 (the branch rule, the phase congruence, and the branch at the eighth frozen diagonal; PROVED, EXACT and MEASURED parts as marked). Session d of the ledger (10.231d I, 10.237d, as corrected by 10.242d A) makes the mechanism behind Conjecture 18.16 explicit. (a) *The branch rule* (PROVED; ledger 10.231d I(ii)). Let d be a frozen diagonal, so $D_d(t) = 0$ for $t \geq \tau_d$ and $D_d(\tau_d - 1) = 1$, and let c^0 be the candidate label vanishing at row 0. The frame recursion gives $D_{d+1}(\tau_d) = D_{d-1}(\tau_d - 1) \oplus 1$ and, for $t \geq \tau_d$, $D_{d+1}(t + 1) = D_{d-1}(t) \oplus D_{d+1}(t)$, which both candidates satisfy; writing $D_{d-1} = r_{d-1} + \Delta_{d-1}$, the defect Δ_{d+1} is eventually the constant $1 + c^0(\tau_d - 1) + \sum_{t \geq \tau_d - 1} \Delta_{d-1}(t)$. Hence $M_{d+1} = M_d$ (Theorem 17.10(d) again), the label taken is the candidate with $r_{d+1}(\tau_d - 1) = 1 + (\text{parity of } \Delta_{d-1} \text{ on rows } \geq \tau_d - 1)$, and nothing below row $\tau_d - 1$ has any part in the branch: it is decided by the phase of the front against the wall and by the tail of one defect. Checked cell by cell at 399, 53 207, 58 286 and 87 866 (EXACT). (b) *The phase congruence* (PROVED with an EXACT step; ledger 10.231d I(iv’), 10.242d A2). Let a left machine whose front stays left of its boundary column have, beyond some diagonal, the seed’s wall up to a translation s_w and the seed’s front up to a translation s_b . Then $s_b \equiv s_w$ modulo the period on every stretch containing 20 000–87 866 or 90 000–800 000: the landing rule of Theorem 17.10(c), transported to the machine, says that the seed’s wall obeys the landing rule shifted by $\delta = s_b - s_w$ at every jump diagonal, and the census of all 37 086 jump diagonals of the period-16 regime and all 385 784 of the period-32 regime finds the shifted rule satisfied for $\delta \equiv 0$ only (the best other residue holds at 26.1% of them). The six left machines of the discussion following Theorem 18.9 confirm it in all twelve cases measured. With (a) this reduces the frozen-diagonal half of Conjecture 18.16 to the universality of the front. That universality fails for some finite starting words: the starting state of [106] takes the other branch after 53 207 (Conjecture 18.16), so its front is not the seed’s up to translation there. For the boundaries and words measured it holds (MEASURED). (c) *The advance criterion* (PROVED; ledger 10.237d A). At a frozen diagonal d with $m = M_{d-1}$ the twin labels $r_{d-1} = r_{d-2} =: r$ make the frame recursion read $D_d(t + 1) = r(t) \oplus (r(t) \vee D_d(t))$ for $t \geq m + 1$, so the front advances at d exactly when $D_d(m + 1) = 1$, that is, when the frozen diagonal is black on the row after the previous front, and then $M_d = \min\{t \geq m + 1 : r(t) = 1\}$. (d) *The*

*seed's branch at d^** (ledger 10.237d D, 10.242d A8–A10; [103, §6]; the fork itself is in [104, 105], which leave the seed's branch undetermined). Given the front's path over the three diagonals before $d^* = 1,420,878,968$ (rows $M_{d^*-3}, M_{d^*-2}, M_{d^*-1} \equiv 29, 31, 31 \pmod{32}$): a move of two at $d^* - 2$, a stall at $d^* - 1$, five label cells force the branch **6cbda353**, the one on which the period doubles at Wolfram's number (PROVED given the path). The path itself is OPEN: it is shared by all eight coin-driven edge machines whose front histories were recorded across the fork, and all twenty-eight such machines take that branch (MEASURED). These are model verdicts and are not calibrated evidence: checked against the exact simulation of the seed on $[4 \times 10^6, 4 \times 10^7]$, the same coin-driven machine with windows up to $W = 256$ was unanimous and wrong at $d = 20\,238\,347$, from 50 000 and from 150 000 diagonals back (ledger 10.164c2). A census over every front history consistent with the landing rule finds both verdicts at every frozen diagonal checked, so the wall alone decides at none of them (EXACT); the seed's own advances at 28 and 58 286 are forced by the labels at no depth up to 32 and are decided by the transient. Every branch of the seed's wall is therefore a finite computation from the labels and the front's phase history, the phase history being the one measured input.

Theorem 18.9 (EXACT). *With the left machine simulated for 20 000 rows and diagonals of edge-aligned index in $[20, 12\,000)$, all 11 980 locked diagonals carry the seed's background pattern up to one global time translation per run, with identical periods and doubling positions, for each of: the seed column with a random initial left state of width 40 and of width 100; a random-coin boundary with empty state, a second coin, and a coin with a random state of width 100; the boundaries $(01)^\infty$ and $(0011)^\infty$ with empty state. The translations are 5, 9, 13, 0, 6, 8, 5 $\pmod{16}$; the $(01)^\infty$ wall is the seed's wall translated by half a period. The range covers the whole period-8 regime and the period-16 regime to 12 000 (it ends at 53 207). Scripts `s3_leftdrive5.py`, `s3_leftdrive7.py`.*

Two further facts from the same runs (EXACT and MEASURED): the lock slope τ_d/d of the left machine takes the same value for every boundary tried in those runs, periodic or random — the value itself, and why the data give no bound on its limit, are in Remark 17.24; and two different initial left states under the same boundary disagree at $x = -1$ on half of all rows forever, so the left machine never forgets its initial state. The slope is not the same for *every* boundary. Driven from the empty left state by the background column $b_B(t) = r_t(t)$, the left machine is the background field $r_{x+t}(t)$ itself, which obeys the rule by Lemma 17.6(b): it has no transient and its front stays at the boundary, with $\tau_d - d \in [-15, 0]$ off the frozen diagonals and $M_d/d = 0.99999$ at $d = 2 \times 10^5$ (EXACT to 2×10^5), so its lock slope is 1. The front's slope is therefore claimed to be a property of the machine, and not of the column, only for boundaries that differ from b_B at a positive rate; every such boundary tried ignited a front with the seed's slope (MEASURED), and that every boundary which ignites a front has that slope is OPEN. For the ignited machines the agreement is stronger than equal slopes: six left machines, driven by $(01)^\infty$, $(0011)^\infty$, a coin and $b \equiv 1$, and by the seed column from random initial left states of widths 40 and 100, have the seed's settling times, after edge alignment, up to one constant translation per machine for every $d > 4\,484$ (EXACT to 2×10^5 ; in a run to 8×10^5 with the seed column for $t < 5\,000$ and b_B afterwards, the translation was reset in episodes where the boundary's influence reached the front, each time by a multiple of the period 32). Consequently the wall is a fixed universal object that the column cannot mark, and every trace of the boundary and of the initial state lives in the transient Δ .

Remark 18.10 (the edge automaton: rule, wall and one input bit; PROVED, MEASURED and EXACT parts as marked). The frame recursion $D_{d+1}(t+1) = D_{d-1}(t) \oplus (D_d(t) \vee D_{d+1}(t))$ shows

that diagonals $d-1$ and d on rows $[M_d-W, M_d+65]$ together with the single cell $D_{d+1}(M_d-W)$ determine diagonal $d+1$ on those rows, and since $M_{d+1} \leq M_d+p_d \leq M_d+64$ by Theorem 17.10(c) they determine M_{d+1} . Hence the front is generated by the wall and exactly one input bit per diagonal, the cell W rows below the front on the new diagonal, for any $W \geq 0$ (PROVED; ledger 10.231d A; run with the true bits the machine reproduces the seed's M_d with zero deviations). With that bit a fair coin — and equally with any biased coin or a constant — the machine reproduces every proved law and every registered frequency of the seed's edge (the near-laws are the exception, below): the move probability 0.5432, the mean move 2.432, the share of cycles with two or more waits 0.1082 and the tight-move fraction 0.3544 change by at most 2×10^{-4} as the input law runs from constant 0 to constant 1 (MEASURED; ledger 10.231d H, 10.233d D). Its speed on the seed's period-32 wall is 1.321294 ± 0.000019 over 2.84×10^{10} diagonals (twenty coins) and 1.321438 ± 0.000007 on the period-64 wall beyond the doubling; within a block of 10^8 diagonals twenty coins agree to ± 0.0001 while the block means have standard deviation 0.00013, seven times the coins' share, so the speed of a stretch of wall is a property of that stretch, the coin only realising it (MEASURED; ledger 10.233d F, 10.242d A16). These are numbers of the coin model on the wall and not a limit of the seed's own front, whose precision is what Remark 17.24 says it is. A SAT encoding of a window two rows below the front, with the frame recursion, the definition of the front and Theorem 17.10(c) as its only constraints, proves 82 of the 86 deterministic two-event contexts and 278 of the 279 three-event contexts of the edge's event table (EXACT; ledger 10.232d C); the remaining patterns are broken by the coin machine at 0.15–1.2% per case, but 142 of 147 exceptions fall while its front is off the seed's 32-lattice, in states the true dynamics never visits, so they are laws of the universal front rather than of rule and wall (EXACT, with the mechanism; ledger 10.233d A).

Proposition 18.11 (gaps between eventually-zero diagonals; PROVED modulo the finite enumerations named in the proof, which are EXACT). *Let $g \geq 1$ and suppose $r_d \equiv 0$ and $r_{d+g} \equiv 0$. Then $g \notin \{1, 2, 3, 4, 6, 7, 9, 10, 11, 13, 15, 17, 18, 19, 20, 22\}$ for any d and any periods; for $g = 5$ the only solution is $d = 2$ and for $g = 21$ the only solution is $d = 7$. Hence the only pairs of eventually-zero diagonals at distance at most 22 are $\{2, 7\}$ and $\{7, 28\}$, with the four distances 8, 12, 14, 16 settled only for joint period at most 16.*

Proof sketch. With $r_d \equiv 0$ the background equation at index $d+2$ becomes a closed affine recursion, and the equations at $d+2, \dots, d+g$ together with $r_{d+g} \equiv 0$ force a deterministic map on $\mathbb{F}_2^{g-1}$. Its recurrent subgraph is functional for every odd $g \leq 21$, so the periodic solutions form a complete finite list, which is then cut by three independent facts: Lemma 17.6(d); that a joint period must be a power of two, which is what removes $g = 11, 15, 17, 19$, whose surviving cycles have lengths 11, 20, 12, 5; and the π -descent of Lemma 18.2, which must reach $v_0 = (0, 1)$. Two independent calibrations agree: brute force over all $(x, 0)$ in V_L^2 for $L \leq 16$ and $g \leq 40$ returns exactly the pairs at distances 5, 21 and 26, namely $d = 2$, $d = 7$ and $d = 2$, with no false positive or negative, and the state machine at $g = 21$ independently returns $d = 7$. $\square$

Remark 18.12 (the orbit is a necessary ingredient). Gap 3 falls to the algebra alone: the only consistent pairs are $(0, 0)$ and $(1, 1)$ and Lemma 17.6(d) kills both. Gap 7 does not. There a family survives every background equation, namely $r_{d+2} \equiv 1$, $r_{d+4} = (10)^\infty$, $r_{d+5} = r_{d+6} = (0110)^\infty$ and $r_{d-1} = (1100)^\infty$, arising from the unique 4-cycle of an eight-state map; it dies only because its π -descent falls into a 28-cycle instead of reaching v_0 . So the background equations alone do not suffice and the backward orbit is needed. The surviving cycle length is non-dyadic

exactly at $g = 11, 15, 17, 19$ and dyadic at $g = 5, 7, 13, 21$; a lemma forcing that dichotomy would settle every g at once, and is OPEN.

Remark 18.13 (the fork law; twin law PROVED, law MEASURED, rate conditional). A frozen diagonal occurs exactly when the two labels before it coincide, $r_{d-1} = r_d$: with $a = b$ the recursion of Lemma 18.2, $c(t+1) = a(t) \oplus b(t) \oplus (\neg b(t) \wedge c(t))$, reads $c(t+1) = \neg b(t) \wedge c(t)$, and c dies at the first black cell of b (PROVED; ledger 10.235d A). The wall is thus the orbit of a deterministic map on pairs of P -bit patterns, 4^P states of which the 2^P twin pairs are the forks; doubling the period squares the state space and only doubles the fork set, which is the shape of the gaps 5, 21, 371, $\approx 3 \times 10^4$, $\approx 10^9$ between forks at periods 2, 4, 8, 16, 32. Measured on the seed (ledger 10.235d B, D): the Hamming distance between consecutive labels over one period is binomial($P, \frac{1}{2}$) to statistical error on all 1,420,788,969 pairs from 90 000 to the fork ($\chi^2 = 29.7$ on 31 classes, every class within two standard deviations, the one fork being the distance-0 count against 0.33 expected) and on 10^{10} pairs past the doubling at period 64, down to distance 8 and up to distance 56 on the two tails ($\chi^2 = 51.7$ on 47 classes) (MEASURED). A fork is the distance-0 class of this law. *If* the law holds down to distance 0 — the single fork at period 32 is the only direct evidence — then forks come once per 2^P diagonals, 1.8×10^{19} at period 64, and, on branch 6cbda353, the next doubling of the edge lies near 10^{19} – 10^{20} (conditional; ledger 10.242d A11). The law is the wall’s own pseudo-randomness statement, of the same kind as P2 for the column; what is proved is the mechanism and Proposition 18.11.

Conjecture 18.14 (M; REFUTED in print, see the remark following). *For every P the left machine’s background follows the shortest maximal path of F_P ; equivalently, the background of the left half of Rule 30 is unique up to time translation for every boundary and every finite initial state. Verified through $d = 12\,000$ as above, and at the branch $d = 53\,208$ for 58 initial words (Conjecture 18.16).*

Remark 18.15 (Conjecture M fails). Two published numbers refute it. Wolfram [88, p. 871] records the doubling depths 3, 8, 29, 400, 87 867 and then 64 at “2,107,985,255 or more” (read as a fork in [103]; the fork and the onset on one branch were public earlier, in [104, 105]); Gravner and Griffeath [67, §8], continuing the sixteen period-32 succession lines, report that “the earliest stopping event was a death at generation 65,154,361”. The seed’s line is one of those sixteen; it reaches the fork at 1,420,878,968 still at period 32, and on branch 6cbda353 its period-32 regime ends exactly at 2,107,985,254, after 2.1×10^9 diagonals, with no further frozen diagonal before 10^{12} , while on branch 93425cac it does not end below 3.34×10^9 (EXACT, ledger 10.232d B, 10.242d A11, 10.243d; the fork and both branch statements, though not the range to 10^{12} , are also in [104, 105]); either way it is not the shortest maximal path of F_{32} : the coincidence observed at $P = 16$, where the seed exits along the shortest branch, does not persist. What survives is the weaker and now fully attributed statement of Proposition 18.6.

Conjecture 18.16 (left-half universality; REFUTED by a finite starting word, see the end of the statement). *For every finite initial word, the eventual cycles of the left diagonals, the frozen diagonals and the doubling positions coincide with the seed’s, and the left half of the evolution is the seed’s left half up to a time delay. Verified for 58 words (lengths 1–40, 38 of them random) through the branch at $d = 53\,208$; Rowland’s “other possibility” at that branch is not realised by any of them. The author of a public gist of 21 August 2026 [106] reports the same for more than 2^{17} random starting states. The gist gives a constructed starting state of 53 209 bits that, its author reports, does realise the other possibility (the gist numbers the diagonal 53 209). We checked that state (ledger 10.235). Evolved under Rule 30 from a single row, its left diagonals*

agree with the seed's up to one translation of rows at every diagonal from 52 800 to 53 199, both have the frozen diagonal 53 207, and from 53 208 on the labels differ, with no translation of rows matching them on [53 300, 53 500] (EXACT). So the conjecture is false as stated: a finite starting word can take the other branch at the first genuine fork. It holds for the 58 words above and the measured boundaries; which finite words take the seed's branch is OPEN.

19 Aperiodic traces inside the ordered region

Kopra [65] places the two-column theorem in a general frame: a cellular automaton is *left expansive* with dimensions (h, d, w) if the contents of a rectangle of those dimensions determine the cell to its left, and *rapidly left expansive* if in addition it is left spreading with speed s and $s < 1/h$. An (m, n) left-permutive rule is left expansive with dimensions $(0, 1, m + n)$, and an elementary rule is left spreading exactly when $001 \mapsto 1$, with speed 1. Rule 30 is therefore rapidly left expansive with $w = 2$, which gives the two-column theorem, and the reason the frame stops there is visible in the inequality: $s = 1$ forces $h = 0$, and at $h = 0$ the width is $m + n = 2$. No choice of parameters reaches $w = 1$, which is Problem 1.

The gap can be stated more sharply than that, and doing so isolates what Problem 1 asserts. The width-one trace is always a factor of the width-two trace, by forgetting the second column. For the 3/2 automaton — Kopra's motivating example, and the one other system where a width-one trace has been proved aperiodic — that factor map is a *conjugacy*, because there one column determines the column to its left [25, Prop. 2.7], so all trace widths carry the same information and aperiodicity descends from width two to width one at no cost. For Rule 30 the same map is a proper factor map. Problem 1 is exactly the assertion that this particular proper factor map does not collapse periodicity. Whether Rule 30 has the property that makes the other case easy is a question we can settle negatively: there is no local map of any support radius from 4 to 10 recovering the column to the left of a single column, at any depth up to 20, so no inverse of radius ≤ 10 exists (EXACT); a conjugacy would need one of some finite radius, and this route to Problem 1 is closed as far as computed.

Width 1 is not hopeless in every direction, though. Our diagonals are width-one traces along slope 1 and they *are* eventually periodic. So eventual periodicity of a width-one trace depends on its slope, and it is natural to ask where the transition is. For a trace along slope a , meaning the cells $x = \lfloor -at \rfloor + \text{const}$, the cell at time t lies on diagonal $d = (1 - a)t + \text{const}$, so slope 1 is exactly the case that stays on one diagonal and every $a < 1$ drifts across infinitely many. The ordered region lies between the light-cone edge, which advances at speed 1, and the front, which advances at about 0.2427, so a trace with $0.2427 < a < 1$ eventually lies *inside* the locked wall.

Measured. Traces along slope a from offset 400, tested to 2×10^5 rows for any period up to 4096: periodic only at $a = 1$ (period 16, preperiod 498); no period at $a = \frac{3}{4}, \frac{1}{2}, \frac{1}{3}, \frac{3}{10}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \frac{1}{8}, 0$. No critical slope at the front speed is visible: slopes well inside the ordered region show no period up to 4096 in 2×10^5 rows (MEASURED).

The slope- $\frac{1}{2}$ case deserves to be stated on its own, because it relocates the difficulty of Problem 1.

Proposition 19.1 (a coin-like trace made of background, with no period up to 20 000; EXACT). *Take the width-one trace of slope $\frac{1}{2}$ from offset 400. Of its first 1.5×10^5 cells, 99.01% are locked, that is, they lie at or beyond the preperiod of the diagonal carrying them, and every diagonal it meets in that range has period exactly 16. On the locked part the trace is reproduced*

by the background alone, $r_d(t) = R_d[(t - \tau_d) \bmod 16]$, in 118 049 cells with no disagreement. Nevertheless it has no period up to 20 000 over a window of 1.14×10^5 rows, and its density of ones is 0.50063.

So the ordered region is not the simple part of the picture. It contains a width-one trace that shows no period up to 20 000 and is coin-like to every test we can apply, although every diagonal it is built from has period 16 and the whole background is, by the results above, seven branch bits together with a deterministic map. The aperiodicity cannot come from the chaotic interior, since the interior contributes 1% of the cells and none after the first few thousand rows; it comes from the phase relationship *between* consecutive diagonals.

That relationship is exactly what the backward orbit controls, which is what makes the following a well-posed problem rather than a restatement of Problem 1. If the slope- $\frac{1}{2}$ trace had period $p = 2m$, then for all large t

$$R_{d(t)+m}[(t + 2m) \bmod 16] = R_{d(t)}[t \bmod 16], \quad d(t) = 400 + \lceil t/2 \rceil,$$

a condition on the sequence $d \mapsto R_d$ alone. The full patterns never coincide: at $m = 1000$ there are 0 coincidences $R_{d+m} = R_d$ among 58 433 diagonals, as Theorem 18.3 requires, since a repeated pattern pair cannot occur. But the trace needs agreement at only one phase per diagonal, and that behaves like a coin, 2486 agreements in 5000 at $m = 1000$.

Problem 19.2. *Prove that the slope- $\frac{1}{2}$ trace of Rule 30 from the single seed is not eventually periodic.*

Unlike Problem 1 this asks for aperiodicity of a sequence every term of which is given by an explicit finite description, and the obstruction is a single-phase coincidence condition on an orbit we can enumerate. We do not know whether it is easier; we record it because it is the only width-one aperiodicity question about Rule 30 we have found in which all of the data is explicit. The background column b_B of the next remark is its slope-one analogue, explicit in the same sense, with no period $\leq 10^7$ found on the last 10^8 of its 10^9 terms (EXACT; ledger 10.233d I).

Remark 19.3 (the background column to 10^9 ; the column EXACT, the tests MEASURED). The whole background, read on the diagonal, is the column $b_B(d) = r_d(d)$: the centre column the stripes would make if the wedge reached it. The label recursion of Lemma 18.2, run forward in C, writes it directly, 10^9 bits in 35 seconds, exact to the fork at 1,420,878,968, on branch 6cbda353 exact to the doubling at 2,107,985,254 (both also in [104, 105]), and known up to a global complement beyond it (ledger 10.233d I; it agrees with the frame's column on all 189 999 diagonals 90 001–279 999). On $d = 90\,001, \dots, 10^9$: density 0.500029, every 10^8 -block within 0.4999–0.5001; no period $p \leq 10^7$ on the last 10^8 bits; run lengths within 0.1% of $N/2^L$ for $L \leq 12$ and within 1.5% to $L = 18$; blocks of 8, 12 and 16 bits each within one standard deviation of a fair coin; the best guess of the next bit from the last $k \leq 16$ bits scores what a coin scores in sample; autocorrelation within 2.1 standard deviations at 26 lags with one exception, lag 64 at -6.2 standard deviations, a trace of the period-32 stripes two periods apart (MEASURED; ledger 10.233d I as corrected by 10.242d A3). Its agreement with the real centre column on rows 90 001–1 130 000 is 0.49973, chance (ledger 10.241d #9). A column with no chaos in its construction that every coin test but one (lag 64) fails to distinguish from a coin, and whose eventual periodicity is the same kind of question as Problem 1, with the same answer so far.

Part IV

The column as an object

Source: *collapse_section.tex* (2026-09-16), labels prefixed *co*; Section 21 is new.

20 Collapse pyramids over the centre column

Let $c = (c_0, c_1, \dots)$ be the centre column of Rule 30 from the single black cell. For an elementary rule R and $N \geq 0$ write the first $2N + 1$ bits as a row and apply R upward, each new cell being R of the three cells beneath it and no assumption being made outside the row. The row loses one cell at each end per step and closes after N steps to a single cell, the *apex*

$$a_N^R = (R^N(c_0c_1 \cdots c_{2N}))_0.$$

Equivalently a_N^R is the leftmost cell of R^N applied to the whole column; it is a function of $c_0, \dots, c_{2N}$ only. The apex sequence $(a_N^R)_{N \geq 0}$ is thus a derived sequence of the column, one bit per two column bits. This section records what these sequences are and what they can and cannot say about c .

20.1 Inheritance

Proposition 20.1 (downstream only; PROVED). *If c is eventually periodic then (a_N^R) is eventually periodic for every R . PROVED*

Proof. Let c have period p from index t_0 , and read each row from its left edge: $X_m[j] := (R^m(c))_{m+j}$, so that $X_{m+1}[j] = R(X_m[j], X_m[j+1], X_m[j+2])$ and the apex is $a_N^R = X_N[0]$. If X_m is p -periodic from relative position t_0 then so is X_{m+1} , and $X_0 = c$ is; hence every X_m is determined by its first $t_0 + p$ cells, a finite set of states, so $m \mapsto X_m$ is eventually periodic and so is $a_N^R = X_N[0]$. (An earlier version read the apex off the periodic part of $R^N(c)$; the apex lies in the transient block, which is why the row must be read from its edge.) $\square$

Hence no apex sequence can be used to prove periodicity of c , and any apex sequence that is provably not eventually periodic proves Problem 1. The converse inheritance fails: for the identity rule $a_N = c_N$, while for the constant rules a_N is constant.

20.2 Two exact apexes: the traffic rules

For a row $r = (r_0, \dots, r_{n-1})$ let $H(k) = \sum_{j < k} (2r_j - 1)$, $0 \leq k \leq n$, be its balance walk, so $H(k) \equiv k \pmod{2}$ and $r_k = [H(k+1) > H(k)]$. Rule 184 is $r'_k = (r_{k-1} \wedge \neg r_k) \vee (r_k \wedge r_{k+1})$.

Lemma 20.2 (height form of Rule 184; PROVED). *Let $r' = R_{184}(r)$ be the interior row $(r'_1, \dots, r'_{n-2})$ and let $H'(k) = \sum_{j=1}^k (2r'_j - 1)$, $0 \leq k \leq n-2$, be its balance walk. Then*

$$H'(k) = \max(H(k), H(k+2)) - C, \quad C = \max(H(0), H(2)) \in \{0, 2\},$$

for $0 \leq k \leq n-2$. PROVED

Proof. Both sides vanish at $k = 0$ and both are walks with steps ± 1 , so it suffices to compare increments. Write $G(k) = \max(H(k), H(k+2))$. Since $H(k+2) = H(k) + (2r_k - 1) + (2r_{k+1} - 1)$, the value of $G(k+1) - G(k)$ is determined by (r_k, r_{k+1}, r_{k+2}) , and checking the eight cases gives $G(k+1) - G(k) = +1$ exactly for $(r_k, r_{k+1}, r_{k+2}) \in \{100, 101, 011, 111\}$, which is exactly the set on which Rule 184 outputs 1, i.e. $r'_{k+1} = 1$. Hence $G(k+1) - G(k) = 2r'_{k+1} - 1 = H'(k+1) - H'(k)$. (Checked exactly on random rows of length 2001.) $\square$

Theorem 20.3 (record parity; PROVED). *Let $S_k = \sum_{j < k} (2c_j - 1)$ be the balance walk of the centre column and $M_N = \max_{0 \leq k \leq 2N+1} S_k$, $m_N = \min_{0 \leq k \leq 2N+1} S_k$. Then*

$$a_N^{184} = M_N \bmod 2, \quad a_N^{226} = 1 - (|m_N| \bmod 2). \quad \text{PROVED}$$

Proof. Iterating Lemma 20.2, after N steps $H_N(k) = \max_{0 \leq j \leq N} H(k+2j) - C_N$ for a constant C_N . The apex is the single remaining cell, whose value is $[H_N(1) > H_N(0)]$, i.e. whether the maximum of H over odd positions in $[0, 2N+1]$ exceeds the maximum over even positions. Since $H(k) \equiv k \pmod{2}$ the two maxima differ by exactly 1, and the odd one is larger exactly when the overall maximum M_N is odd. Rule 226 is the complement conjugate of Rule 184, $R_{226}(x) = 1 - R_{184}(1 - x)$ (equivalently its mirror image); complementing the row sends H to $-H$ and the maximum to the minimum. $\square$

Remark 20.4 (provenance). Rule 184 is the ultradiscrete Burgers equation [47], which is the theorem-bearing form of the statement that follows. The height-function (max-plus) representation of Rule 184 is standard in the traffic and ballistic annihilation literature (Belitsky–Ferrari [78]; Krug–Spohn [79]). Theorem 20.3 is its immediate corollary for the collapse geometry and is recorded here for the criterion below, not as a new result.

20.3 A two-sided criterion

Corollary 20.5 (two-sided criterion; PROVED). *If both apex sequences (a_N^{184}) and (a_N^{226}) change value infinitely often, then c is not eventually periodic.*

Proof. a_N^{184} changes only when M_N increases, a_N^{226} only when m_N decreases. If c is eventually periodic with period p and δ the net balance over one period, then for $\delta = 0$ the walk is bounded and both records are eventually constant, and for $\delta \neq 0$ the walk drifts, so one record is eventually constant. $\square$

The hypothesis is that the balance walk of c is unbounded above and below; it is neither implied by nor implies Problem 2 (a bounded walk has density $\frac{1}{2}$; a one-sided drift of order $\sqrt{k}$ has density $\frac{1}{2}$), and it is not implied by Problem 1. EXACTto $k = 200,001$: the walk's record high is 419, first reached at $k = 24,989$ (last new high at $N = 12,494$), and its record low -257 at $k = 172,735$ ($N = 86,367$); the Rule 184 apex has changed 196 times, the Rule 226 apex 127 times. Whether either changes again is OPEN, and a long freeze is ordinary behaviour at the $\sqrt{k}$ scale.

Proposition 20.6 (why the predictor measurements cannot be upgraded; PROVED). *Suppose the word frequencies $d(u) = \lim_N N^{-1} \#\{i < N : c_i \cdots c_{i+|u|-1} = u\}$ exist, and for $w \geq 1$ let $A_w = \sum_{|u|=w} \max(d(u0), d(u1))$ be the limiting accuracy of the best window- w predictor. Then*

$$A_w = \frac{1}{2} + \frac{1}{2} \sum_{|u|=w} |d(u0) - d(u1)|,$$

and consequently $A_w = \frac{1}{2}$ for a single $w \geq 1$ already forces $d(1) = \frac{1}{2}$, which is P2.

Proof. $\max(a, b) = \frac{1}{2}(a+b) + \frac{1}{2}|a-b|$ and $\sum_{|u|=w} (d(u0) + d(u1)) = 1$ give the identity. If the sum vanishes then $d(u0) = d(u1) = \frac{1}{2}d(u)$ for every u , so $d(1) = \sum_{|u|=w} d(u1) = \frac{1}{2} \sum_{|u|=w} d(u) = \frac{1}{2}$. The identity was checked against the direct computation of A_w on 2^{18} terms for $w = 1, \dots, 14$, agreeing to 10^{-12} at every w . $\square$

Remark 20.7 (what this says about the rest of this Part). The measurements of this Part say that no bounded-window predictor of the centre column beats a coin, and we have been careful to label them MEASURED. Proposition 20.6 explains why they must stay that way: proving the statement in its natural limiting form, at even one window length, would prove P2. So the gap between our measurements and a theorem here is not a gap in effort or in sample size — it is exactly Problem 2, and any programme that proposes to close it by computing harder is mistaken about what it is computing. Across all w the same statement is base-2 normality.

This also fixes the status of the predictor question in the other direction. The learning-theoretic machinery that looks as though it should apply does not: mistake bounds and the Littlestone dimension are min-max statements quantified *over* sequences — the dimension of the window- w class is 2^w , whatever sequence is presented — and regret bounds are relative to a comparator class, so neither says anything about one fixed sequence. The one branch of that literature which does face a single sequence, the finite-state predictability of Feder, Merhav and Gutman [102], delivers a universal algorithm and never a lower bound, a gap those authors state themselves.

20.4 What the pyramids cannot do

Every apex, and every cell of every collapse pyramid, is a function of a bounded window of the column. A single computation therefore bounds the whole family. Let P_w be the majority predictor of c_n from $(c_{n-w}, \dots, c_{n-1})$ fitted on the first 2^{19} bits and evaluated on the second 2^{19} . MEASURED (out of sample; EXACT as counts on that split): its accuracy lies in 0.4973–0.5011 for every $w \leq 22$ (standard error ≤ 0.002), with conditional empirical entropy 1.000 bit per symbol through $w = 14$. Consequently:

- no rule R , height k and offset d gives a^R or any edge cell of the pyramid a correlation with c_{2N+1+d} beyond the chance maximum of the number of tests (256 rules, 40 heights, 9 offsets: best in-sample 3.2σ ; the right-edge family: best 4.6σ in-sample, -0.1σ out of sample);
- the pyramid's own centre column matches c at no position and shift beyond chance (6.5×10^7 tests, best 5.2σ , expected 5.3σ);
- rules applied to the locked left wall at time t correlate with the causally reachable centre bits $c_{\lfloor 3t/4 \rfloor + d}$ at the chance level out of sample.

These are measurements, not theorems; the theorem-level statement is Proposition 20.1: the collapse pyramids are downstream of the column, and information does not flow back.

21 Where the information enters the column

Proposition 21.1 (parent-instance split; PROVED). *Let $\ell_t = u_{-1}^t$ and $u_t = u_1^t$ be the columns adjacent to the centre. Then:*

- (a) ℓ_t is a function of $c_0, \dots, c_{t-1}$ alone: the half-plane $x \leq -1$ evolves by Rule 30 with column 0 as its boundary input.
- (b) If $c_t = 1$ then $c_{t+1} = 1 + \ell_t$; if $c_t = 0$ then $c_{t+1} = \ell_t + u_t$.
- (c) $u_t = 1 + c_{t-1}$ exactly when $u_1^{t-1} \vee u_2^{t-1} = 1$.

Proof. (a) A cell at $x \leq -1$ and time t depends on cells at $x-1, x, x+1 \leq 0$ at time $t-1$; induct. (b) is the rule at $x = 0$. (c) is the rule at $x = 1$, where c_{t-1} enters by XOR. $\square$

So after every black centre cell the next bit is determined by the column's own past, through an unbounded-memory computation; after every white centre cell one bit enters from the right, and that bit is the complement of the previous centre bit three times in four (MEASURED: 0.7499 on 16384 rows). Parent instances of c_{t+1} are equidistributed over the eight triples (EXACT, counts 1953–2107 of 16383). A bounded-context model on the eight-colour parent sequence predicts the next colour at 0.37–0.44 against 0.125, entirely from (b) and (c), and the next centre bit at 0.48–0.50 (MEASURED, train/test halves). Combined with Theorem 3.1, P1 is equivalent to: *if c were eventually periodic, the bits u_t at the white times t would have to be eventually periodic*. Part I settles this family by family; the general case is OPEN.

Part V

Computational record, and Problem 3

22 Computational record

Each item names its script in the author's working folder (not all of them are public); the ledger section is in brackets.

The two-sided kill instrument (Exact, then Measured)

Fix the tail $w = 01$. Let a survivor at (N, T) be a configuration with $a(0, i) = 0$ for $i < -N$ whose centre follows w on rows $0..T-1$; $T_{\text{death}}(N)$ is the first T with no survivor and $f(N) = T_{\text{death}}(N) - N$. PROVED (one line): f bounded implies w is not a tail of c . EXACT: a survivor at (N, T) is an admissible interface whose reconstructed row 0 vanishes on $(N, T]$; minimal unsatisfiable cores use nearly every centre row but only the 7–8 deepest vacuum cells; the kill depth $m(T)$ (the least number of consecutive forced zeros that no admissible interface can carry at depth T) was computed to $T = 120$. MEASURED: $m(T) \approx 1.7-2 \log_2 T$; the record $f = 18$ from $N = 19$ to 120 broke to $f = 20$ at $N = 130, 140$. Locality test: forcing the centre only on rows $[t_0, T_{\text{death}})$ keeps the instance impossible only for $t_0 \leq 6$. The bounded- f route is not supported by the data: f has already increased once and no finite record can establish boundedness (MEASURED; whether f is bounded is OPEN), and the kill is global [10.47–10.50].

The left machine alone, for the tail 01 (Exact; ledger 10.96–10.106)

At a black centre cell the OR is already 1, so a 01 tail forces $u_{-1}^t = 1$ at every odd t : a condition on the left machine only. With column 0 alternating from $t = 0$, column -1 forced at odd times and free at even times, left-permutivity reconstructs the left half, and number-likeness with support N requires the row-0 cells $P_k = u_{-k}^0$ to vanish for $k > N$. The odd

equations $P_{2m+1} = 0$ fix the later free bits (each free bit enters its anti-diagonal by XOR), the even ones are checks, and the candidates are the $2^{\lfloor (N-1)/2 \rfloor + 1}$ initial free-bit strings. Exhaustive enumeration (validated against SAT at $N \leq 30$) gives the first depth with no survivor, $T_{\text{death}}(N) = 8, 18, 30, 32, 38, 44, 54, 66, 78, 88, 98$ for $N = 4, 8, 12, 16, 20, 24, 30, 32, 40, 44, 50$ (all even $N \leq 50$ computed; full table in ledger 10.106). The death depth is not an independent law but the unicity distance implied by the halving: against $N + 2n_{\text{free}}$ with $n_{\text{free}} = \lfloor (N-1)/2 \rfloor + 1$ the residuals lie in $[-8, +2]$ with mean -3.7 (over $16 \leq N \leq 50$; over all even $N \leq 50$ they lie in $[-8, +6]$ with mean -2.5) and vanish at $N = 16$ and $N = 44$, so quoting a ratio T_{death}/N as if it were a constant overstates the content (an outside proposal $T_{\text{death}} = N p/(p-z)$, exactly $2N$ for 01, is refuted: the ratio is not constant). Survivors halve at every check (the checks are exact coins conditional on all earlier ones, $m \leq 22$), and the longest-lived candidates form classes riding a stripe zone in which column -1 runs in phase with the centre, the $\dots 0101 \dots$ fixed point of Rule 30. Every such champion at $N \geq 8$ is right-inadmissible within 18 interface symbols, most by the forbidden factor 11: the right half's language removes exactly the striped candidates, which is why the two-sided instrument above dies with delay $\sim \log N$ while the left machine alone needs $\sim N$. We do not claim that the surviving family is polynomially small: the count of right-admissible completions has the same non-polynomial signature in the ensemble setting of [82, Table 1] as $|L_k|$ does here. PROVED alongside: $T_{\text{death}}(N) \geq T_{\text{death}}(N-2)$; an eventually periodic left output is never number-like; survival is invariant under the two-row shift; the row-0 polynomials satisfy $P_{k+2} = \sigma(P_k) + (P_{k+1} \vee P_k) + (S_k \vee S_{k-1})$ with σ the index shift and $S_j = P_{j+1} + (P_j \vee P_{j-1})$, an exact recursion whose state is the anti-diagonal pair; every check has odd weight. Two ingredients of this instrument are KNOWN and we claim neither: the forcing at a black centre cell, that u_{-1}^t is the complement of the next centre bit, is stated one paragraph after equation (4) of [82], repeated in [88, p. 1087] and proved as Proposition 3.4 of [84]; and the count of free bits, about $N/2$ for a balanced word, is Spencer's bound of $n-j-1$ coins for a word with j ones [84, §3.2], also given as $2^{n/2}$ in [85]. What is new here is the use of the row-0 equations as checks, the exhaustion of the candidate set and the measurement of a death depth; Meier and Staffelbach explicitly turned away from the left branch, since for key recovery the left restrictions are an obstacle rather than a tool, and Spencer proposed the left programme without developing it. The target, no candidate surviving forever, is OPEN. The obstruction is that every check is a coin conditional on everything computable, and it is sharpened by a reformulation that is already in print: left-permutivity makes the map from the row-0 cells at $x = -1, \dots, -T$ to the centre bits at $t = 1, \dots, T$ a bijection, which is in substance Meier and Staffelbach's completion backwards [82, eq. (4)] (KNOWN; the two-column triangle reconstruction is already in [87], credited there to C. and R. Feynman), and of which [91, Lemma 1] is the injectivity half. The reconstruction is therefore unique, there is no ensemble over which to equidistribute, and the question is properly: for the single string that bijection returns, is the density of ones bounded below? That is why no measure-level theorem can reach it.

The fibre over the alternating column (Exact)

$|L_k(01)|$ for $k = 1..44$: 2, 3, 5, 8, 12, 17, 25, 36, 50, 68, 91, 119, 156, 199, 251, 316, 393, 487, 596, 721, 875, 1054, 1255, 1493, 1780, 2111, 2483, 2904, 3378, 3908, 4502, 5153, 5875, 6664, 7541, 8534, 9649, 10876, 12231, 13730, 15384, 17196, 19197, 21389, extended by an independent encoding to $|L_{45..60}| = 23809, 26482, 29422, 32656, 36193, 40047, 44280, 48878, 53894, 59333, 65240, 71672, 78632, 86169, 94361, 103220$; the local exponent rises without a bend from 4.3 at $k = 30$ through 5 at $k = 50$ to 5.34 at $k = 60$, its slope in $\ln k$ itself rising. Out of

sample on $k = 45..60$, fitted on $k \in [K_0, 44]$ and with K_0 swept over 8, 12, 16, 20, 24 so that no conclusion rests on one window, and with every family given the same number of parameters, exactly two families are excluded and they are excluded by orders of magnitude: a free-degree polynomial Ck^d misses by 21.8–43% and the pure exponential $C\lambda^k$ by 84–402%. Nothing else is separated. The quasi-polynomial $\exp(c\ln^2 k)$ and the stretched exponentials $k^a \exp(ck^\gamma)$ for every γ tried in $[0.25, 0.70]$ all sit between 1.48% and 12.90%, and their *ranking* reverses with the window — $\gamma = 0.25$ is the best of them at $K_0 = 8$, $\gamma = 0.50$ at $K_0 = 12$, $\gamma = 0.70$ at $K_0 = 16, 20, 24$ — so an earlier phrasing that put the quasi-polynomial “at errors of 5% or more” and left “only a stretched exponential” surviving is withdrawn: it is true at $K_0 = 12$ (5.13%) and false at $K_0 = 8$, where the quasi-polynomial at 2.63% beats $\gamma = 0.50$ at 6.70%. Note also that the excluded pure exponential is excluded *as a fit*, which says nothing about h : the proved bounds are upper bounds — $h \leq 0.278$ from the counts (Proposition 9.3) and the sharper block-automaton bound $h \leq 0.132$ quoted below — and $h > 0$ is not excluded. The exponent is not determined, and we withdraw three stronger phrasings used earlier. It is not true that every pure $\exp(ck^\gamma)$ with $\gamma \geq 0.3$ is excluded: $\exp(5.91k^{0.28})$ predicts $k = 46..60$ to 1.31% with one parameter *fewer* than $k^{1.9} \exp(0.87\sqrt{k})$ at 1.38%, and beats it on the longer window, 0.60% against 1.08%. Nor, and this is the third withdrawal, does that comparison exclude $\gamma \geq 0.35$, because it is exactly a two-parameter form beating a three-parameter one; see the end of this remark. Nor is $k^{1.9} \exp(0.87\sqrt{k})$ stable across fitting windows: moving the left endpoint gives (1.10, 1.10) at 0.45% and (0.98, 1.14) at 0.25%, so the fit lives on a degenerate ridge and no value of c is pinned. What matters is that γ is exactly the quantity that decides between mechanisms — a partition mechanism requires $\gamma = \frac{1}{2}$ exactly, a hierarchical one requires $\log_\ell r = \frac{1}{2}$ — and by Cassaigne’s Theorem 3, which realises $\log p(n) \sim n^\alpha$ for *every* $\alpha \in (0, 1)$ with a uniformly recurrent word, no theorem will pin it for us. So γ is OPEN — and, correcting a narrower reading of $[0.25, 0.35]$ given earlier, with no interval pinned at all. That reading set a two-parameter pure $\exp(ck^\gamma)$ against a three-parameter $k^a \exp(ck^\gamma)$, and the exclusion does not survive equalising the count: at three parameters throughout, $k^{1.91} \exp(0.866\sqrt{k})$ predicts 46..60 from 12.45 to 1.19%, *better* than the pure $\gamma = 0.28$ form’s 1.60% on the same window, and the held-out optimum over a grid $\gamma \in [0.15, 0.85]$ drifts monotonically with the fit window, taking the values 0.150, 0.150, 0.250, 0.525, 0.700, 0.750 as the window ends at $k = 34, 38, 42, 46, 50, 54$, while the in-sample optimum on sliding windows of 20 visits both ends of that grid. An estimate that sweeps its whole grid as the window moves is not an estimate, so what the data supports is superpolynomial growth with γ unconstrained, and the exclusions that follow are the ones that do not go through γ . A cleaner exclusion is available than the follower-set census, which exhibited 3591 distinct follower sets: a monomial algebra whose basis language is *regular* has growth either polynomial or exponential, and no intermediate growth is possible (Ufnarovskii; stated in this form by [8]). So intermediate growth alone excludes regularity, hence soficity, with no bound on follower sets needed. One scope note, since the graph form of the same statement assumes a finite forbidden set and our language is not of finite type: it is the automata-algebra form, which needs only regularity, that applies. The inference is conditional on the growth being intermediate, which is MEASURED on $k \leq 60$ only; the stream of Remark 9.6 puts the asymptotic growth near $2^{0.07k}$, and if that holds the exclusions of sofic, finite-type and context-free languages here lapse, while the follower-set bound 3591 does not. In the same direction: a sofic language is regular, and the factor-counting function of a regular language is exactly a polynomial times an exponential on each congruence class for large n (Berstel; see [1, Thm. V.3] and [2]), so *superpolynomial together with subexponential* growth already excludes sofic, of finite type and

group shift, with no bound on follower sets needed. The context-free case is stronger than we had it: by Bridson and Gilman [5, Thm. 1.1] a context-free language has growth eventually at least r^n or else bounded, so superpolynomial growth excludes *all* context-free languages with no unambiguity hypothesis, which answers a question of Flajolet (see also [3]). Topological rank two is excluded as well [6, Cor. 6.6]. What is *not* excluded is the indexed, ET0L and EDT0L levels, and that is where the one published language with partition growth sits. One correction to how we have described $L(01)$: it is an *anchored* language, not the language of a subshift — at $k = 59$ some 24.4% of its words occur only at row 0 — so the absence of dead ends holds on the right only, while factoriality and submultiplicativity were verified at all sixty depths and all 1770 ordered pairs with $m + n \leq 60$. These exclusions are therefore conditional exactly on the growth law, and on nothing else (and the growth law is now in question: positive entropy is MEASURED on random right halves, Remark 9.6); block-automaton entropy bounds reach $h \leq \log_2 1.0958 = 0.132$ bits (proved in the ledger, not reproduced in this paper) [10.16, 10.31, 10.54, 10.105, 10.117]. PROVED by hand: constraint (A), no 11, gaps at most 4, factoriality, no dead ends. PROVED finite, on the realisability of constant gap words by temporally periodic half-planes: 1^∞ , 2^∞ , 4^∞ and $(14)^\infty$ are realised, at row periods 4, 6, 10 and 14 and spatial cycles 7, 84, 155, 728, each certified and re-verified by independent forward simulation, the period-14 cycle length 728 being pinned between two independent searches (a direct sweep finds no cycle of length ≤ 700 , a wide solution scan finds the repeat at exactly 728); 3^∞ , $(13)^\infty$, $(23)^\infty$, $(24)^\infty$, $(34)^\infty$, $(244)^\infty$, $(1134)^\infty$ and $(24^4)^\infty$ are not. Of the 3779 primitive necklaces of width ≤ 24 , four are unbounded, 3767 are proved bounded and eight are OPEN, the sharpest being $(2444)^\infty$ at row period 36. A conjectured row period 26 is REFUTED. Zero entropy and the growth law remain OPEN, and the literature now says three separate things about them. First, our shape is realised: Cassaigne [4, Thm. 1] exhibits an explicit binary word with $2^{\lceil \sqrt{n} \rceil} \leq p(n) \leq n^2 2^{\lceil \sqrt{n} \rceil}$, which is our form exactly, $\gamma = \frac{1}{2}$ with a positive polynomial prefactor, Mitchell [10] gives a second such subshift with the polynomial prefactor explicit, and Greenfeld, Moreira and Zelmanov [9] realise the entire subexponential range by subshifts, even strictly ergodic ones. So no general theorem forces subexponential growth and none forbids our shape; in particular there is no polynomial-or-exponential dichotomy to refute it.

Second, there *is* a usable sufficient condition, contrary to what we said in earlier notes. By Donoso, Durand, Maass and Petite [6, Cor. 6.5] an S -adic subshift whose alphabet growth is negligible against its word lengths — in particular one of finite topological rank — has zero entropy, and finite rank does not bound complexity, so it is compatible with any subexponential growth. That is a constructive target rather than a hope, and it is the route worth building. Two cautions attach to it: word entropy is vacuous for subexponential functions, and the monotone decrease of $\frac{1}{k} \log_2 |L_k|$ is not evidence for the limit being zero, since $3 \cdot 2^k$ decreases monotonically to 1; the bound $h \leq 0.278$ from the counts stands as an infimum, but no computation at finite depth can decide h .

Third, the mechanism. The partition heuristic for the $\exp(c\sqrt{k})$ form is half right. Structurally it holds: the ratio of $|L_k|$ to the number of domain-length multisets is 4.91 at $k = 60$ with increments +0.39, +0.16, +0.07, admissible orderings are nearly forced, only 3.6×10^{-6} of the combinatorially available ones occurring, and the multiset is a genuine partition with 5.56 distinct part sizes per word growing like $0.72\sqrt{k}$ against the random-partition value $0.780\sqrt{k}$. These parts are *domain lengths* in the configuration and not the run lengths of the interface word; the latter take at most the four values $\{1, 2, 3, 4\}$ at every depth to 60, so the two part sets in this paper are different objects and neither figure bounds the other. The magnitudes

confirm it: partitions of 60 into parts ≤ 4 number exactly 1906 against $|L_{60}| = 103\,220$, a factor 54.2, where the domain-length ratio quoted just above is 4.91. As an explanation it fails, and the decisive reason is not the one we first gave. The fitted constant sits on a degenerate ridge where every cell agrees to 1.3–1.7% and sparse-part models at $\gamma \approx 0.25$ fit equally well, so “ $c = 0.87$ against Meinardus’ 2.607” proves little on its own. What does refute it is the measured *positive density* of the part set: every slot of type other than 3 is realised contiguously to the space cap, which forces $\gamma = \frac{1}{2}$ *together with* a prefactor exponent at most $\frac{1}{4}$, and it is the pair that fails. Fitting $k^a \exp(c\sqrt{k})$ on $k \in [K_0, 44]$ and testing on 45..60, with a held at $\frac{1}{4}$, misses by 13.5–46% as K_0 runs over 8, 12, 16, 20, 24 (and by 16–55% with $a = 0$); releasing a brings the error down to 1.5–11.8% but only by demanding $a = 1.49, 1.89, 2.36, 2.89, 3.05$, an order of magnitude above the $\frac{1}{4}$ the density argument allows. So what is refuted is the *conjunction*. The value $\gamma = \frac{1}{2}$ on its own is not refuted — with a free prefactor it is the best-fitting exponent at the longest windows — and nothing in the withdrawal of the interval $[0.25, 0.35)$ above revives sorting, whose refutation runs through the bounded part set and the prefactor, not through γ . The structural half also undercuts the explanatory half, since a correspondence bijective up to a bounded factor cannot itself generate growth.

One line of reasoning we pursued has to be withdrawn. We argued that since free segmentation is never intermediate, the exponent must be forced by the ordering constraint, and that identifying the mechanism would pin γ . It will not, and the obstruction is a theorem rather than a gap. The universal realisation constructions — Bell and Zelmanov’s, and the subshift realisations of [9] — all run on a single mechanism, a gap threshold tending to infinity, and the shape $An^a \exp(cn^\gamma)$ satisfies their conditions for *every* $\gamma \in (0, 1)$; our own measured table satisfies them with room to spare, the largest ratio $|L_m|/|L_n|^2$ being 0.889 over all 595 pairs. So no theorem of the form “exponent γ requires mechanism X ” can exist. Nor are the exponents quantised: Bergman’s gap remains the only known gap in the space of growth functions, and sorting realises every exponent too — over a part set of density exponent r it gives $\gamma = r/(r+1)$ by Meinardus [98]. The consequence for us is that γ separates branching from everything else and nothing finer; the quantity that would discriminate sorting is the *density of the part set*, not the exponent. The sorting mechanism does have a published witness — Grigorchuk and Machi’s [7] language $\{ab^{i_1} \cdots ab^{i_k} : i_1 \leq \cdots \leq i_k\}$, whose growth is exactly Euler’s partition function — and several classes remain excluded by our own measurements: deterministic substitution, capped at $\Theta(n^2)$ by Pansiot [11], Toeplitz, and primitive random substitution. Non-primitive random substitution is not on that list, and the density half of that route is dead in any case: the frequency theory for random substitutions is primitive-only and almost-every-only, and a non-primitive subshift carries uncountably many ergodic measures, so even an exact substitution would not by itself give a density for our particular word. The mechanism is also partly settled: the switching law between domains is bounded-memory, a function of the outgoing type, the run length truncated at 8 and the incoming type, with every threshold at most 7, yet it is *not* a function of type, phase and type alone ($1^2 4$ is realised while $c 1^2 4$ is realised for no letter c), and $L(01)$ is not left-extendable ($100101 \in L_6$ but $0100101 \notin L_7$). The domain periods do *not* grow with k ; what grows is the number of domains needed under any fixed period cap. So neither branch of the earlier dichotomy applies, and since a regular language grows either polynomially or exponentially and never in between, the measured intermediate growth is inconsistent with any finite state space — conditional on that measurement, which is why the growth law itself is still OPEN.

Every bounded-window predictor of the column (Measured)

See Section 21 (“What the pyramids cannot do”), which carries the numbers: accuracy 0.4973–0.5011 for every $w \leq 22$ on the held-out half, entropy 1.000 bit through $w = 14$, and the same for all 256 rules applied to the column and to the locked left wall [10.84–10.87].

Other coin readings (Measured)

Density 0.4996 to 8000 rows and 0.5006 to 2^{20} ; firing rate of the OR gate 0.750; subword complexity 2^n to $n = 16$; partial-sum Hurst exponent 0.506; exponential sums coin-like at all rational frequencies tested; lag-1 autocorrelation along the diagonals through the centre -0.504 (the wire) and 0 for the centre column itself; the Rule 30/Rule 86 overlay has four colours at density $\frac{1}{4}$ each with an isotropic agreement field [10.73, 10.77, 10.78]. Smooth (multilinear, spin and wave) extensions of the rule collapse to $\frac{1}{2}$ [10.70]. The single-column Hurst figure above is noisy, because fitting a power law to one realisation of a random walk is dominated by wherever that path sits near zero; averaging across columns fixes it. Taking the 201 columns $x \in [-100, 100]$ to $T = 2^{20}$ (column 0 validated bit-for-bit against the stored centre column on all 1 048 576 rows) and writing $D_x(T)$ for the excess of blacks over whites, corrected for the $|x|$ rows in which the column lies outside the light cone, the root-mean-square of D across columns tracks the fair-coin value $\sqrt{T}/2$ at every scale: the ratio reads 1.105, 1.034, 0.964, 0.888, 0.954, 0.927, 0.963, 0.971, 1.039, 1.004, 0.939 at $T = 2^{10}, \dots, 2^{20}$, and the fitted exponent is 0.4932 over the whole range and 0.4974 over the last five checkpoints. The largest $|D|$ across the 201 columns stays between 2.5 and 3.5 standard deviations throughout, against the 2.9 expected for that many independent walks, and the pooled density over all columns is 0.499999.

Two things follow. First, the columns exhibit square-root cancellation with constant 1, which is the random-model value — the same statement that, for the primes, is the *conclusion* of the Riemann Hypothesis rather than a route to it. Achieving the expected size of cancellation is what a proof would have to deliver, so measuring it settles nothing; P2 needs only $D_0(T) = o(T)$, which is far weaker, and no finite computation reaches either. Second, and this is the new negative: *there is no chiral signature in the column densities*. The root-mean-square of D over the hundred left columns against the hundred right ones gives ratios 0.913, 0.928, 0.828, 0.974, 0.951, 1.021 at the last six scales, and the mean of D is +65.2 on the left and -24.6 on the right against a standard error of 51.2, so neither is significant. Rule 30’s chirality is plain in the front geometry — the left edge advances at speed 1 and the right at about 0.2427 — and is invisible in the first-order statistics of the columns. That matches what Remarks 15.8 and 15.9 found from two other directions: chirality organises the geometry and does not organise the statistics.

The rule chart (Exact; criterion Known)

From the seed, the aperiodic centre columns among the 256 rules number thirteen (Remark 15.12); excluding the five degenerate ones (126, 129, 161, 167, 181; Remark 15.11 for the type) leaves exactly 30, 45, 75, 86, 89, 101, 135, 149: the one-wire rules $l \oplus g(c, r)$ with gate weight 3. Rule 135 from the seed is Rule 30’s orbit complemented and shifted by $(1, 1)$ (EXACT, 2.2×10^6 cells); 75 is 45 from the word 11 [10.36, 10.41].

Remark 22.1 (the one hypothesis Rule 90 fails, and why it still does not separate the rules). Throughout this paper Rule 90 has satisfied every structural hypothesis we could state while

failing every conclusion we want, which is what closes route after route. The hot-spot bound is the first exception, and it is worth recording precisely because the exception is instructive rather than useful. Rule 90's centre column is $1, 0, 0, 0, \dots$, so the all-zero word of length m occupies almost the whole prefix and $C_m = 2^m$ exactly: it fails the bound as badly as any sequence can, while Rule 30 tracks a coin. So here at last is a computable quantity that separates them.

It does not give P2, and the reason is structural. Bailey and Misiurewicz's theorem is an *equivalence*: the hot-spot bound holds if and only if the sequence is normal. So the bound is not a route to the conclusion, it is the conclusion in other words, and exhibiting a quantity that separates the two rules is exactly as hard as P2 itself. What the published proofs actually verify the bound with is self-similarity — for the Stoneham constants, the orbit is trapped on a grid of about n points — and that is the ingredient we lack. The correction to record is that the missing ingredient is in the *verification* and not in the theorem: nothing in the hypothesis of the hot-spot lemma requires a finite k -kernel, a digit formula or a functional equation, and an earlier draft of ours said otherwise.

One further negative closes the obvious alternative. One might look for a third family in the fractal-geometry literature [32], where a finite automaton recognising an address relation yields a self-similar quotient space. It is not a third family. The implication runs the wrong way — the automaton is the hypothesis and the geometry the conclusion — and where that literature does produce frequencies, for the Rauzy tiles of a primitive substitution, the vector of subtile measures *is* the Perron eigenvector of the incidence matrix. That is family (i) in geometric coordinates, with the same Perron datum doing the same work.

23 What Problem 3 asks, and what is provable about it

Problem 3 asks whether the n -th cell of the centre column can be computed with substantially less than $O(n)$ effort. We have said least about it, so we set out what the relevant literature settles, what it forbids, and why the one route it appears to leave open does not reach the centre column. Nothing in this section is ours except the computation reported at the end.

Remark 23.1 (what the decision procedures for automatic sequences can and cannot do). Büchi's theorem identifies the sets definable in $\langle \mathbb{N}, +, V_2 \rangle$ with those whose base-2 representations are automaton-recognised, and Walnut [100] mechanises the resulting decision procedure, eventual periodicity included. It settles Rule 90 outright, whose entire space-time diagram from the seed is the definable set $\{(t, i) : ((t + i)/2) \wedge ((t - i)/2) = 0\}$ by Kummer's theorem, and it cannot be aimed at Rule 30 at all: its input would be an automaton generating the centre column, and the existence of that automaton is Problem 1 by Christol's theorem. Every decision procedure we have found consumes the presentation we are trying to establish. On the difficulty of the statement itself, eventual periodicity of a computable sequence is Σ_2^0 -complete, so non-periodicity is Π_2^0 and each pair (period, preperiod) is finitely refutable; what is missing is uniformity in the pair, which is what Part I supplies for the families it covers.

Remark 23.2 (the formalisation, and a category error to avoid). Problem 3 is not the statement that predicting Rule 30 is P-complete, and the two are formally independent in both directions. P-completeness bounds parallel *depth*, while Problem 3 bounds total *work*, and CIRCUIT VALUE is P-complete and computable in linear time; conversely there are cellular automata with P-complete prediction and communication complexity $O(n^{1/k})$. The inputs differ as well: the universality results give the time parameter in unary, so their input length is $\Theta(t)$, whereas

Problem 3 gives n in binary, input length $\Theta(\log n)$. The correct frame is therefore non-uniform — circuit size for one fixed sequence — and not **P** versus **NC**.

Remark 23.3 (status: open in both directions, and citably so). Rule 110 is Turing-universal [34], and prediction for it is logspace-complete for **P** [35, Thm. 1]. Rule 30 is one of 33 elementary rules left unresolved by the only systematic complexity classification of all 256 [39, §4.1], and Moore and Pnini [101], whose method predicts the motion of non-colliding defects in Rule 18, write that they have been unsuccessful in applying it to rules 22, 30 and 110. So there is no lower bound of any kind for Rule 30 and no $O(n^{2-\epsilon})$ algorithm either. On the upper side, Wolfram’s request for any bound below $O(n^2)$ — a “nice partial result” in his words [68], not Problem 3 itself — has in fact been met in print: there is an $O(n^2/\log n)$ -time algorithm, at the cost of $O(n^2/(\log n)^3)$ space [40]. No sublinear-space method is known, and the generic space-efficient simulations are worse here than the trivial bound.

Remark 23.4 (the barrier ladder). It is worth knowing which forms of Problem 3 are unreachable rather than merely unproved. A circuit lower bound of $\Omega(n)$ is *false*: every Boolean function of n variables has circuits of size $O(2^n/n)$, so the n -th bit of any sequence has circuits of size $O(n/\log n)$, and that ceiling applies to us. A bound of n^ϵ would, through Impagliazzo and Wigderson [99], give **BPP** = **P**, and lies behind the natural-proofs barrier. Even $(\log n)^{\omega(1)}$ would give $\mathbf{E} \not\subseteq \mathbf{P}/\text{poly}$ and hence $\mathbf{EXP} \neq \mathbf{MA}$. Two reformulations are foreclosed rather than hard: Levin’s Kt of the prefix is $O(\log N)$, since a short program plus the time to run it describes the sequence, so no incompressibility statement is available there; and KT complexity is the circuit question relabelled. Relativisation and algebrisation do not apply here, there being no class separation in the statement.

One distinction belongs here, because it is easy to conflate two different obstacles. In restricted models, lower bounds for *explicit* objects do exist and are not barrier-bound: Nisan’s bound for algebraic branching programs is the Fliess–Schützenberger theorem in another notation, and it gives exponential lower bounds for explicit polynomials over any field, $\mathbb{F}_2$ included. What that argument consumes is a witness uniform in the degree, which comes from a formula for the object and never from its initial segment. So in the weighted or branching-program setting our obstacle is not the circuit-lower-bound barrier of the table above but the prefix obstruction of Remark 14.5, and the two should not be quoted as one.

Remark 23.5 (the propagator carries no hardness; the source carries all of it). This is where Part II’s decomposition earns its keep, in the negative direction. One might hope the Rule 150 propagator is hard to compute, and matrix rigidity is the tool for that. It is not: the Pascal-mod-2 matrix is exactly the Kronecker power $(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix})^{\otimes m}$, which is the canonical object of Alman’s non-rigidity theorem [41, Thm. 1.8] and is therefore not Valiant-rigid. Independently and more simply, $\Lambda^t = \prod_{j \in \text{bits}(t)} \Lambda(z^{2^j})$ by Frobenius, a product of at most $\log_2 N$ three-sparse factors, giving an $\mathbb{F}_2$ -circuit of size $O(N \log N)$ and depth $O(\log N)$. And the two linear rules settle it from the other side: Rule 90’s centre column is $1, 0, 0, 0, \dots$ and Rule 150’s is identically 1, both computable in constant time. So all of the difficulty of Problem 3 sits in the quadratic source, exactly where Corollary 15.5 says the chirality lives, and none of it in the kernel.

Remark 23.6 (the centre column is not a verifiable delay function). Problem 3 has the shape of a cryptographic assumption, so it is worth saying why the centre column is not a candidate for the one deployed primitive of that kind. A verifiable delay function needs a challenge with min-entropy; a single fixed sequence has none, so the preprocessing phase of the standard definition wins trivially. Moreover the sequentiality of iterated squaring is *trapdoor* hardness, broken in groups of known order, and no unconditional sequentiality bound exists for any explicit map.

Proposition 23.7 (the window monoid is non-solvable; EXACT). *For each pair $(c, f) \in \{0, 1\}^2$ the map $\Phi(c, \cdot, f)$ of Section 3 is a transformation of the 2^K windows; let T_K be the monoid they generate. Then $|T_2| = 113$ and $|T_3| = 41\,197$, neither is a group, both have minimum rank 1, and their largest group $\mathcal{H}$ -classes below full rank are $\text{Sym}(3)$ and $\text{Sym}(4)$. At $K = 4$ the largest such class is $\text{Sym}(5)$: a group of order 120 acting faithfully on a five-element image, with element orders $\{1, 2, 3, 4, 5, 6\}$. Hence T_4 contains A_5 and is not solvable, and its Krohn–Rhodes complexity is at least 1. By contrast the corresponding monoid for Rule 90 is a solvable group at even K , of order 8 at $K = 2$ and 96 at $K = 4$.*

Computation. T_2 and T_3 were enumerated in full by closure under left multiplication. At $K = 4$ the enumeration was capped at 6×10^6 elements, which suffices: the elements of a group $\mathcal{H}$ -class are those whose restriction to their own image is a bijection, and any set of them found within the monoid generates a subgroup, since a monoid is closed under products and products within a group $\mathcal{H}$ -class stay in it. Closing the 67 rank-five elements found under composition gives a group of order exactly 120 on five points, which is $\text{Sym}(5)$. $\square$

Remark 23.8 (where this puts Rule 30 in Moore’s classification). This is the sharpest algebraic statement we know about the rule, and it is the same invariant that settled the nearest comparable case. Moore’s parallel-prediction theorem [101] places a cellular automaton in ACC^1 when the blocked local rule’s algebra is a *solvable* group, and that is exactly how single-defect motion in Rule 18 was shown to be computationally reducible; he conjectures that automata built on non-solvable groups such as A_5 are P-complete. Our corpus previously recorded only that Rule 30’s algebra is not a group. Proposition 23.7 replaces that with a positive statement: it contains A_5 . So Rule 30 sits on the side of that dichotomy which Moore conjectures hard, while Rule 90 — which satisfies more of our hypotheses than Rule 30 throughout this paper — sits on the side where his theorem applies and gives a polylogarithmic-depth algorithm. The conjecture is a conjecture, so this is evidence and not a bound.

There is, however, a *theorem* at exactly the level our data reaches, and it is the sharper thing to cite. Barrington and Thérien [31] prove a dichotomy for the word problem of a finite monoid: if the monoid contains a non-solvable group the word problem is NC^1 -complete under AC^0 reductions, and if the monoid is solvable the word problem lies in ACC^0 . Proposition 23.7 is precisely the checked hypothesis of the first branch, so the word problem of our window monoid is NC^1 -complete, unconditionally, while Rule 90’s sits in ACC^0 . Three limits belong with it and we state them rather than let the result be overread: NC^1 is a small class and completeness in it is far from the superlinear work bound Problem 3 asks about; the statement is about the monoid T_K , not about the centre column; and AC^0 reductions are coarse. What it does is replace a conjecture with a theorem on the one question where we have the hypothesis in hand.

The Rule 90 side of the comparison can be strengthened from measurement to proof. Its four window generators are affine over $\mathbb{F}_2$ with a common linear part, so every subgroup of the monoid they generate is metabelian, for every K — which is why the orders 8, 96, 896 we computed at $K = 2, 4, 6$ are not a small- K accident. Rule 30’s generators are affine at no K .

Remark 23.9 (communication rank: a measurement, and the control it fails; MEASURED). What remains is communication complexity. For a cellular automaton that is intrinsically universal, the deterministic communication complexity of prediction is $\Omega(n)$ [39, Thm. 3.1], and the standard lower bound on that is the logarithm of the rank of the prediction matrix. Taking the maximum over all cuts, rules 90, 150 and 105 give rank exactly 2 at every $n \leq 10$ — a clean instance of the filter that has governed this paper — while Rule 30 gives rank about 1.59^n , a $\log_2$ -rank

slope of 0.668. Three caveats belong with it: the range is only $n \leq 10$, log-rank is a lower bound on communication and not an upper one, and Rule 110's own slope declines over the same range. Note also that Rule 110's intrinsic universality is itself open — Open Problem 5 of [38], with Ollinger recording that Cook's construction cannot be used to settle it — so the hypothesis of the theorem above is not known to hold for any elementary rule. Meunier's results put this more sharply than mere openness: intrinsic universality is *refuted* for 55 of the 88 minimal rules, and 33 remain open, including both Rule 110 and Rule 30. An earlier version of this remark called a bound $\text{rank} \geq 2^{\delta n}$ for Rule 30 the first hardness theorem about the centre column itself, and the only such target not barred in principle; that is **WITHDRAWN**, because the statistic fails our own control (Remark 23.10). Rules 22 and 54 have trivially computable centre columns — $1, 1, 0, 0, \dots$ (Remark 15.17) and periodic with period 4 (Remark 15.7) — and their communication rank over $\mathbb{F}_2$ is of the same order as Rule 30's or larger at every n computed (**EXACT**). Taking the maximum over cuts at $n = 6, 7, 8, 9$, Rule 30 gives 19, 26, 44, 79, Rule 22 gives 36, 58, 94, 147 and Rule 54 gives 21, 31, 46, 69 (below Rule 30 only at $n = 9$); at the centre cut, $n = 10, 11, 12$, Rule 30 gives 53, 79, 104, Rule 22 gives 231, 355, 555 and Rule 54 gives 100, 145, 208. Rule 126 exceeds Rule 30 at each of these n too, but its column is identified only for $t \leq 2^{14}$ (Remark 15.11), so the argument does not rest on it. The measurement above is of real rank, which for a 0/1 matrix is at least the rank over $\mathbb{F}_2$ (an odd minor is non-zero); for Rule 30 at the centre cut the two agree at $n = 10, 11$, so there the real rank fails the control in the same way. The reason is that the rank measures the Boolean function F_n of Proposition 14.6 over all 2^{2n+1} inputs, as Meunier's communication complexity of prediction does, and not the single-seed column: a rank lower bound would be a theorem about predicting Rule 30 and not about the centre column, just as the NC^1 -completeness above concerns the window monoid and not the column. Over $\mathbb{F}_2$ the rank of the prediction matrix equals the rank of the matrix of ANF coefficients $a_{S,T}$ of F_n split across the cut, since the monomial-evaluation matrices are unitriangular in the subset order (**PROVED**; ledger 10.230d A). Computed to $n = 14$ (**EXACT**; 10.230d B): the maximum over cuts sits left of centre and moves left with n , and at a fixed block of n_L leftmost variables the rank saturates in n at about $2^{n_L}/3$ (87, 172, 343, 684 for $n_L = 8, \dots, 11$), so the slope 0.668 is a reading on a range where the cut is still moving and not a constant of the rule; whether the slope tends to 1 is **OPEN**.

Remark 23.10 (a control on inferring hardness from a complexity measure; **EXACT**). The caveats above are not hypothetical, and the witness is inside the family of Part II. The centre column of Rule 126 is the Fredholm–Rueppel sequence for every $t \leq 2^{14}$ (Remark 15.11), which covers every bit measured below. Berlekamp–Massey on its first 2048 bits returns linear complexity 1024, exactly $n/2$ — the perfect-profile value — and exactly what a random control returns on the same length; Rule 30's centre column returns 1025 and Rule 150's returns 1. So on this measure Rule 30 is indistinguishable both from a random sequence and from Rule 126 — whose t -th bit on that range is 1 if and only if $t + 1$ is a power of two, and is therefore computable in $O(\log t)$ time from t alone. A perfect linear-complexity profile is compatible with constant-depth computability. We claim nothing new in that observation: this sequence is the standard textbook witness for a perfect linear complexity profile, and its primary source is Rueppel [94]; although we have not located an earlier source for it in our own corpus, it should be assumed known and is used here only as a control. The consequence for Problem 3 is a restriction on what any such measurement can establish: a complexity statistic that Rule 126 also maximises cannot, on its own, be evidence that Rule 30's column is hard to compute. The rank measurement above is of this kind as well, with Rules 22 and 54 in the role of Rule 126: rules 90, 150 and 105 give rank

exactly 2 at every $n \leq 10$, which separates the affine rules, but Rules 22 and 54, whose columns are eventually 0 and periodic with period 4, have communication rank of the same order as Rule 30's or larger at every n computed (Remark 23.9), so the statistic does not separate the tractable cases. Rules 22, 54 and 126 should be run against any future candidate before it is reported, and they are the controls we now use.

Part VI

Closed routes, verdict, and the measure-level obstruction

24 Closed routes

Most of the work behind this paper was the elimination of routes, and the result is more useful as a table than as prose. Each row names a route, the reason it fails for this object, and where the reason lives. “Rule 90” as a reason means the firewall of Proposition 26.1: the rule satisfies the hypothesis and fails the conclusion. “Prefix” means the obstruction of Remark 14.5: every finite word extends to an automatic sequence, so no statistic of a prefix can decide the question.

Route	Why it fails here	Where
Measure-level randomisation	Rule 90 satisfies strictly more and has density 0	26.1
Density at width w , then descend	Rule 90's width- w trace has density 0 at every w and is aperiodic for $w \geq 2$	§26
Spread above $\frac{1}{2}$	Rule 90 saturates the threshold exactly	§26
Substitutive route to a density	needs linear recurrence; ours is $R(13) \geq 105\,901$	26.2
Self-similarity route to a density	hypothesis consistent, no known verification	22.1
Fractal address-relation route	family (i) in geometric coordinates; implication runs backwards	22.1
Kernel census as aperiodicity	prefix; a padded prefix is automatic with 107 047 classes	14.5
k -regularity as a weaker hypothesis	over $\mathbb{F}_2$ it coincides with automaticity	14.12
Hankel rank	prefix, dominated by the antichain census, and a coin saturates it	14.11
Boolean spectral invariants	the designated coefficient is invisible to the Walsh multiset	14.10
Sublattice linearisation	the invariant set is three points	15.14
Conservation laws, solitons, ultradiscrete-integrable systems	no additive invariant of order 1 or 2, exhaustively	25.1
Integrable “Rule 54/150”	those are reversible two-step automata, not these maps	25.1
Mixing for the uniform measure	holds, and says nothing about either problem	25.1
Solvable-circuit duality	needs bipermutivity; and the class is empty of informative members	13.5
2-adic ergodicity	not ergodic, and the obstruction is the diagonal read	13.6
Operator-algebraic reformulation	equivalent by definition; the invariant is the suffix count	§25

Route	Why it fails here	Where
Coupling, Kingman, Fekete	the increment is an exact cocycle	17.27
(a, τ) -blockers	cost 2^a at numerator a , published only to $a \leq 4$; sup constant infinite	17.26
Front as a finite max-plus system	band width grows; every bounded control is flat	17.25
Bounded-window prediction	provable at one width would be P2 itself	20.6
Learning-theoretic mistake bounds	quantified over sequences, not about one	20.7
Decision procedures for automaticity	each consumes the presentation we lack	23.1
Matrix rigidity of the propagator	the kernel is the canonical non-rigid object	23.5
Verifiable-delay-function framing	the challenge has zero min-entropy	23.6
Communication rank of the prediction matrix	Rules 22 and 54 (centre columns eventually 0 and of period 4) and Rule 126 have rank of the same order as Rule 30's or larger at every n computed: the rank measures F_n over all inputs, not the column	23.9
Černý-type bounds	wrong quantifier, and vacuous by a factor of 949	6.6
Topological classification	one bit of discriminating power, and it is bipermutivity	15.16
Preperiod theory	for nonlinear rules the only mechanism we have found is a blocking word, which needs inertness	17.4

Two entries deserve emphasis because they are the sharpest form of the obstruction. The first is the observation recorded in Remark 14.11 (MEASURED): every instrument tried here separates Rule 30 from Rule 90, and none tried separates it from a fair coin. The second is that the one hypothesis Rule 90 does fail, the hot-spot bound, is equivalent to the conclusion we want, so exhibiting a separating quantity there is exactly as hard as P2.

25 Verdict

PROVED Part I: three infinite families and two thresholds of excluded period words, for every number-like configuration; the reach of the method characterised (Proposition 9.2). At that boundary: the exponential clause cannot be discharged by enumeration (Corollary 9.4); the branching criterion for uncountability (Proposition 9.7); a finite row-0 block that traps the interface (Proposition 9.9); and the diagonal cap, $g(N) = 6$ on every even diagonal $N \geq 58$ and $g(N) = 2$ on every odd $N \geq 5$ (Theorem 9.11). Part II: the Duhamel formula, exact decimation of its class, the closed form in α , the chiral-only corollary, the centre-cell degree law $\deg F_t = 2t - 1$ with its unique top monomial (Theorem 14.7), the right-edge influence law $\text{Inf}_t = \text{Inf}_{t-1} = 2^{-t}$ (Proposition 14.8), and the ring surjectivity defects of Rules 45, 150, 90 and 30 from the transfer monoid, Rule 30's exponent being $\log_2 \lambda - 1$ with λ the real root of $x^3 - x^2 - 2$ (Proposition 15.10). Part III: reset formula, degree formula, monotone

front, exact failure condition, window slaving, Theorem H* and the light-cone corollary. Unbounded diagonal periods (Theorem 18.3) is KNOWN, Gravner–Griffeath [67, Prop. 3.3]; it is reproved here only to fix the diagonal-frame notation. Part IV: the downstream-only proposition, the record-parity theorem, the two-sided criterion, the parent-instance split.

EXACT The kernel bounds of Theorem 14.1; the chiral recursion verified on 126 words; the front data to $d = 90\,000$ (E1–E6), with E1–E3 and the (C) half of E4 extended to 8×10^5 by the third frame of E7, the (U) half of E4 only to 90 000; the frozen diagonals and doubling positions to 8×10^5 (E7); τ_d exact to 8×10^5 (Remarks 17.28, 17.29) and per diagonal, with a certificate, to 4×10^6 , so Conjecture H holds on $[20, 4 \times 10^6]$ (E10), with bin medians of M_d/d to 4×10^6 (E9, MEASURED); the fibre sizes to $k = 60$ ($|L_{60}(01)| = 103\,220$); the forward trees F_P to $P = 16$; the kill depth to $T = 120$.

OPEN P1 and P2. Conjecture H ($\tau_d \geq d$ for $d \geq 20$, Conjecture 17.18) is also open, and on its own it decides neither: it is a statement about the geometry of the ordered region, and Rule 60 has $\tau_d = d + 1$ for every $d \geq 0$ with centre column constantly 1, periodic and of density 1. Theorem 17.20 reduces Conjecture H to two inequalities on the front, whose margins are measured and wide (Remark 17.21) and which no proof reaches; every sufficient statement on the front row we have found is equivalent to them, and what a proof needs is a lower bound on the density of ones in the transient interior. Conjecture H settles no Prize problem (ledger 10.153c2 D). The automaticity route needs an invariant of the chiral Cartier recursion taking infinitely many values; none is known. The kernel criterion of Allouche–Shallit–Yassawi [92, Thm. 1] needs only one residue class modulo 2^k at each level k , with the resulting kernel elements pairwise distinct; in the instances collected there the residues are given in closed form, and no dynamical recursion has yet supplied such a family. The zero-run criterion (Remark 16.2; observed to hold at the coin’s rate; MEASURED) and the two-sided criterion of Corollary 20.5 are honest bridges between P1 and P2 without a proof on either side. Conjecture 18.16 (left-half universality) is refuted by a 53 209-cell starting word that takes the other branch after 53 207 (EXACT); which finite words take the seed’s branch is OPEN. Whether $L(01)$ contains an interface that is not eventually periodic: one such interface already puts 01 beyond Theorem 6.3 at every K (proof of Proposition 9.2), and uncountability of $L(01)$, the hypothesis stated there, is one sufficient condition for it and not a necessary one, since a countable closed shift-invariant set can contain a point that is not eventually periodic. Counting cannot reach uncountability (Corollary 9.4), the branching criterion is the only route to it here, and the trap shows that route must succeed despite atoms of the push-forward (Remark 9.10). Random right halves driven by $(01)^\infty$ settle into a stationary stream of positive measured entropy (Remark 9.6), so the zero-entropy expectation is withdrawn and positive entropy of $L(01)$ is MEASURED; uncountability remains OPEN.

Closed Every bounded-window or bounded-state route to P1 (Section 22); the constructive zero-run route (halving law, ledger 10.76; not reproduced in this paper); small linear lifts (Proposition 16.3). The bounded- f kill route is not closed but unsupported by the data (Section 22). From the wider literature, also closed: the sublattice linearization that underlies every rigorous Rule 18 result, by Proposition 15.14; every method resting on a conservation law, by exhaustion over the elementary rules (Remark 25.1); the measure-level routes of Section 26, at every trace width, not only at width one; and, for Problem 3, the communication-rank route (Remark 23.9).

closed, by equivalence The operator-algebraic reformulation. The chain representation of the Cuntz algebra $\mathcal{O}_2$

(KNOWN: Kawamura [95], Bratteli–Jorgensen [96]) attached to a binary word is irreducible exactly when the word is not eventually periodic, so P1 is equivalent to an irreducibility question; but the equivalence is definitional, the only numerical invariant in that theory is the count of distinct suffixes, which is the Myhill–Nerode count whose finiteness *is* eventual periodicity, and K -theory is unavailable since $K_*(\mathcal{O}_2) = 0$. Likewise the learning-theoretic route (Remark 20.7).

Decision points Two questions now have the property that either answer is informative. The front speed is rational or it is not, and the second answer splits in two (Remark 17.25). If it is rational, a universal (a, τ) -blocker would certify it, but that route is closed twice, by its cost (Remark 17.26) and by its infinite sup-over-environments constant (Remark 17.27). If it is not, the front’s discrepancy from a line is either bounded, as for a Beatty sequence, or unbounded, the third branch, which is the one the growing band of Remark 17.25 points to (MEASURED); and the unbounded periods of Theorem 18.3 are not the mechanism, since irrationality needs no unbounded system, the dividing line being autonomy rather than dimension (Remark 17.25). And the centre column is substitutive or it is not: the recurrence function argues against the primitive substitutivity that family (i) of Proposition 26.2 needs (MEASURED), which removes one of the two families of technique behind every single-sequence density- $\frac{1}{2}$ theorem we know; the column is consistent with the hypothesis of family (ii), and what is missing there is any known way to verify that hypothesis for one orbit (Section 26). The cheapest unclaimed step in the paper is neither of these but the near-diagonal almost-subadditivity of the front (Remark 17.27): proving $M_{d+e} \leq M_d + M_e + f(d+e)$ for $d \leq e \leq 2d$ with $\sum f(n)/n^2 < \infty$ would give the existence of $\lim M_d/d$. We have been aiming at a constant $f = 326$, which the measurement supports but which is strictly harder than necessary: $\sum n^{\alpha-2}$ converges for every $\alpha < 1$, so any $f(n) = O(n^\alpha)$ with $\alpha < 1$ suffices, and Remark 17.28 reduces the whole statement to a bound on a single index. The coupling route we proposed in an earlier draft is REFUTED: the increment is an exact cocycle, so every subadditivity hypothesis holds with equality.

Remark 25.1 (families closed by exhaustion over the elementary rules). Three families of technique are unavailable for reasons that are themselves theorems, proved by exhaustion over the elementary rules. Rule 30 is not number-conserving — only 204, 170, 240, 184 and 226 are [58] — and has no additive invariant of second order either, the Hattori–Takesue conditions leaving only 12, 14, 15, 34, 35, 42, 43, 51, 140, 142, 200 and the trivial cases [56, 57]. Every soliton, ultradiscrete-integrable, height-function and hydrodynamic method begins with a conservation law, so this closes them together, and it is why the Rule 184 identity of Theorem 20.3 transfers as an identity and not as a method. The exactly solvable “Rule 54” and “Rule 150” of the statistical-mechanics literature [59, 60] are reversible two-step automata rather than these maps. And the strongest measure-level theorem that does apply, that every one-sided permutative rule is mixing for the uniform measure [62], says nothing about either problem.

26 Why no measure-level argument can give P2

Rule 30 is surjective (Hedlund [14, Thm. 6.6]) and the balance property behind that — the preimage of a length- k cylinder is a disjoint union of exactly $|A|^{r+\ell}$ cylinders of length $k + r + \ell$ — is the Blankenship–Rothaus theorem, first published as [14, Thm. 5.4]. So the uniform Bernoulli measure is invariant, and it is tempting to read P2 as a consequence. It is not, and the obstruction can be stated as a proof rather than as a caution.

Proposition 26.1 (the seed is invisible to measure-level hypotheses; KNOWN). *No hypothesis of the form “ F is surjective, preserves the uniform measure, is (bi)permutive, is linear, or asymptotically randomizes every harmonically mixing measure” can imply that the centre column of F from a single seed has density $\frac{1}{2}$.*

Proof. Rule 90 satisfies all of them: it is bipermutive, surjective, preserves the uniform measure, is a linear cellular automaton, and therefore asymptotically randomizes every harmonically mixing measure in the sense of [15]. Its centre column from the single seed is $1, 0, 0, 0, \dots$, of density 0. $\square$

Rule 90 thus satisfies strictly more than Rule 30 does and fails the conclusion outright, which is the sharpest form of the remark that permutivity alone cannot separate the two rules ([65, §4]; [91, §1]). Four further points fix the status of this route.

First, the randomization theorems do not merely fail on the initial measure, they never engage: every one of them quantifies over *linear* automata, and Rule 30 is not linear, so the question of which measures they randomize does not arise. Pivato and Yassawi do not discuss point masses at all; what they say is that all known sufficient conditions require the measure to have full support, and a point mass violates that at every rank. Their §8 also refutes the conjecture that positive entropy suffices, so “the obstruction is that the seed has zero entropy” would be the wrong diagnosis. Pivato’s survey [16] contains nothing about single-seed or Dirac initial conditions.

Second, the routes that have worked for a single deterministic orbit rest on extra arithmetic structure, and they sit at the two ends of one complexity axis (Proposition 26.2). At the low-complexity end are the decimation identity $t_{2n} = t_n$ that gives the Thue–Morse density exactly $\frac{1}{2}$ and the Perron-eigenvector frequencies of primitive substitutions; a finite k -kernel belongs only to the constant-length (automatic) case, since the Perron argument needs primitivity rather than constant length and so covers primitive substitutive words of non-constant length, the Fibonacci word for instance. At the high-complexity end is the normality of the Stoneham constants [18], through the hot-spot lemma: its hypothesis is a one-sided bound on block frequencies with no finite k -kernel in it, and self-similarity enters only in verifying that bound (Remark 22.1). Our kernel figures give a scale, not an exclusion from either end: $A_2^c(2^{21} - 1) = 138\,672$ (Theorem 14.1; a census at 2^{24} terms reported more than 10^6 in [81], not re-derived here) says that an automaton generating c least-significant digit first would need at least that many states, not that none exists, since every finite prefix extends to an automatic sequence (Remark 14.5). What separates c from the low-complexity end is its recurrence function, and that separation is MEASURED.

Third, the measure-rigidity template that replaced symmetry in the quantum setting is unavailable in principle here, not merely missing. Rule 30 is left-permutive, hence closing [42, §3], and for closing maps the jointly (σ, F) -periodic points are dense (Boyle–Kitchens [17, Thm. 4.4]); the invariant simplex is therefore large, and no rigidity statement of the Host–Maass–Martínez kind can hold.

What does constrain a single orbit is small but real: for every finite configuration the width-2 traces are never eventually periodic and the orbit has infinitely many limit points ([65, Thms. 3.5, 4.7]), and the right Lyapunov exponent is exactly 1 at *every* point, seed included, for a left-permutive rule with negative left radius [42]. The left exponent, the one that carries our front, has no theorem at any point. The measured values of that speed, ours and Wolfram’s two [88], are collected in Remark 17.24, which also gives their precision honestly. Wolfram’s are not the first in print: his own CRYPTO ’85 extended abstract [87] already gives the left Lyapunov

exponent as 0.25, and Meier and Staffelbach [82] report that the effect of a bit change propagates to the left with speed roughly $\frac{1}{4}$, an observation they attribute to an earlier paper of Wolfram's; Remark 17.26 says what the literature can and cannot do with such a number either way.

Fourth, and this is the one that cuts against our own programme rather than against the literature. Every proved statement that a single explicit deterministic sequence has letter density exactly $\frac{1}{2}$ — and we have not found an exception — belongs to one of two families, and the two sit at opposite ends of a single complexity axis. Our own measurements place the centre column at the high-complexity end (MEASURED): outside the first family, and consistent with the hypothesis of the second, though out of reach of every known way of verifying it.

Proposition 26.2 (the two families, at opposite ends of one axis; KNOWN chain, MEASURED placement of c). *The published single-orbit density theorems divide as follows.*

- (i) Low complexity, through substitutivity. *Thue–Morse through the decimation identity $t_{2n} = t_n$; primitive substitutive sequences through the Perron eigenvector; automatic sequences; Toeplitz words; and the Thue–Morse word in base 3/2 [63, Thm. 18]. Each requires the sequence to be substitutive, and the frequency theorem requires primitivity, since a non-primitive substitution need not have letter frequencies at all. Primitivity forces linear recurrence, hence $p(n) \leq Kn$ and $R(n) \leq Kn$ for a fixed K , where R is the recurrence function.*
- (ii) High complexity, through a block-frequency bound. *The normal numbers: Champernowne's constant by explicit construction, and the Stoneham constants through the hot-spot lemma [18]. The hypothesis is a one-sided upper bound on block frequencies — for every y , $\liminf_m \limsup_n 2^m A(x, y, n, m)/n < \infty$ — and nothing else.*

The two families sit at opposite ends of a single axis. Family (i) forces $p(n) \leq Kn$. Family (ii) forces the reverse: a uniform hot-spot bound C makes every word of length m occur often enough that $p(m) \geq 2^m/C$, so its hypothesis requires exponential factor complexity.

Where c sits is a measurement, not part of the proposition: as far as computed it is at the (ii) end, not between them (MEASURED, from the measurements that follow).

Measurements. For (i): on 2^{18} terms, the recurrence function satisfies $R(13) \geq 105\,901$, so any linear-recurrence constant for the centre column exceeds 8 146; the growth is geometric, with $R(n)/2^n$ tracking $n \log 2$, which is the value for an i.i.d. coin, and $p(n) = 2^n$ exactly for $n \leq 16$ on 2^{20} terms. A gap between consecutive occurrences found in a prefix is a genuine gap in the infinite word, so the lower bound on K is not an artefact; what a finite table cannot do is refute an asymptotic $O(n)$ bound, so this is evidence together with a quantitative constraint — any primitive substitution generating this word needs linear-recurrence constant above 8 146, which bounds its alphabet from below — rather than a proof. For the claim about family (ii): a uniform bound $C_m \leq C$ forces at least $2^m/C$ distinct words of length m , since no word may carry more than C times its fair share of a prefix. Our measured profile $C_m = \max_w 2^m \text{count}(w)/N$ at $N = 2^{18}$ is 1.00, 1.00, 1.01, 1.01, 1.02, 1.04, 1.06, 1.10, 1.13, 1.21, 1.34, 1.58, 1.72, 2.19, 2.75, 3.50 for $m = 1, \dots, 16$, tracking an actual coin sequence on the same prefix to within 0.25 at every m , the growth being the finite- N floor and nothing else. Rule 90 gives $C_m = 2^m$ exactly. $\square$

The honest conclusion is therefore not that we fall between the two families but that we sit squarely at the high-complexity end and cannot use its methods. We are excluded from family (i) by the recurrence function. We are *not* excluded from family (ii)'s hypothesis — we are consistent with it, and the measured profile is the evidence — but we are excluded from

every known way of *verifying* that hypothesis. That is the precise sense in which P2 has no available technique, and it is sharper than the caution we gave earlier.

One subtlety is worth spelling out, because it is the obvious objection. Substitutive is strictly broader than automatic — automatic means substitutive of constant length — and a primitive substitutive word of non-constant length, the Fibonacci word for instance, is aperiodic *and* has letter frequencies given by the Perron eigenvector. So primitive substitutivity would deliver P1 and P2 together, and our Part II’s argument against automaticity would not by itself rule it out. What rules it out is the complexity, not the automaticity: linear recurrence is incompatible with a recurrence function growing like $n2^n$. The exclusion therefore rests on the recurrence function alone, and the kernel census could not carry it in any case: no prefix census can imply non-automaticity (Remark 14.5).

Two further facts close the obvious workarounds. One might hope to prove a density statement at width 2, where Theorem 3.1 does bite, and then descend to the centre column. Rule 90 kills that at every width, not just at width 1: counting the rows whose width- w window is not identically zero, from the single seed, gives

T	$w = 1$	$w = 2$	$w = 4$	$w = 8$	$w = 16$
10^3	1	10	25	50	91
10^4	1	14	37	78	151
10^5	1	17	46	99	196
10^6	1	20	55	120	241

which is $O(\log T)$ at each fixed width, so Rule 90’s width- w trace is aperiodic for every $w \geq 2$ by Theorem 3.1 and has density 0 for every w . No theorem of the form “rapidly left expansive and number-like implies the width- w trace has density bounded below” can be true, at any width. The same test kills the weakest quantitative form of P1 as well. Reading a row as a binary expansion, $r_t = \sum_{i \geq 0} u_i^t 2^{-i-1}$, the first digit of r_t is the centre cell, so $\limsup r_t - \liminf r_t > \frac{1}{2}$ would force the centre column not to be eventually constant — the exact analogue of Vijayaraghavan’s remark on $(3/2)^n \bmod 1$. Over three thousand steps Rule 30 gives spread 0.99906; Rule 90 gives $\liminf = 0$, $\limsup = \frac{1}{2}$, spread exactly $\frac{1}{2}$. The threshold is saturated but not exceeded inside the class, so nothing above it is provable there, and the conclusion it would have given is in any case now a theorem by a Rule-30-specific argument [91, Cor. 5].

A second single-orbit family exists, and although it concerns linear rules it changes what P2 should be asked to prove. Kawaharada studies the cumulative count of ones along one orbit of one elementary rule from a single-site seed, with no measure anywhere, and finds that the limiting density profile is a *singular function* rather than a constant [24]. That is a proved single-orbit density statement outside both families of Proposition 26.2 — no contradiction, since that classification concerns density exactly $\frac{1}{2}$ — and its conclusion is a negative one: for those rules the natural density does not converge to a number at all. So P2 is not only unproved but formulated in a way that fails for the nearest rules where the question has been answered, and any route to it must explain why Rule 30 should be better behaved than they are.

What P2 would need, then, is a technique that is neither measure-level nor substitutive, and exactly one such technique exists in print. Gravner and Griffeath prove that a two-dimensional Packard box rule fills the lattice from a single seed with asymptotic density $\frac{4}{9}$ [30, 51], and their own assessment is that exact global asymptotics for a nontrivial local interaction is rare in nonlinear spatial dynamics. The recipe is: a *finite* alphabet of nested domain types, the density from a linear recursion over that alphabet, then a refinement argument for uniformity. It needs

monotone solidification, which Rule 30 does not have, and an additive component, which Part II supplies; but it breaks at the first step for us, because Theorem 18.3 makes the periods of the left wedge unbounded, so the domain alphabet is infinite, and the centre column lies in the region where no domain structure is known at all. If a finite domain alphabet is ever found for any part of this orbit, that is the paper to imitate.

Where the difficulty sits, in one sentence each way. From the front: the stall run is decided by parities of transient cells, and a certificate carried at the front keeps losing them, since a bit of a diagonal it does not carry survives only until the first 1 of the diagonal to its left — by the erasure-cone theorem of the referee dialogue (PROVED there and summarised in ledger 10.61, Chat B, round 5; the argument is not reproduced here) a certificate holding the two front diagonals on a window of rows, the background and boundedly many further diagonals loses at least one row of that window per plateau (Definition 17.15); on the seed’s run, with no further diagonals, the actual loss outruns the refill from the advancing front by 0.46 to 0.70 rows per diagonal (EXACT), but the proved one row per plateau does not, and a net loss in general would need a lower bound on the density of ones in the transient interior (OPEN). From the column: Section 21 — after every white centre cell one bit enters from the right, and after every black one the next bit is computed from the column’s whole past.

27 Sources and reproduction

Ledger `SESSION_FINDINGS.md` (§§1–10.229, the entries suffixed b, c and c2, and the session-d entries 10.229d–10.243d; entries 10.199–10.229 record the boundary-section, degree-law, influence and ring-defect work of this version, and 10.150c the ledger-to-paper audit) and `HANDOFF.md`; proofs `PROOFS_theorems_6_to_9.md`; the component documents `rule30_excluded_words.tex`, `duhamel_decimation_section.tex`, `front_section.tex`, `collapse_section.tex`; the automaticity document [81] and its independent re-run `agent_certificate.md`; the inventory `rule30_totality_paper`. Scripts are the `s3_*.py`, `ag1_*`, `ag2_*` files in the working folder, together with the agent working directories `agent_reach/` (the trap, the cap witness `cap/ring_cycle.txt`, the literature sweep) and `agent_topcoef/` (the degree law); the certificates of this version are `s3_trap_repro.py`, `s3_cap_verify.py`, `s3_cap_upper.py`, `s3_cap_plateau.py`, `s3_cap_odd.py`, `s3_runcap.py`, `s3_gate_fast.py`, `s3_verify_topcoef.py`, `s3_verify_general.py`, `s3_influence.py`, `s3_ring_defect.py` and `s3_ring_defect2.py`, each with its log. This document compiles with Tectonic (XeTeX engine).

References

- [1] P. Flajolet, R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [2] A. Szilard, S. Yu, K. Zhang, J. Shallit, Characterizing regular languages with polynomial densities, *MFCS 1992*, LNCS 629, 494–503.
- [3] R. Incitti, The growth function of context-free languages, *Theoret. Comput. Sci.* 255 (2001) 601–605.
- [4] J. Cassaigne, Constructing infinite words of intermediate complexity, *Developments in Language Theory* (DLT 2002), LNCS 2450, 173–184.

- [5] M. R. Bridson, R. H. Gilman, Context-free languages of sub-exponential growth, *J. Comput. System Sci.* 64 (2002) 308–310.
- [6] S. Donoso, F. Durand, A. Maass, S. Petite, Interplay between finite topological rank minimal Cantor systems, S -adic subshifts and their complexity, *Trans. Amer. Math. Soc.* 374 (2021) 3453–3489.
- [7] R. I. Grigorchuk, A. Machì, An example of an indexed language of intermediate growth, *Theoret. Comput. Sci.* 215 (1999) 325–327.
- [8] A. Ya. Belov, R. Mikhailov, Free subalgebras of Lie algebras close to nilpotent, *Groups Geom. Dyn.* 4 (2010) 15–29; the growth dichotomy for automata algebras is Ufnarovskii, *Mat. Zametki* 31 (1982) 465–472.
- [9] B. Greenfeld, C. G. Moreira, E. Zelmanov, On the complexity of subshifts and infinite words, arXiv:2408.03403 (2024); and arXiv:2508.16840.
- [10] A. Mitchell, On word complexity and topological entropy of random substitution subshifts, arXiv:2305.04817 (2023).
- [11] J.-J. Pansiot, Complexité des facteurs des mots infinis engendrés par morphismes itérés, *ICALP 1984*, LNCS 172, 380–389.
- [12] E. Jen, Global properties of cellular automata, *J. Stat. Phys.* 43 (1986) 219–242.
- [13] E. Jen, Aperiodicity in one-dimensional cellular automata, *Physica D* 45 (1990) 3–18.
- [14] G. A. Hedlund, Endomorphisms and automorphisms of the shift dynamical system, *Math. Systems Theory* 3 (1969) 320–375.
- [15] M. Pivato, R. Yassawi, Asymptotic randomization of sofic shifts by linear cellular automata, *Ergodic Theory Dynam. Systems* 26 (2006) 1177–1201.
- [16] M. Pivato, The ergodic theory of cellular automata, survey.
- [17] M. Boyle, B. Kitchens, Periodic points for onto cellular automata, *Indag. Math.* 10 (1999) 483–493.
- [18] D. H. Bailey, M. Misiurewicz, A strong hot spot theorem, *Proc. Amer. Math. Soc.* 134 (2006) 2495–2501.
- [19] J. Kopra, Lyapunov exponents of multiplication automata, arXiv:2001.09675 (2020), Thm. 5.5.
- [20] V. Cyr, J. Franks, B. Kra, The spacetime of a shift endomorphism, *Trans. Amer. Math. Soc.* 371 (2019) 461–488, Question 5.7.
- [21] E. Coven, M. Pivato, R. Yassawi, Prevalence of odometers in cellular automata, *Proc. Amer. Math. Soc.* 135 (2007) 815–821, Thm. 4.
- [22] J. Gravner, X. Liu, Weak robust periodic solutions in one-dimensional cellular automata, *J. Cell. Autom.* 17 (2023) 251–280.

- [23] W. Li, M. G. Nordahl, Transient behavior of cellular automaton rule 110, *Phys. Lett. A* 166 (1992) 335–339.
- [24] H. Kawaharada, Singular functions arising from cellular automata, arXiv:2008.13217 and arXiv:2203.05236.
- [25] J. Kopra, On the trace subshifts of fractional multiplication automata, *Theoret. Comput. Sci.* 851 (2021) 92–110; arXiv:2005.05112 (2020), Prop. 2.7 (numbering of arXiv v1).
- [26] J. M. Baetens, J. Gravner, Stability of cellular automata trajectories revisited: branching walks and Lyapunov profiles, *J. Nonlinear Sci.* 26 (2016) 1329–1367.
- [27] M. Pivato, Defect particle kinematics in one-dimensional cellular automata, *Theoret. Comput. Sci.* 377 (2007) 205–228.
- [28] K. Sutner, Almost periodic configurations on linear cellular automata, *Fund. Inform.* 58 (2003) 223–240, Thm. 3.2.
- [29] J. Gravner, D. Griffeath, The one-dimensional Exactly 1 cellular automaton: replication, periodicity, and chaos from finite seeds, *J. Stat. Phys.* 142 (2011) 168–200.
- [30] J. Gravner, D. Griffeath, Asymptotic densities for Packard Box rules, *Nonlinearity* 22 (2009) 1817–1846.
- [31] D. A. Barrington, D. Thérien, Finite monoids and the fine structure of NC^1 , *J. ACM* 35 (1988) 941–952.
- [32] C. Bandt, Elementary fractal geometry 4: automata-generated topological spaces, *Comm. Math.* 33 (2025), arXiv:2312.01486.
- [33] F. Green, NP-complete problems in cellular automata, *Complex Systems* 1 (1987) 453–474.
- [34] M. Cook, Universality in elementary cellular automata, *Complex Systems* 15 (2004) 1–40.
- [35] T. Neary, D. Woods, P-completeness of cellular automaton Rule 110, *ICALP 2006*, LNCS 4051, 132–143.
- [36] M. Schüle, R. Stoop, A full computation-relevant topological dynamics classification of elementary cellular automata, *Chaos* 22 (2012) 043143.
- [37] E. Kohlberg, Invariant half-lines of nonexpansive piecewise-linear transformations, *Math. Oper. Res.* 5 (1980) 366–372.
- [38] J. Kari, Theory of cellular automata: a survey, *Theoret. Comput. Sci.* 334 (2005) 3–33, Open Problem 5.
- [39] P.-É. Meunier, Unraveling simplicity in elementary cellular automata, arXiv:1406.5306 (2014).
- [40] J. Natal, O. Al-saadi, Fast simulation of cellular automata by self-composition, arXiv:2409.07065 (2025).

- [41] J. Alman, Kronecker products, low-depth circuits, and matrix rigidity, *STOC 2021*, 772–785.
- [42] P. Kůrka, Languages, equicontinuity and attractors in cellular automata, *Ergodic Theory Dynam. Systems* 17 (1997) 417–433; and *Topological and Symbolic Dynamics*, Soc. Math. France, 2003.
- [43] G. J. Martínez, A note on elementary cellular automata classification, arXiv:1306.5577 (2013), §4.3; after A. Wuensche, M. Lesser, *The Global Dynamics of Cellular Automata*, Addison-Wesley, 1992.
- [44] G. Christol, T. Kamae, M. Mendès France, G. Rauzy, Suites algébriques, automates et substitutions, *Bull. Soc. Math. France* 108 (1980) 401–419.
- [45] F. Baccelli, G. Cohen, G. J. Olsder, J.-P. Quadrat, *Synchronization and Linearity*, Wiley, 1992.
- [46] V. D. Blondel, S. Gaubert, J. N. Tsitsiklis, Approximating the spectral radius of sets of matrices in the max-algebra is NP-hard, *IEEE Trans. Automat. Control* 45 (2000) 1762–1765.
- [47] K. Nishinari, D. Takahashi, Analytical properties of ultradiscrete Burgers equation and rule-184 cellular automaton, *J. Phys. A* 31 (1998) 5439–5450.
- [48] N. G. de Bruijn, P. Erdős, Some linear and some quadratic recursion formulas II, *Indag. Math.* 14 (1952) 152–163.
- [49] Z. Füredi, I. Z. Ruzsa, Nearly subadditive sequences, *Acta Math. Hungar.* 161 (2020) 401–411.
- [50] V. Anashin, *Applied Algebraic Dynamics*, de Gruyter Expositions in Mathematics 49, Berlin, 2009.
- [51] J. Gravner, D. Griffeath, Cellular automaton growth on $\mathbb{Z}^2$: theorems, examples, and problems, *Adv. Appl. Math.* 21 (1998) 241 ff.
- [52] K. Eloranta, E. Nummelin, The kink of cellular automaton Rule 18 performs a random walk, *J. Stat. Phys.* 69 (1992) 1131–1136.
- [53] A. Rupe, J. P. Crutchfield, Spacetime symmetries, invariant sets, and additive subdynamics of cellular automata, arXiv:1812.11597 (2018).
- [54] H. Fukś, Solving the initial value problem for cellular automata by pattern decomposition, arXiv:2512.24337 (2025), §2.
- [55] H. Fukś, *Solvable Cellular Automata: Methods and Applications*, Springer, 2023, Ch. 12.
- [56] H. Fukś, Second order additive invariants in elementary cellular automata, arXiv:nlin/0502037 (2005).
- [57] T. Hattori, S. Takesue, Additive conserved quantities in discrete-time lattice dynamical systems, *Physica D* 49 (1991) 295–322.

- [58] N. Boccara, H. Fukś, Number-conserving cellular automaton rules, *Fund. Inform.* 52 (2002) 1–13.
- [59] B. Buča, K. Klobas, T. Prosen, Rule 54: exactly solvable model of nonequilibrium statistical mechanics, *J. Stat. Mech.* (2021) 074001.
- [60] J. W. P. Wilkinson, T. Prosen, J. P. Garrahan, Exact solution of the Rule 150 reversible cellular automaton, arXiv:2110.15085 (2021).
- [61] F. Hübner, S. W. P. Kim, Influence-solvability: a systematic theory of $(1 + 1)$ D solvability and its application to brickwork circuits, arXiv:2606.12538 (2026).
- [62] R. Kleveland, Mixing properties of one-dimensional cellular automata, *Proc. Amer. Math. Soc.* 125 (1997) 1755–1766.
- [63] J. Cassaigne, B. Espinoza, M. Rigo, M. Stipulanti, Symbols frequencies in the Thue–Morse word in base $3/2$ and related conjectures, arXiv:2602.21895 (2026), Thm. 18.
- [64] O. Martin, A. M. Odlyzko, S. Wolfram, Algebraic properties of cellular automata, *Comm. Math. Phys.* 93 (1984) 219–258.
- [65] J. Kopra, Rapid left expansivity, a commonality between Wolfram’s Rule 30 and powers of p/q , *Theoret. Comput. Sci.* 946 (2023) 113668; arXiv:2202.13809 (2022), under the title *A natural class of cellular automata containing fractional multiplication automata, Rule 30, and others*. Def. 2.4, Thms. 3.5 and 4.7 and Problem 4.8 are cited in the numbering of arXiv v1.
- [66] E. S. Rowland, Local nested structure in Rule 30, *Complex Systems* 16 (2006) 239–258.
- [67] J. Gravner, D. Griffeath, Robust periodic solutions and evolution from seeds in one-dimensional edge cellular automata, *Theoret. Comput. Sci.* 466 (2012) 64–86.
- [68] S. Wolfram, Announcing the Rule 30 Prizes, *Stephen Wolfram Writings*, 1 October 2019, <https://writings.stephenwolfram.com/2019/10/announcing-the-rule-30-prizes/>; the three problems are also posted at <https://rule30prize.org/>.
- [69] S. Eilenberg, *Automata, Languages and Machines*, Vol. A, Academic Press, 1974.
- [70] B. Adamczewski, R. Yassawi, A note on Christol’s theorem, arXiv:1906.08703 (2019).
- [71] J. Shallit, Automaticity IV: sequences, sets, and diversity, *J. Théor. Nombres Bordeaux* 8 (1996) 347–367.
- [72] V. Nesme, arXiv:2207.13062.
- [73] J.-P. Allouche, J. Shallit, *Automatic Sequences*, Cambridge University Press, 2003.
- [74] E. Krawczyk, C. Müllner, Automaticity of uniformly recurrent substitutive sequences, arXiv:2111.13134 (2021), Thm. B (numbering of arXiv v3).
- [75] A. Bridy, Automatic sequences and curves over finite fields, *Algebra Number Theory* 11 (2017) 685–712.

- [76] B. Litow, P. Dumas, Additive cellular automata and algebraic series, *Theoret. Comput. Sci.* 119 (1993) 345–354.
- [77] J.-P. Allouche, F. von Haeseler, H.-O. Peitgen, G. Skordev, Linear cellular automata, finite automata and Pascal’s triangle, *Discrete Appl. Math.* 66 (1996) 1–22.
- [78] V. Belitsky, P. A. Ferrari, Ballistic annihilation and deterministic surface growth, *J. Stat. Phys.* 80 (1995) 517–543.
- [79] J. Krug, H. Spohn, Universality classes for deterministic surface growth, *Phys. Rev. A* 38 (1988) 4271–4283.
- [80] N. Berzai, The 0^m1 family and the quench lemma, working document, 2026 (quench_writeup.md).
- [81] N. Berzai (with Claude), The Rule 30 centre column: an exact Duhamel formula and certified bounds on automaticity, working document rule30_automaticity.pdf, 2026-09-16.
- [82] W. Meier, O. Staffelbach, Analysis of pseudo random sequences generated by cellular automata, *Advances in Cryptology — EUROCRYPT ’91*, LNCS 547, Springer, 1991, 186–199.
- [83] C. K. Koç, A. M. Apohan, Inversion of cellular automata iterations, *IEE Proc. Comput. Digit. Tech.* 144 (1997) 279–284.
- [84] J. Spencer, Cellular automata as cryptographic random number generators, arXiv:1306.3546 (2013).
- [85] León et al., cryptanalysis of cellular-automaton keystream generators, 2003.
- [86] S. Wolfram, Statistical mechanics of cellular automata, *Rev. Mod. Phys.* 55 (1983) 601–644.
- [87] S. Wolfram, Cryptography with cellular automata (extended abstract), *Advances in Cryptology — CRYPTO ’85*, LNCS 218, Springer, 1986, 429–432.
- [88] S. Wolfram, *A New Kind of Science*, Wolfram Media, 2002.
- [89] L. Mariot, Insights from a decade of cellular automata in cryptography, arXiv:2405.02875 (2024).
- [90] E. Chan-López, A. Martín-Ruiz, Symmetric nonlinear cellular automata as algebraic references for Rule 30, arXiv:2604.00165 (2026).
- [91] D. L. Condrey, Finite configurations cannot generate a constant trace in Rule 30, arXiv:2609.09431 (2026).
- [92] J.-P. Allouche, J. Shallit, R. Yassawi, How to prove that a sequence is not automatic, *Expo. Math.* 40 (2022) 1–22.
- [93] E. Rowland, R. Yassawi, A characterization of p -automatic sequences as columns of linear cellular automata, *Adv. Appl. Math.* 63 (2015) 68–89.
- [94] R. A. Rueppel, *Analysis and Design of Stream Ciphers*, Springer, 1986.

- [95] K. Kawamura, Polynomial representations of the Cuntz algebras arising from permutations, arXiv:math/0508273 (2005).
- [96] O. Bratteli, P. E. T. Jorgensen, Iterated function systems and permutation representations of the Cuntz algebra, *Mem. Amer. Math. Soc.* 139 (1999), no. 663.
- [97] M. Fekete, Über die Verteilung der Wurzeln bei gewissen algebraischen Gleichungen mit ganzzahligen Koeffizienten, *Math. Z.* 17 (1923) 228–249.
- [98] G. Meinardus, Asymptotische Aussagen über Partitionen, *Math. Z.* 59 (1954) 388–398.
- [99] R. Impagliazzo, A. Wigderson, $P = BPP$ if E requires exponential circuits: derandomizing the XOR lemma, *STOC 1997*, 220–229.
- [100] H. Mousavi, Automatic theorem proving in Walnut, arXiv:1603.06017 (2016).
- [101] C. Moore, Predicting nonlinear cellular automata quickly by decomposing them into linear ones, *Physica D* 111 (1998) 27–41; arXiv:patt-sol/9701008 (1997), by C. Moore and T. Pnini.
- [102] M. Feder, N. Merhav, M. Gutman, Universal prediction of individual sequences, *IEEE Trans. Inform. Theory* 38 (1992) 1258–1270.
- [103] N. Berzai, The left edge of Rule 30 without Rule 30: the stripe recursion, Wolfram’s number 2,107,985,255, and the forks beyond it, manuscript, 2026; with its code and data at <https://github.com/nateberzai-svg/rule30-left-edge>.
- [104] Dibujaron (D. McPolstra), rule30, public code repository, <https://github.com/Dibujaron/rule30>; commit 94833d21c1efc1e826df3c137777b580817c5e16, 8 September 2026, file blueprint/crystals.md, item 55.
- [105] Patto1155, rule30-foundry, public code repository, <https://github.com/Patto1155/rule30-foundry>; commit 04f30f6085210b476e64d742c77dc7c69a075791 (pull request 32), merged 8 September 2026, from a run of about 30 August 2026, files docs/experiment-logs/2026-09-08-b1-recovery.md and runs/b1-recovered-2026-09-08/pattern_map_walk32.original.json.
- [106] szymon-lania, public gist with a starting state for the other version of left diagonal 53209 of Rule 30, 21 August 2026, <https://gist.github.com/szymon-lania/2535d62468a7912137b5002d900ebdd6>; described in its author’s comment on E. Rowland’s video *The Hidden Structure of Rule 30*, <https://www.youtube.com/watch?v=HcGjCcLEgN4>.

A The certificate engine (independent implementation)

```
import numpy as np, sys
K=10; N=1<<K; idx=np.arange(N,dtype=np.int64); bits=[(idx>>i)&1 for i in range(K)]
def succ(c,f):
    left=[np.full(N,c,dtype=np.int64)]+bits+[np.full(N,f,dtype=np.int64)]
    out=np.zeros(N,dtype=np.int64)
    for i in range(1,K+1): out|=(left[i-1]^(left[i]|left[i+1]))<<(i-1)
    return out
S0=[succ(0,0),succ(0,1)]; S1=[succ(1,0),succ(1,1)]; u0=(bits[0]==0)
```

```

def step(mask,c):
    nxt=np.zeros(N,dtype=bool); tab=S0 if c==0 else S1
    src=np.nonzero(mask)[0]; nxt[tab[0][src]]=True; nxt[tab[1][src]]=True; return nxt
def key(m): return np.packbits(m).tobytes()
def graph(word):
    ALL=np.ones(N,dtype=bool); ids={key(ALL):0}; states=[ALL]; todo=[0]; edges=[]
    while todo:
        s=todo.pop(); fr={():states[s]}
        for ch in word:
            nx={}
            for lab,T in fr.items():
                if ch=='0':
                    for u,sel in ((0,u0),(1,~u0)):
                        Tb=T&sel
                        if Tb.any(): nx[lab+(u,)] = step(Tb,0)
                    else: nx[lab] = step(T,1)
            fr=nx
        for lab,T in fr.items():
            if not T.any(): continue
            k=key(T)
            if k not in ids: ids[k]=len(states); states.append(T); todo.append(len(states)-1)
            edges.append((s,ids[k],lab))
    return states,edges
# Tarjan SCC on (states, edges); criterion: for every SCC with an edge,
# #edges (both ends inside) == #vertices.

```